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Exhibit 78C

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February 23, 2023

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via Certified Mail, Return Receipt

Mr. Larry Smith
President
Barcalounger Corporation¹
PO Box 2128
Morristown, TN 37816

Re: Barcalounger's Infringement of Ultra-Mek's Patent Rights

Dear Mr. Smith:

We represent Ultra-Mek, Inc. I am writing to you about Barcalounger's infringement of Ultra-Mek's patent rights.

Barcalounger is

[REDACTED]

[REDACTED]

¹ "Barcalounger" is used herein to refer to Barcalounger Corporation and its owners and/or entities controlling Barcalounger, including HPC3 Furniture Holdings, LLC

PUBLIC VERSION

Mr. Larry Smith
February 23, 2023
Page 2

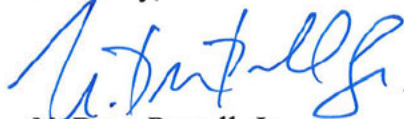
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Sincerely,



N. Dean Powell, Jr.

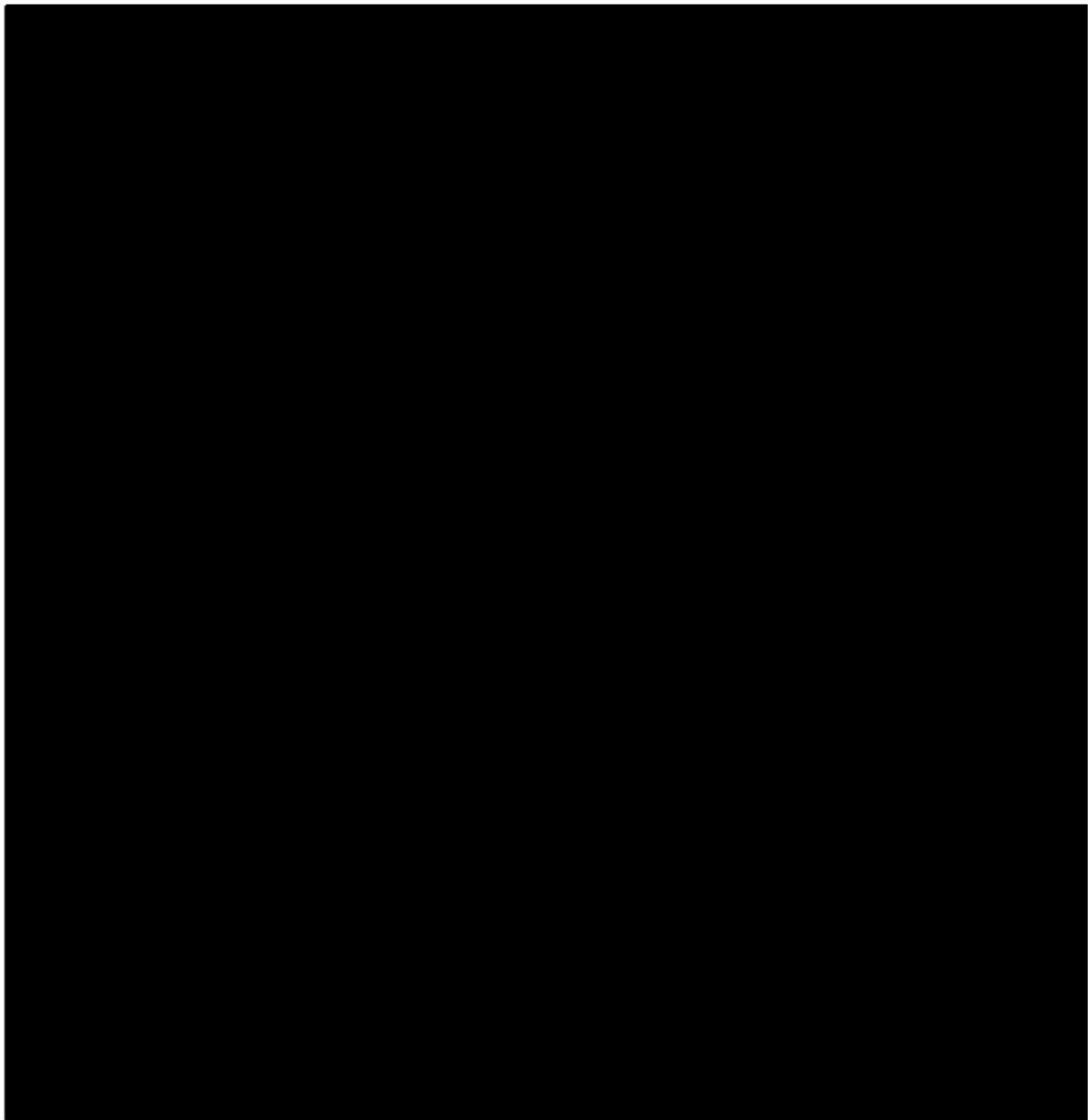
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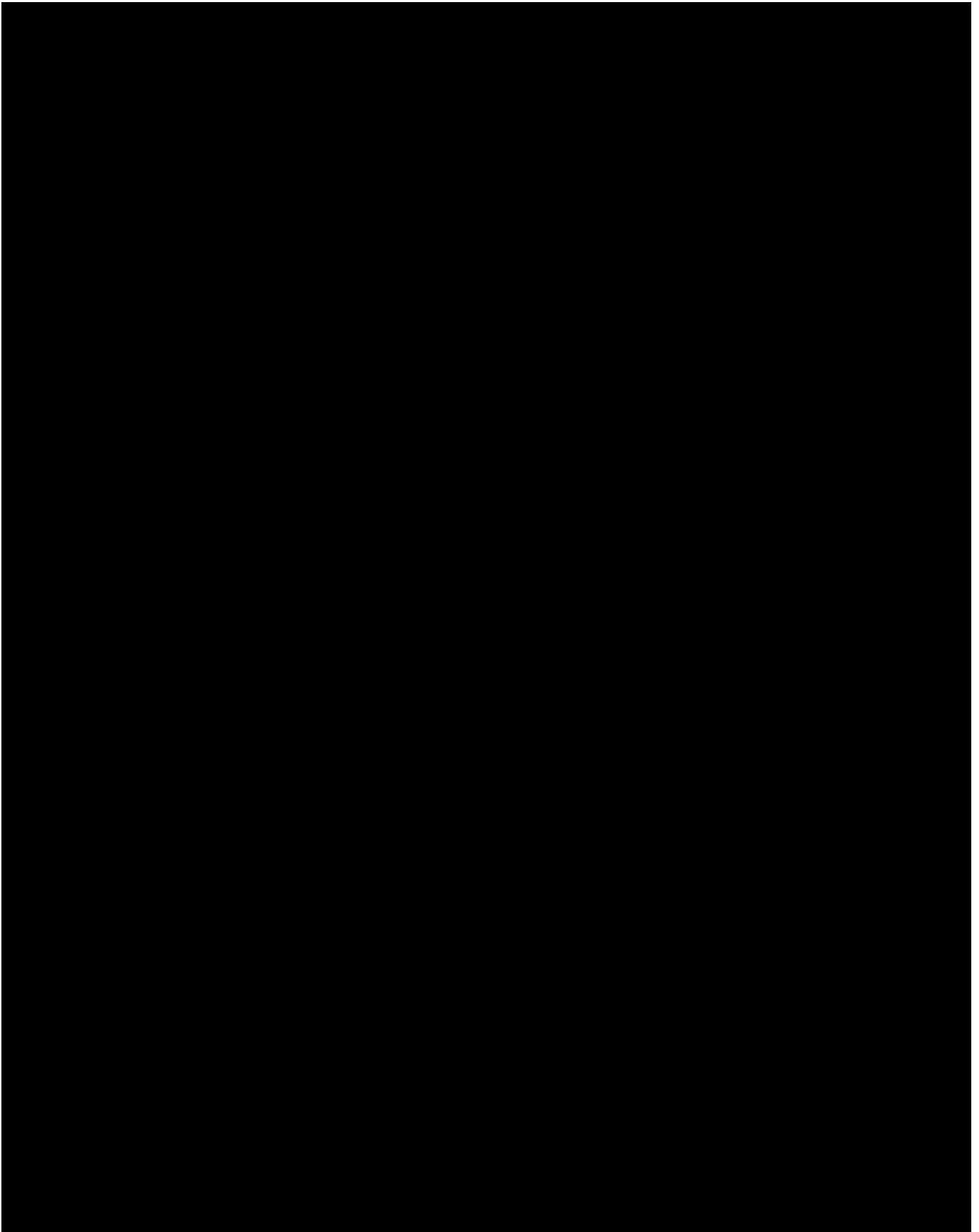
cc: Ultra-Mek, Inc.

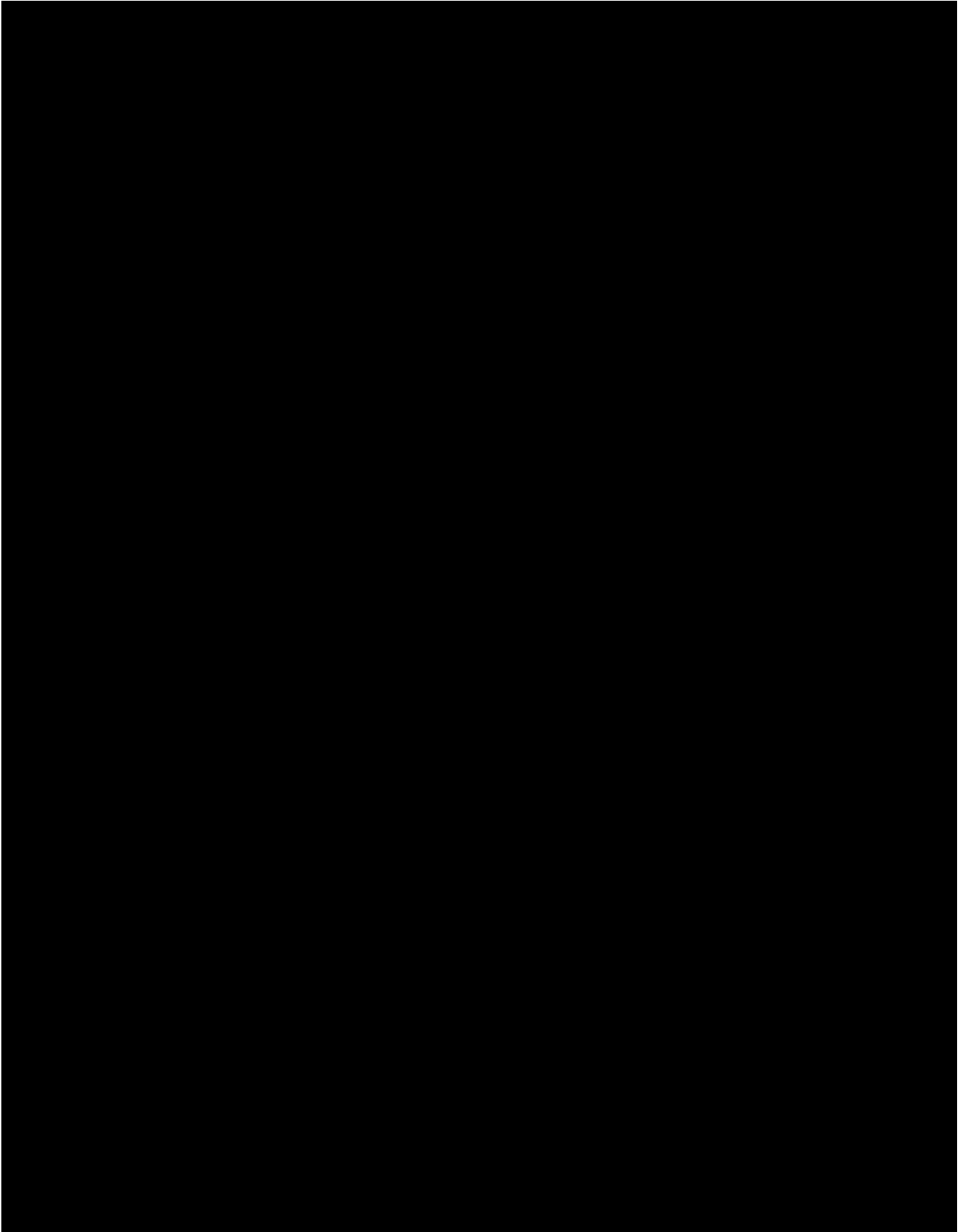


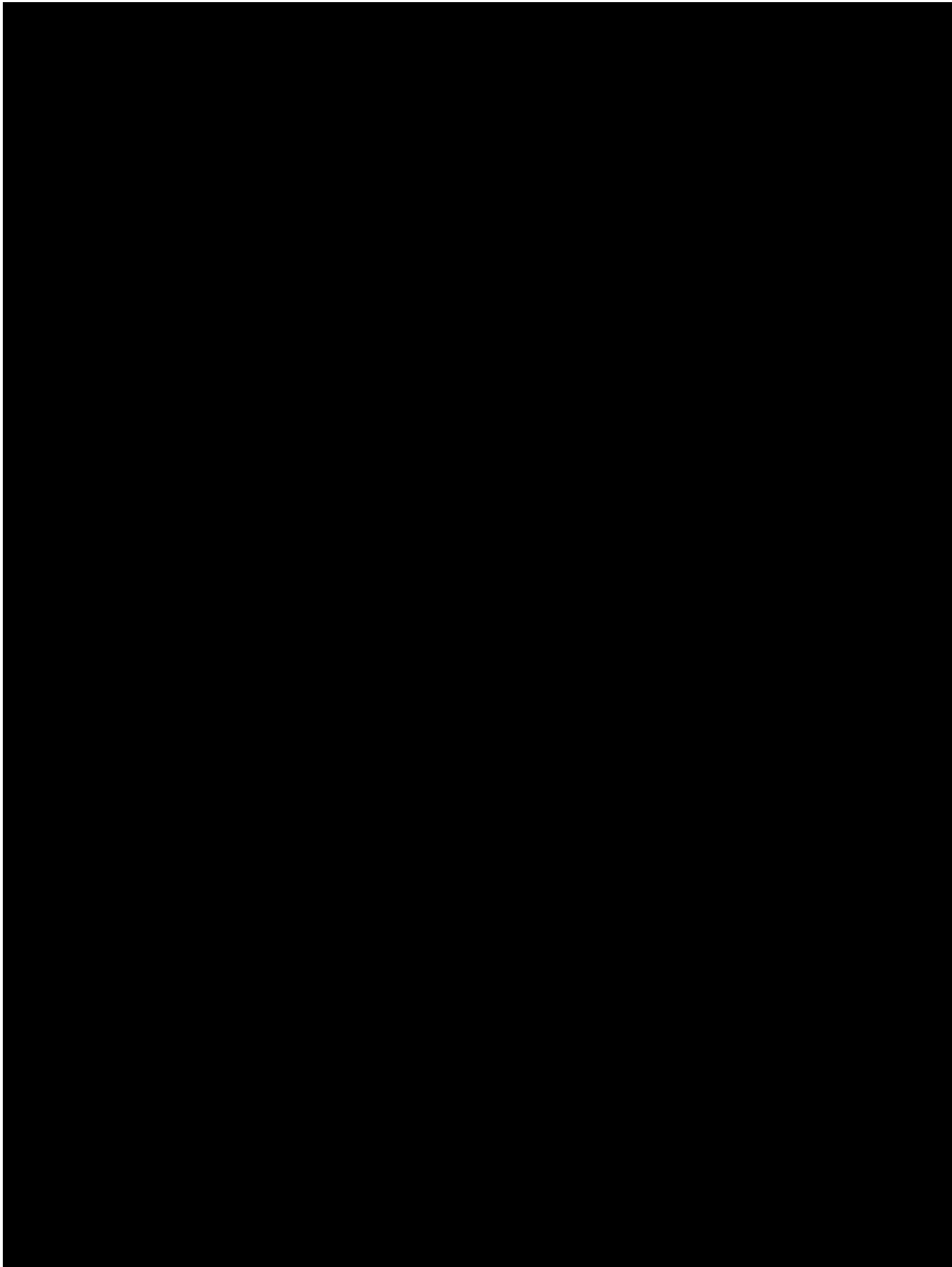


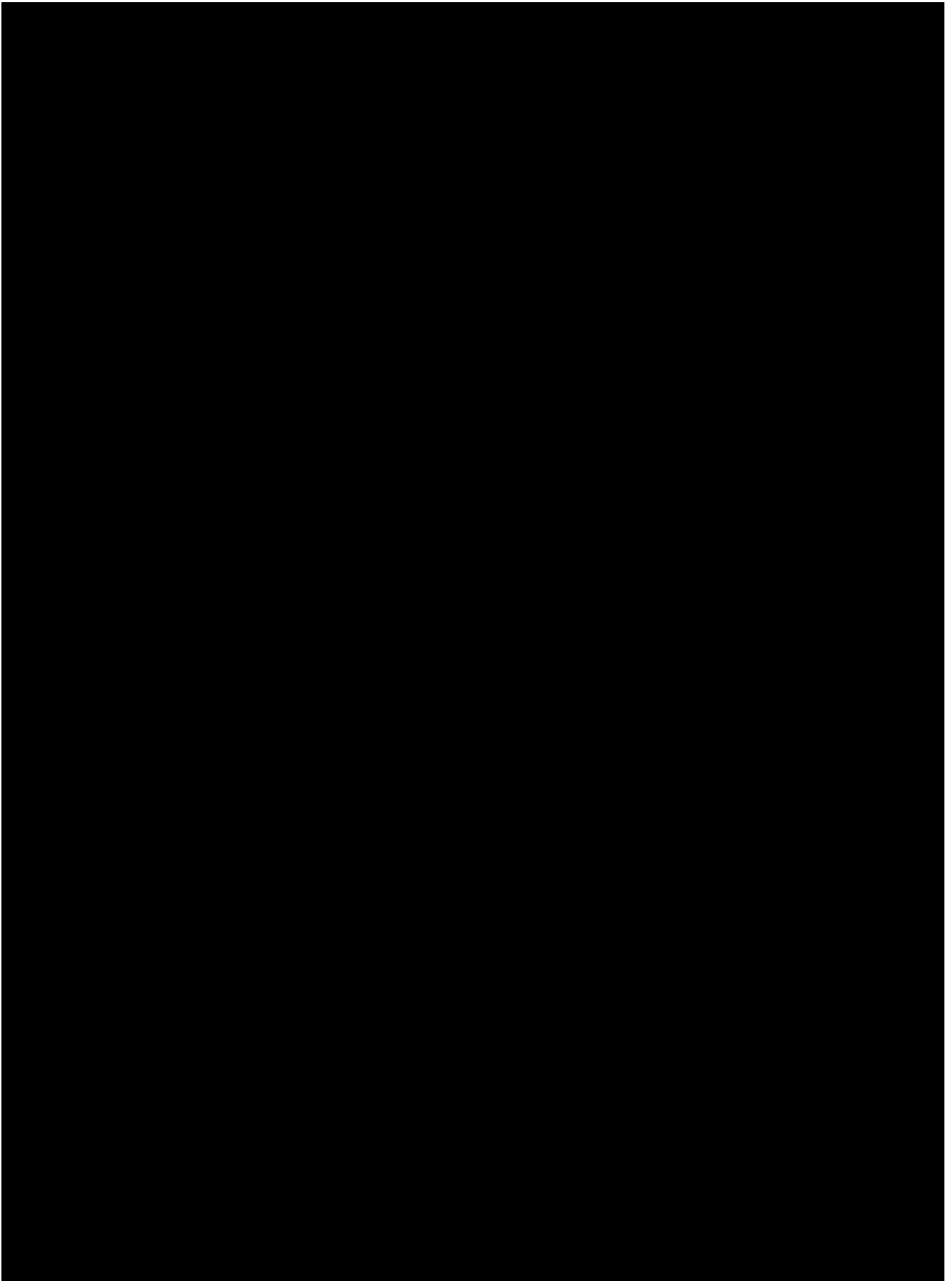














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US008016348B2

(12) **United States Patent**
Hoffman et al.

(10) **Patent No.:** **US 8,016,348 B2**
(45) **Date of Patent:** **Sep. 13, 2011**

(54) **RECIPROCATING SEATING UNIT WITH
POWER ACTUATOR**

(75) Inventors: **D. Stephen Hoffman**, High Point, NC
(US); **Marcus L. Murphy**, Lexington,
NC (US)

(73) Assignee: **Ultra-Mek, Inc.**, Denton, NC (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 207 days.

(21) Appl. No.: **12/508,901**

(22) Filed: **Jul. 24, 2009**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 12/276,567,
filed on Nov. 24, 2008.

(51) **Int. Cl.**
A47C 1/031 (2006.01)

(52) **U.S. Cl.** **297/85 M**

(58) **Field of Classification Search** **297/85 M,**
297/68, 342, 330, 344.21, 344.24, 344.26,
297/84

See application file for complete search history.

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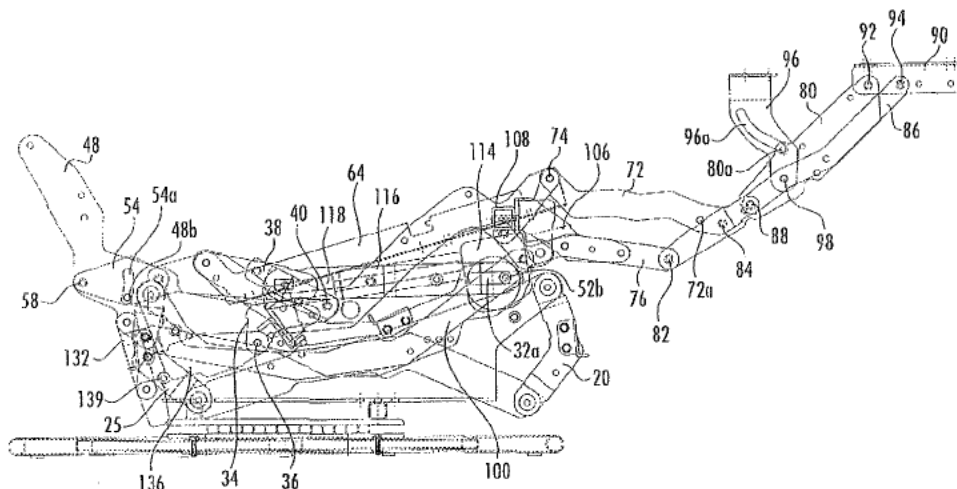
Primary Examiner — Milton Nelson, Jr.

(74) *Attorney, Agent, or Firm* — Myers Bigel Sibley &
Sajovec

(57) **ABSTRACT**

A reciprocating and reclining seating unit includes: a base
unit with a first bearing surface; a generally horizontally-
disposed seat positioned above the base; a generally upright
backrest positioned above the base and substantially rearward
of the seat; an extendable ottoman; a reclining mechanism
attached to the seat, the backrest, the ottoman and the base
unit, the reclining mechanism comprising a plurality of piv-
otally interconnected links; a reciprocating mechanism
attached to the base unit and the reclining mechanism, the
reciprocating mechanism being configured to enable the seat,
backrest and reclining mechanism to reciprocate relative to
the base unit along a longitudinal path responsive to a longi-
tudinally-directed force; and a power actuating unit attached
to the reclining mechanism.

15 Claims, 8 Drawing Sheets



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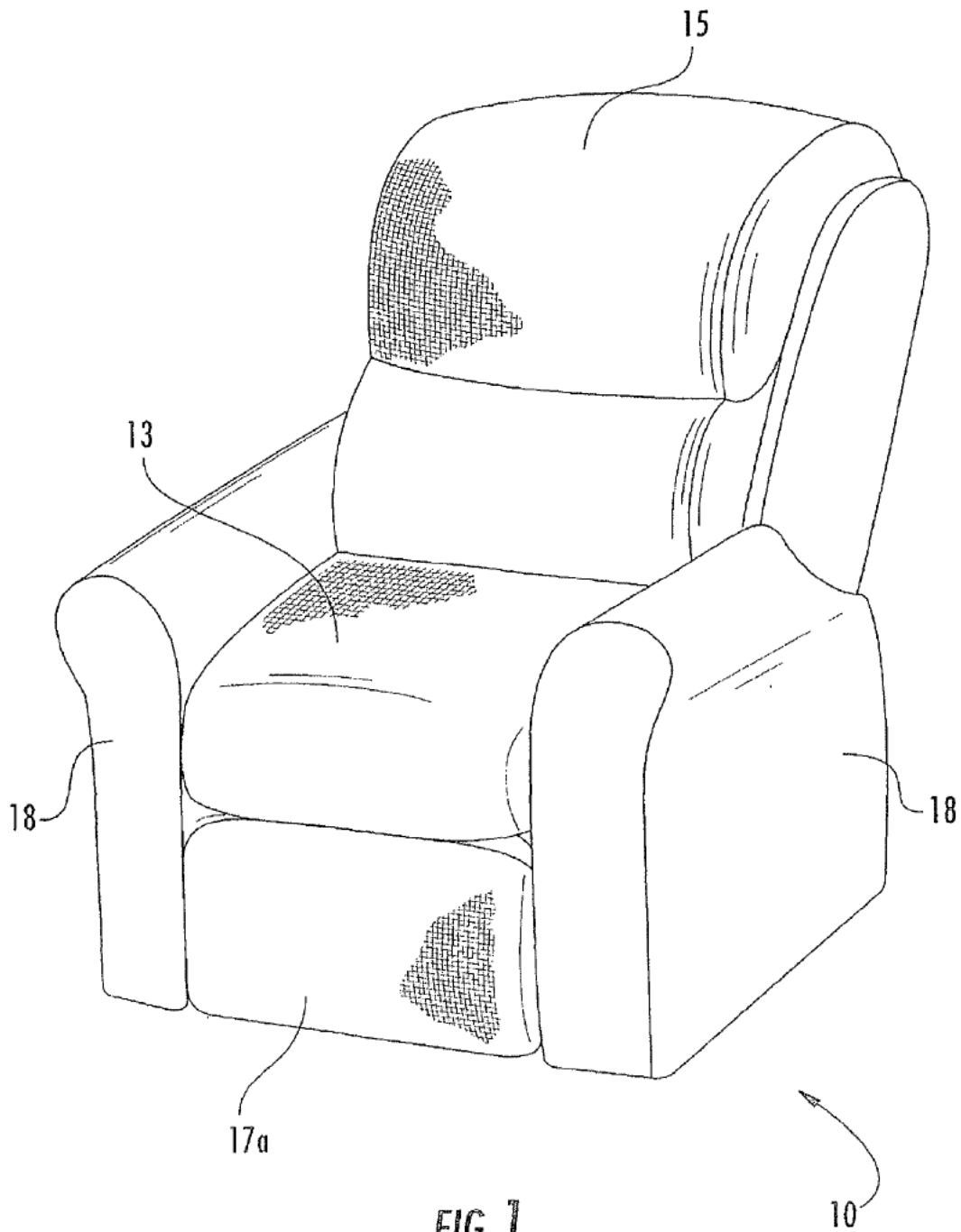
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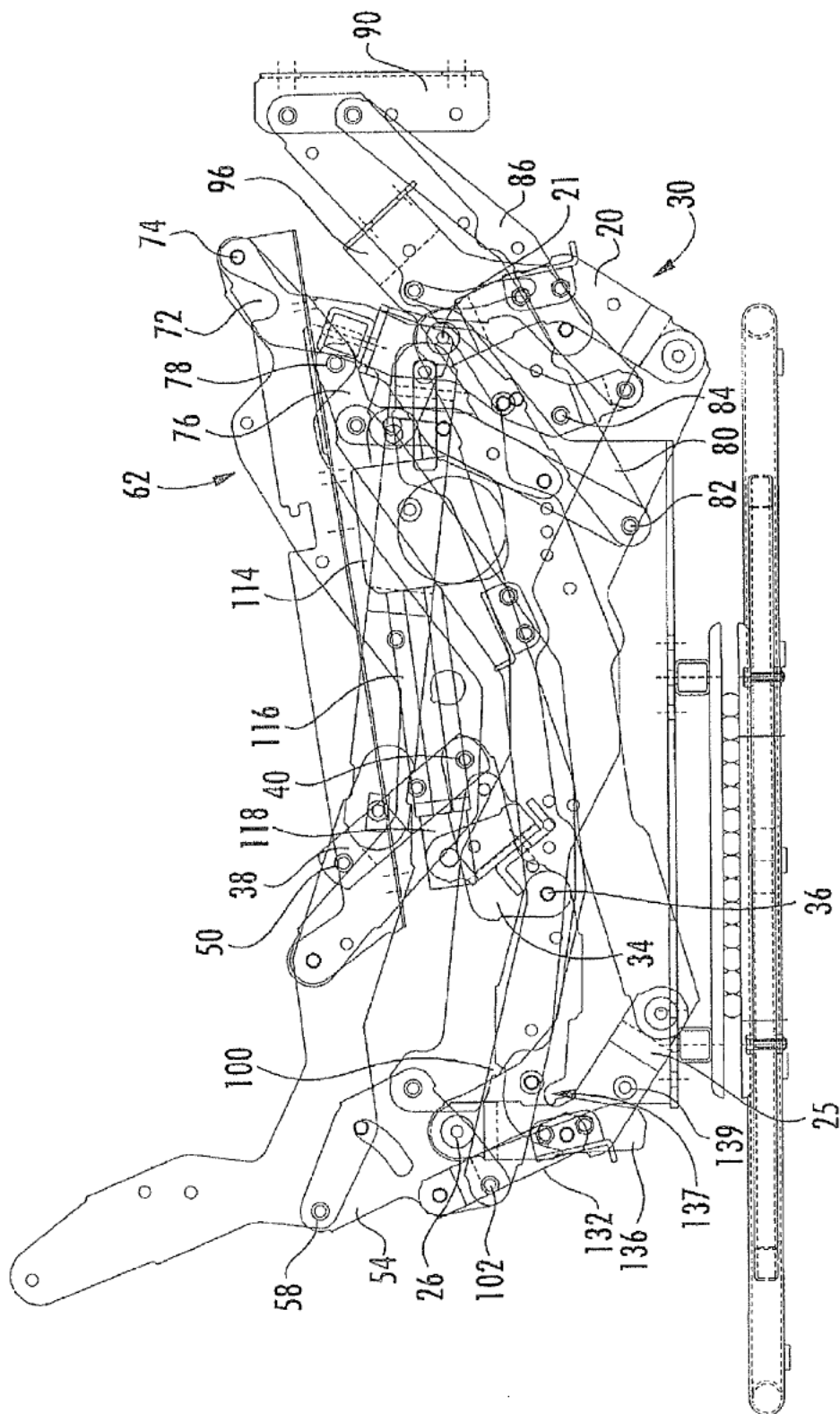
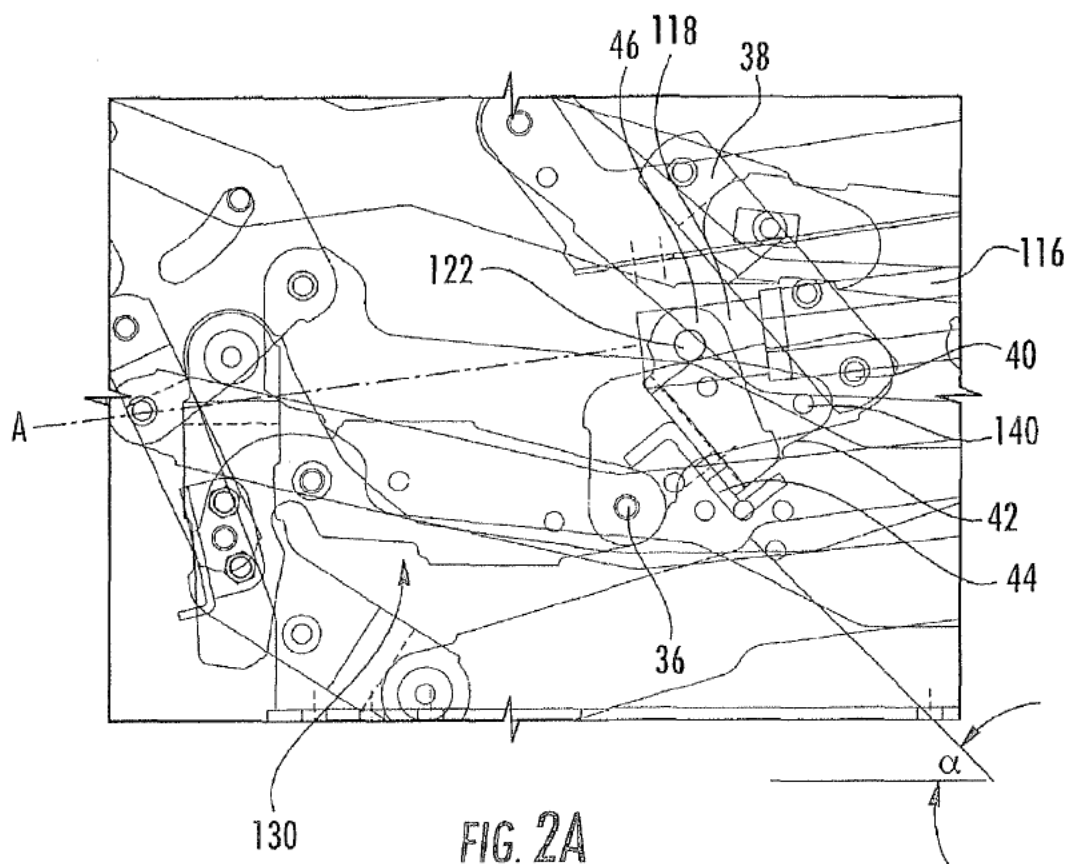
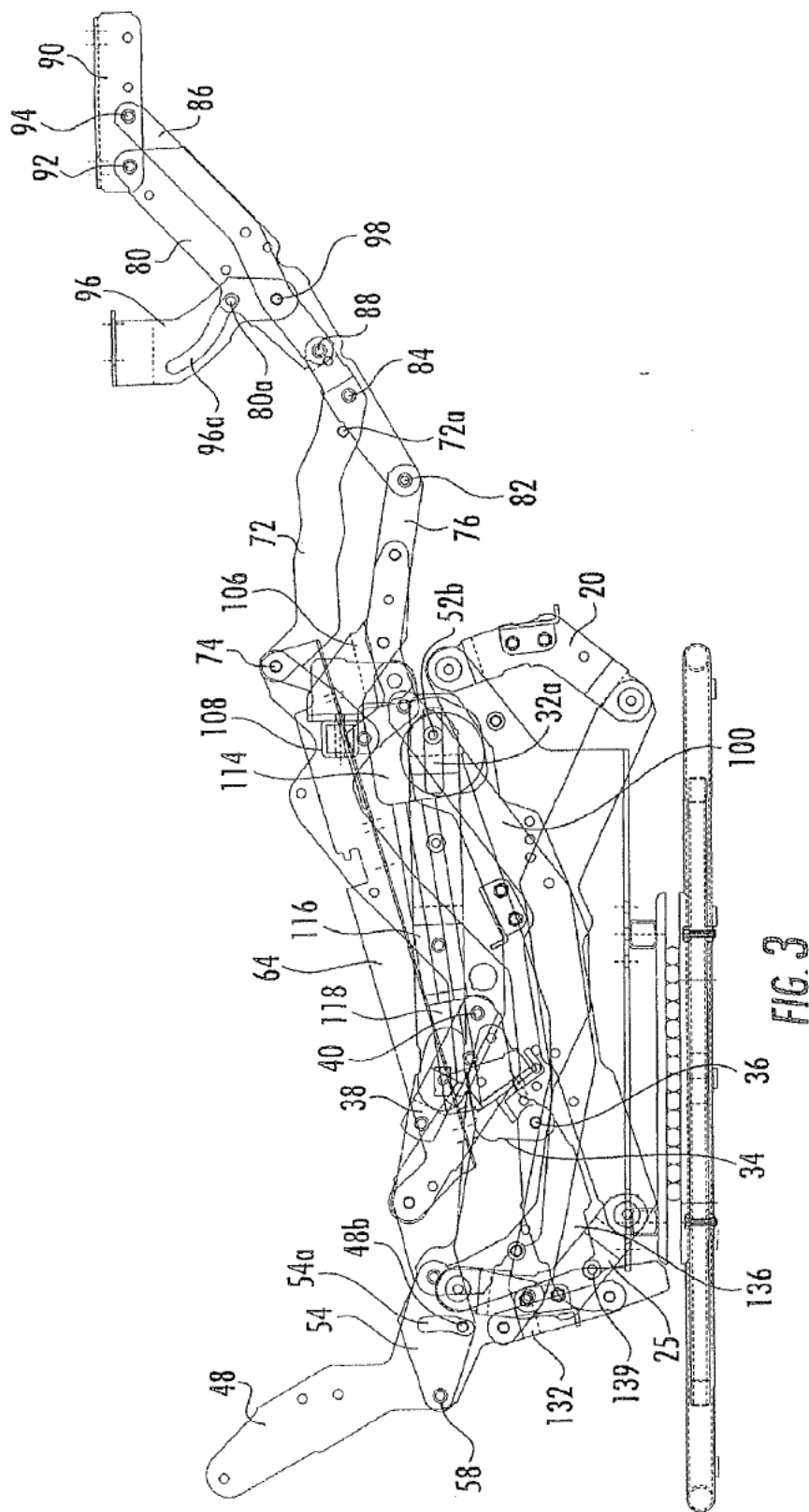


FIG. 2





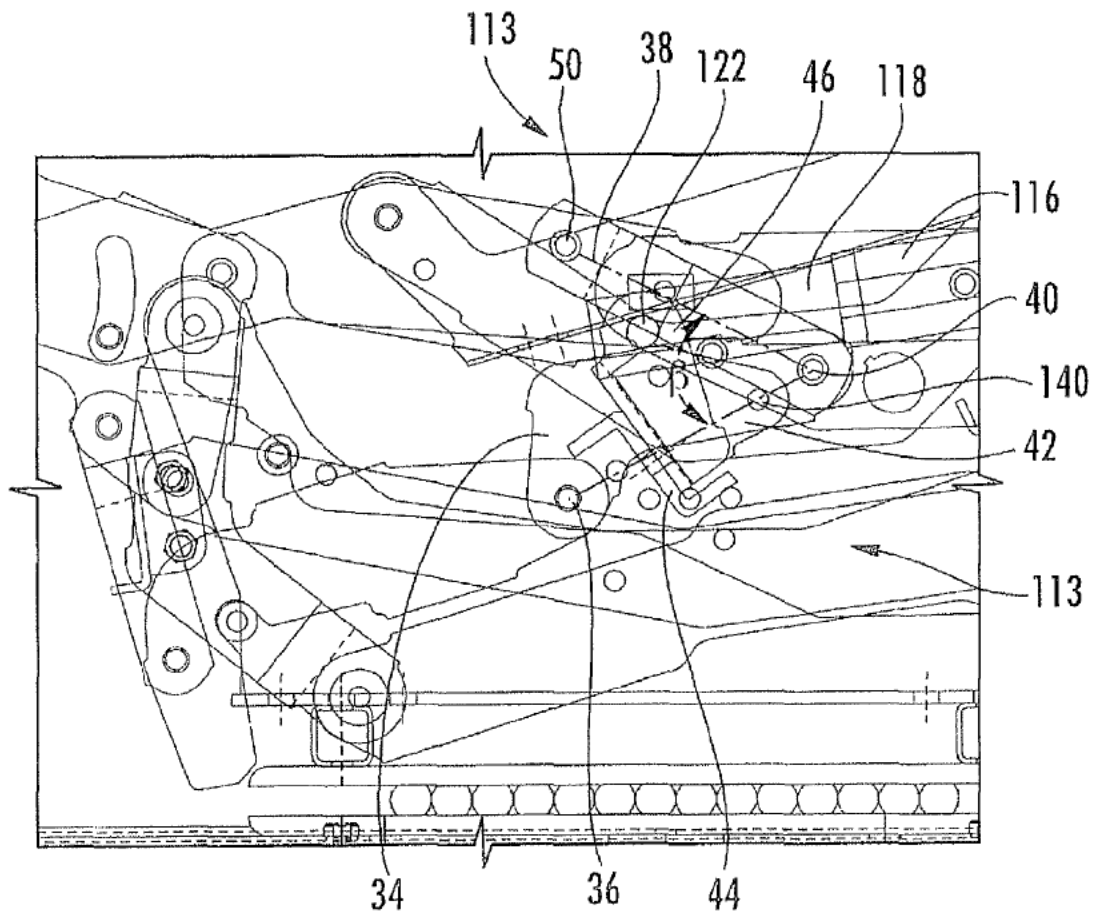
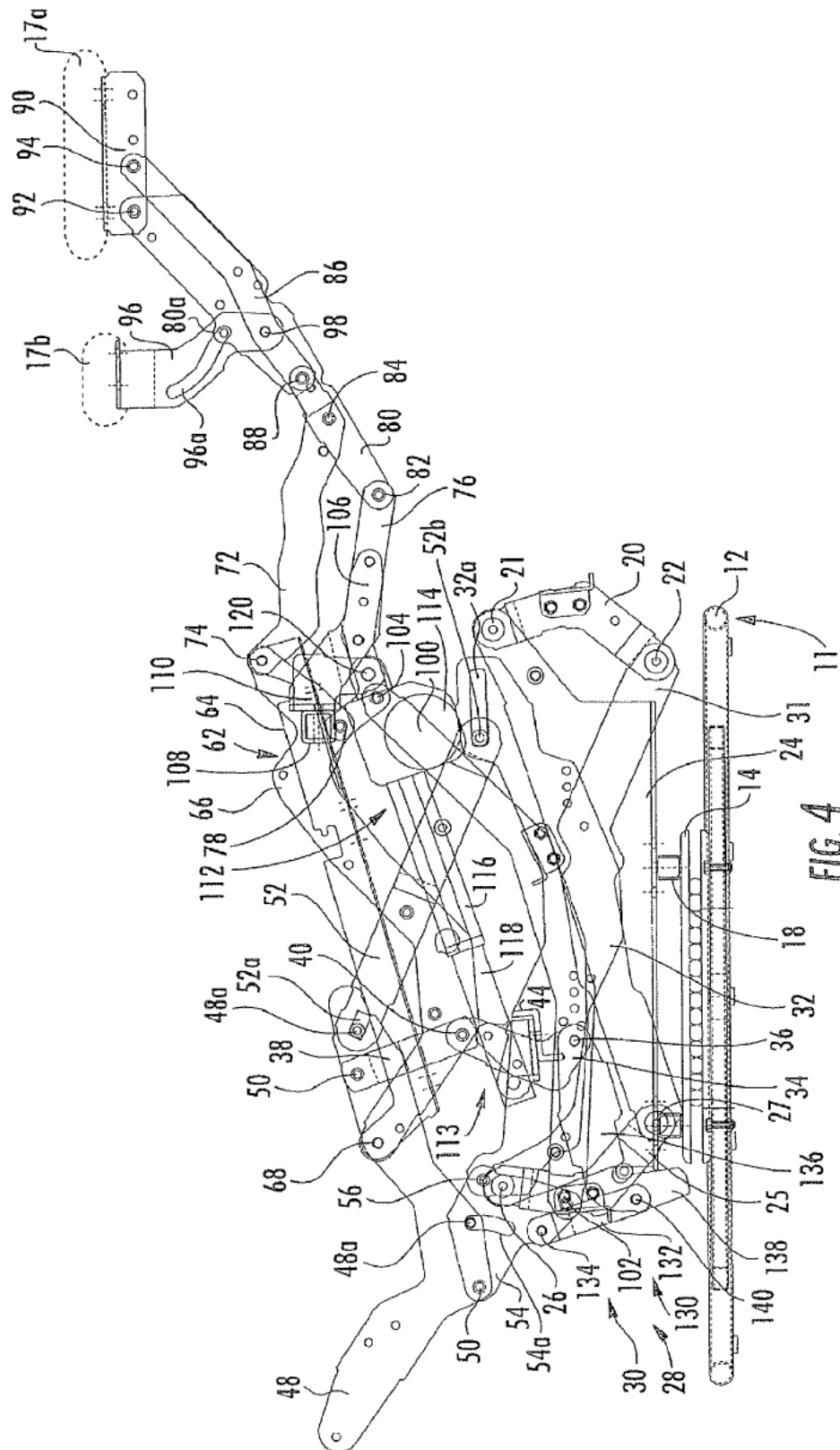


FIG. 3A



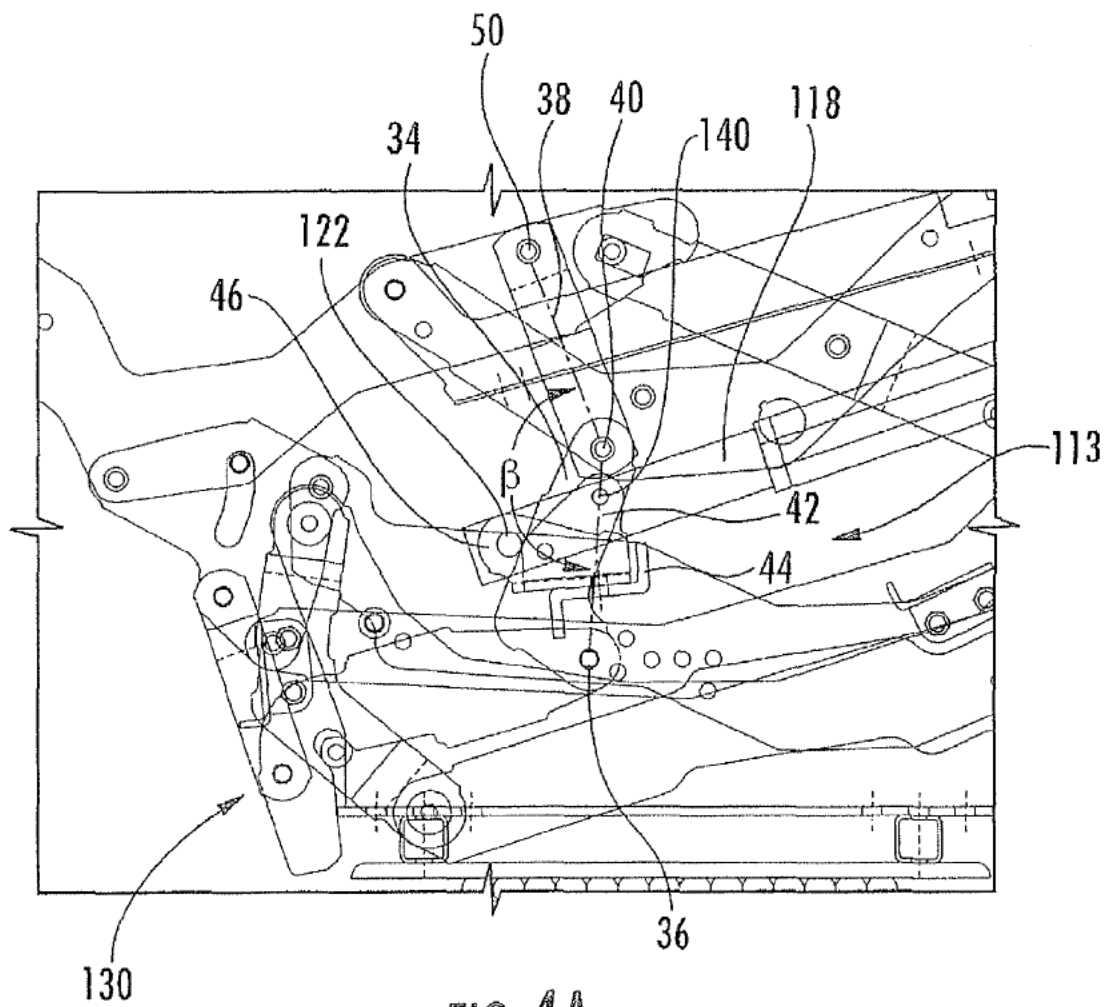


FIG. 4A

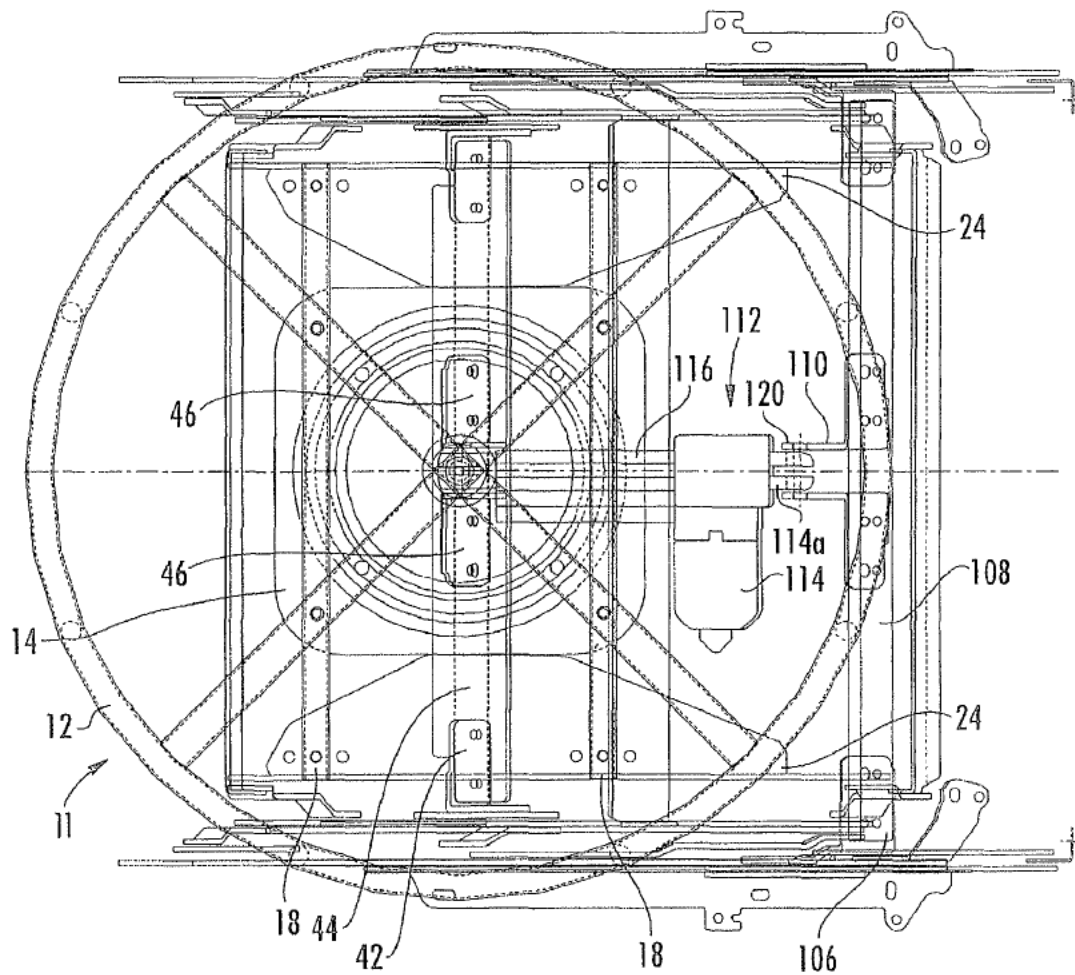


FIG. 5

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RECIPROCATING SEATING UNIT WITH POWER ACTUATOR

RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 12/276,567, filed Nov. 24, 2008, the disclosure of which is hereby incorporated herein in its entirety.

FIELD OF THE INVENTION

This invention relates generally to seating units, and relates more particularly to reclining seating units with rocking capability.

BACKGROUND OF THE INVENTION

Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

In recent years, furniture designers have looked for alternatives to rocking chairs that can provide a similarly relaxing repetitive motion. One alternative has been the gliding chair, or "glider", which includes structure that enables the seat portion of the chair to "glide" forwardly and rearwardly relative to its base to mimic generally the rocking motion of a rocking chair. Often the gliding structure comprises a set of swing links (usually two at the front of the chair, and two at the rear) that are pivotally attached at their upper ends to the base and extend downwardly therefrom to attach to a structure, such as a mounting bracket, that is attached to the seat. In this configuration, the seat is suspended from the base and is free to swing forwardly and rearwardly in a double pendulum-type motion in response to a forwardly or rearwardly-directed force applied by a seated occupant. The gliding path of the chair is controlled by the configuration and mounting of the swing links. These chairs can be constructed to resemble traditional rocking chairs and thus are quite popular.

Reclining capability has been combined with gliding capability in a single unit to provide a chair that both reclines and glides. This chair includes a reclining mechanism that enables it to move between upright and one or more reclined positions, and further includes the aforementioned swing links attached between the base and the seat, armrests, or mechanism itself to enable the chair to glide. Examples of such chairs are illustrated and described in U.S. Pat. Nos. 4,536,029 and 4,544,201, both to Rogers, Jr., the disclosures of which are hereby incorporated herein by reference in their entireties.

Although they are already popular seating units, it may be desirable to provide additional functionality to glider-recliners.

SUMMARY OF THE INVENTION

As a first aspect, embodiments of the present invention are directed to a reciprocating and reclining seating unit. The seating unit comprises: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base

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and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The actuating unit is attached to an actuating mechanism, the actuating mechanism comprising a cross-member extending transversely across the seating unit and first and second projections fixed to the cross-member. The first projection is pivotally attached to the actuating unit. The second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism.

As a second aspect, embodiments of the present invention are directed to a reciprocating and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power linear actuating unit attached to the reclining mechanism, the actuating unit including a motor and an extendable rod and configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The actuating unit is attached to an actuating mechanism, the actuating mechanism comprising: a cross-member extending transversely across the seating unit; and first and second projections fixed to the cross-member. The first projection is pivotally attached to the actuating unit. The second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism. The actuating unit includes opposed first and second ends, wherein the first end of the actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the

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second end of the actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a perspective view of a glider-recliner chair according to embodiments of the present invention.

FIG. 2 is a side section view of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 2A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 3 is a side section view of the chair of FIG. 1 with the chair in its TV position.

FIG. 3A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its TV position.

FIG. 4 is a side section view of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 4A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 5 is a top cutaway view of the frame of the chair of FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

The present invention now is described more fully herein-
after with reference to the accompanying drawings, in which
embodiments of the invention are shown. This invention may,
however, be embodied in many different forms and should not
be construed as limited to the embodiments set forth herein;
rather, these embodiments are provided so that this disclosure
will be thorough and complete, and will fully convey the
scope of the invention to those skilled in the art.

Like numbers refer to like elements throughout. In the
figures, the thickness of certain lines, layers, components,
elements or features may be exaggerated for clarity. Broken
lines illustrate optional features or operations unless specified
otherwise.

The terminology used herein is for the purpose of describ-
ing particular embodiments only and is not intended to be
limiting of the invention. As used herein, the singular forms
“a”, “an” and “the” are intended to include the plural forms as
well, unless the context clearly indicates otherwise. It will be
further understood that the terms “comprises” and/or “com-
prising,” when used in this specification, specify the presence
of stated features, integers, steps, operations, elements, and/
or components, but do not preclude the presence or addition
of one or more other features, integers, steps, operations,
elements, components, and/or groups thereof. As used herein,
the term “and/or” includes any and all combinations of one or
more of the associated listed items. As used herein, phrases
such as “between X and Y” and “between about X and Y”
should be interpreted to include X and Y. As used herein,
phrases such as “between about X and Y” mean “between
about X and about Y.” As used herein, phrases such as “from
about X to Y” mean “from about X to about Y.”

Unless otherwise defined, all terms (including technical
and scientific terms) used herein have the same meaning as
commonly understood by one of ordinary skill in the art to
which this invention belongs. It will be further understood
that terms, such as those defined in commonly used dictio-
naries, should be interpreted as having a meaning that is
consistent with their meaning in the context of the specifica-

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tion and relevant art and should not be interpreted in an
idealized or overly formal sense unless expressly so defined
herein. Well-known functions or constructions may not be
described in detail for brevity and/or clarity.

It will be understood that when an element is referred to as
being “on”, “attached” to, “connected” to, “coupled” with,
“contacting”, etc., another element, it can be directly on,
attached to, connected to, coupled with or contacting the
other element or intervening elements may also be present. In
contrast, when an element is referred to as being, for example,
“directly on”, “directly attached” to, “directly connected” to,
“directly coupled” with or “directly contacting” another ele-
ment, there are no intervening elements present. It will also be
appreciated by those of skill in the art that references to a
structure or feature that is disposed “adjacent” another feature
may have portions that overlap or underlie the adjacent fea-
ture.

This invention is directed to seating units that have a sta-
tionary base, a seat portion, and a backrest. As used herein, the
terms “forward”, “forwardly”, and “front” and derivatives
thereof refer to the direction defined by a vector extending
from the backrest toward the seat parallel to the underlying
surface. Conversely, the terms “rearward”, “rearwardly”, and
derivatives thereof refer to the direction directly opposite the
forward direction; the rearward direction is defined by a vec-
tor that extends from the seat toward the backrest parallel to
the underlying surface. The terms “lateral,” “laterally,” and
derivatives thereof refer to the direction parallel with the
floor, perpendicular to the forward and rearward directions,
and extending away from a plane bisecting the seating units
between their armrests. The terms “medial,” “inward,”
“inboard,” and derivatives thereof refer to the direction that is
the converse of the lateral direction, i.e., the direction parallel
with the floor, perpendicular to the forward direction, and
extending from the periphery of the seating units toward the
aforementioned bisecting plane.

The seating units illustrated and described herein comprise
a plurality of pivotally interconnected links. Those skilled in
this art will appreciate that the pivots between links can take
a variety of configurations, such as pivot pins, rivets, bolt and
nut combinations, and the like, any of which would be suit-
able for use with the present invention. Also, the shapes of the
links may vary as desired, as may the locations of certain of
the pivots. Moreover, in some instances combinations of pivot
points may be replaced by equivalent structures, such as
“slider-crank” configurations, like those described in B. Paul,
Kinematics and Dynamics of Planar Machinery 4-21 (1979).
In other embodiments, another mechanism that provides
reciprocating motion along a longitudinal path, such as a
rocking unit like that discussed in co-assigned and U.S. patent
application Ser. No. 12/276,559, the disclosure of which is
hereby incorporated herein in its entirety, may also be
employed.

Referring now to the figures, a swiveling glider-recliner
chair, designated broadly at 10, is illustrated in FIGS. 1-5. The
chair 10 includes a base unit 11, a seat 13 that is generally
horizontally disposed above the base unit 11, a backrest 15
that is generally vertically and disposed substantially above a
rear portion of the seat 13, and two ottomans 17a, 17b, which,
in the upright position of FIG. 1, are generally vertically
disposed below a front portion of the seat 13. Arms 18 are
positioned on either side of the seat 13 and move in concert
with the seat 13.

A pair of mirror image reclining mechanisms 30 (only one
of which is shown herein in FIGS. 2-4) are attached to the
backrest 15, the seat 13, and the ottomans 17a, 17b. The
reclining mechanisms 30, which comprise a plurality of inter-

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connected links, move the chair 10 between (a) an upright position (FIGS. 1 and 2), in which the seat 13 is generally horizontally disposed, the backrest 15 is generally vertically disposed, and the ottomans 17a, 17b are generally vertically disposed and positioned below the seat 13, (b) an intermediate TV position (FIG. 3), in which the ottomans 17a, 17b are generally horizontally disposed in front of the seat 13 and the backrest 15 and the seat 13 substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position (FIG. 4), in which the angle between the backrest 15 and the seat 13 increases. These components are described in greater detail below.

The base unit 11 includes a circular lower base 12. A swivel unit 14 is mounted onto the top of the lower base 12. Cross-members 18 are mounted atop the swivel unit 14. Those skilled in this art will recognize that the base unit 11 may take other forms that provide mounting locations for the remaining components of the chair 10; for example, the swivel unit 14 may be omitted.

Glide foundation plates 24 are mounted to the top surfaces of the cross-members 18. A front glide link 20 is attached at a pivot 21 to the front end of the glide foundation plate 24, and a rear glide link 25 is attached at a pivot 26 to the rear end of the glide foundation plate 24. The front and rear glide links 20, 25 are suspended from the glide foundation plate 24 and together form a gliding mechanism 28 that provides a gliding motion to the chair 10 when it is in its upright position (FIG. 2). Those skilled in this art will appreciate that the gliding mechanism may take other forms; it may include glide links of different shapes, or it may include a "track"-based gliding mechanism.

Because the reclining mechanisms 30 are mirror images of each other, only one reclining mechanism will be described in detail herein, with the understanding that the discussion is equally applicable to the mirror image mechanism 30. Also, for clarity the mechanism will be described first with reference to FIG. 4 (with the mechanism 30 in its reclined position), then will be described in its upright and TV positions.

Turning now to FIG. 4, the reclining mechanism 30 includes a generally arcuate foundation link 31, which is attached at either end to the front and rear glide links 20, 25 at pivots 22, 27, respectively. A generally crescent-shaped mounting bracket 32 is fixed to the foundation link 31. A slot 32a is present at the front end of the mounting bracket 32. An L-shaped lower swing link 34 is attached to a lower region of the mounting bracket 32 at a pivot 36 and extends upwardly and rearwardly, then upwardly and forwardly therefrom. A straight upper swing link 38 is attached at its lower end to the upper end of the lower swing link 34 at a pivot 40 and extends upwardly therefrom.

A backpost 48 is fixed to the backrest 15 and extends generally forwardly from the lower front end of the backrest 15. The upper swing link 38 is attached to a forward portion of the backpost 48 at a pivot 50. A straight coupling link 52 having a slot 52a is attached to the backpost 48 via a pin 48a located forwardly of the pivot 50. The coupling link 52 extends forwardly and downwardly from the pin 48a to attach to the front end of the mounting bracket 32 via a pin 52b inserted into a slot 32a in the mounting bracket 32. A trapezoidal sequencer plate 54 is attached to the backpost 48 at a pivot 58, and also interacts with the backpost 48 via a slot 54a that receives a pin 48b. The sequencer plate 54 is also attached to the rear end of the mounting bracket 32 at a pivot 56.

Still referring to FIG. 4, a seat frame 62, to which the arms 18 are mounted, underlies the seat 13. A seat adapter 66 is fixed to the seat frame 62. The seat adapter 66 is attached to the backpost 48 at a pivot 68 that is located just rearward of the

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pivot 50. An upper ottoman swing link 72 is attached to a front region of the seat adapter 66 at a pivot 74 and extends downwardly and forwardly therefrom. A tripartite lower ottoman swing link 76 is attached to a pivot 78 that is located rearwardly and downwardly from the pivot 74; the lower ottoman swing link 76 extends generally forwardly from the pivot 78. An upper ottoman extension link 80 is attached to the forward end of the lower ottoman swing link 76 at a pivot 82 and extends forwardly and upwardly therefrom. Also the upper ottoman extension link 80 is attached to the upper ottoman swing link at a pivot 84. A lower ottoman extension link 86 is attached to the forward end of the upper ottoman swing link at a pivot 88 that is positioned above and forward of the pivot 84 and extends upwardly and forwardly therefrom generally parallel with the upper ottoman extension link. An outer ottoman bracket 90 is generally horizontally disposed and is attached to the upper and lower ottoman extension links 80, 86 at pivots 92, 94 respectively. The ottoman 17a is mounted on the outer ottoman bracket 90.

A bi-angled inner ottoman bracket 96 is attached at its lower, forward end to the lower ottoman extension link 86 at a pivot 98. At its opposite end, the inner ottoman bracket 96 supports the ottoman 17b. The inner ottoman bracket 96 also includes a slot 96a that receives a pin 80a located on the upper ottoman extension link 80.

A connecting link 100 is attached to the sequencer plate 54 at a pivot 102. The connecting link 100 extends forwardly and slightly upwardly from the pivot 102 to a pivot 104 with the lower ottoman swing link 76.

A locking mechanism 130 is attached to the reclining mechanism 30 to prevent gliding of the chair 10 when it is in the TV or fully reclined positions. The locking mechanism 130 includes a drive link 132 that is pivotally interconnected at one end to the sequencer plate 54 at a pivot 134. The drive link 132 is a straight link that slopes downwardly and slightly forwardly from the pivot 134. The opposite end of the drive link 132 is pivotally interconnected with a downwardly-extending projection 138 of a hook-shaped locking link 136 at a pivot 140. The locking link 136 is attached to the mounting bracket 32 at the pivot 36.

Referring to FIG. 4A, The chair 10 includes a power unit 112 that drives the chair 10 between its upright, TV and fully reclined positions. The power unit 112 includes a motor unit 114, to which is attached a sleeve 116. The sleeve extends rearwardly from the motor unit 114 and receives a retractable rod 118. The motor unit 114 is electrically connected with an actuator (not shown), such as a toggle switch or the like, that energizes the motor unit 114 upon actuation.

The motor unit 114 is attached to an actuating mechanism 113 that includes a mounting bracket 110 at a pivot 120 via a mounting tab 114a (see FIG. 5). The mounting bracket 110 is then attached to a cross-member 108 that extends between the reclining mechanisms 30. A bracket 106 is fixed to each of the lower ottoman swing links 76 and to the cross-member 108.

The rod 118 of the power unit 112 is attached to a projecting bracket 46 of the actuating unit 113 (FIGS. 4, 4A and 5) at a pivot 122. The bracket 46 is then attached to a Z-shaped cross-member 44 that extends between the reclining mechanisms 30. The cross-member 44 is fixed to a second projecting bracket 42 that is pivotally attached to the lower swing link 34 of each reclining mechanism 30 at a pivot 140.

As can be seen in FIG. 2A, in the upright position, the rod 118 of the power unit 112 is retracted into the sleeve 116. In this position, the "internal leg" of the Z-shaped cross-member 44 is disposed at an angle α of approximately 45 degrees to horizontal, such that both of the pivots 40, 140 are positioned

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below the axis A defined by the rod 118 of the power unit 112. The upper segment of the lower swing link 34 is generally parallel with the axis A.

Because the rod 118 is in its retracted position, a pantographic linkage formed by the upper and lower ottoman swing links 72, 76 and the upper and lower ottoman extension links 80, 86 is folded under the seat frame 62, which positions the ottoman 17b underneath a forward portion of the seat frame 62 and the ottoman 17a just forward of and below the seat frame 62 in a vertical orientation. Also, the lower and upper swing links 34, 38 are oriented such that the upper segment of the lower swing link 34 is generally horizontal, which positions the backpost 48 and, in turn, the backrest 15, in a generally upright position.

Additionally, the sequencer plate 54 is oriented such that the pivot 102 is located downwardly and rearwardly, such that the locking link 136 of the locking mechanism 130 is raised, with a pocket 137 formed at the vertex of the locking link 136 facing downwardly and positioned above a post 139 on the rear glide link 25. In this position, the front and rear glide links 20, 25 are free to swing on the pivots 21, 26, thereby enabling the reclining mechanisms 30, the seat 13, the backrest 15 and the ottomans 17a, 17b to move fore-and-aft relative to the base unit 11. During the gliding motion, the majority of the locking mechanism 130 is stationary relative to the reclining mechanisms 30. Because the locking link 136 does not move relative to the reclining mechanism 30 as the chair 10 glides, it does not interact with the post 139 to impede the gliding motion.

To move the chair 10 from the upright position to the TV position of FIG. 3, an occupant of the chair 10 actuates the actuator, which causes the rod 118 to begin to extend from the sleeve 116. Extension of the rod 118 attempts to force the lower swing link 34 to rotate slightly counterclockwise (from the vantage point of FIGS. 2 and 3) about the pivot 36. However, the weight of the seated occupant and the geometry of the lower swing link 34, the upper swing link 38, and the actuation mechanism 113 prevent any substantial rotation of the lower swing link 34 at this point. Because there is little to no fore-and-aft movement of the rear end of the rod 118, the motor unit 114 moves forwardly relative to the base unit 11, thereby driving the lower ottoman swing link 76 counterclockwise about the pivot 78. Rotation of the lower ottoman swing link 76 forces the upper ottoman extension link 80 forward, which in turn draws the upper ottoman swing link 72 counterclockwise around the pivot 74. Also, the lower ottoman extension link 86 moves forwardly more than the upper ottoman extension link 80, such that the outer ottoman bracket 90 and the inner ottoman bracket 96 rotate counterclockwise (rotation of the inner ottoman bracket 96 causes the pin 80a to move in the slot 96a toward the pivot 98). The rotation of the outer and inner ottoman brackets 90, 96 induces the ottomans 17a, 17b to rotate from a vertical orientation to a horizontal orientation. Motion of the ottomans 17a, 17b ceases when a pin 72a on the upper ottoman swing link 72 strikes the upper edge of the upper ottoman extension link 80.

In addition, the rotation of the lower ottoman swing link 76 draws the connecting link 100 forward. Movement of the connecting link 100 rotates the sequencer plate 54 counterclockwise about the pivot 58 with the backpost 48. The drive link 132 of the locking mechanism 130 is driven down and and rotates the locking link 136 counterclockwise. The movement of the locking link 136 ceases when the pocket 137 of the locking link 136 descends sufficiently to receive the post 139 on the rear glide link 25. In this position, relative movement between the front and rear glide links 20, 25 and the

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reclining mechanisms is restrained, thereby preventing the reclining mechanisms 30, and in turn the chair 10, from gliding relative to the base unit 11.

Rotation of the sequencer plate 54 and movement of the connecting link 100 also lower the backpost 48 relative to the base unit 11, which forces the seat link 64 to increase in pitch angle relative to its underlying surface. The upper swing link 38 rotates counterclockwise about the pivot 48 as a result.

To move the chair 10 from the TV position of FIG. 3 to the fully reclined position of FIG. 4, the occupant can continue to operate the actuator, which extends the rod 118 farther out of the sleeve 116. When the ottomans 17a, 17b are fully extended, the pivots 82, 84 and 88 are in a near "over-center" condition that helps to lock the ottomans 17a, 17b in an extended position. Thus, as the rod 118 continues to extend from the sleeve 116 when the chair is in the TV position, the motor unit 114 cannot move forward relative to the base unit 11 any farther, so the rear end of the rod 118 overcomes the resistance provided by the occupant's weight and the geometry of the actuating mechanism 113 and the lower and upper swing links 34, 38 and begins to move rearwardly relative to the base unit 11, resulting in counterclockwise rotation of the lower swing link 34 about the pivot 36. This rotation drives the upper swing link 38 clockwise about the pivot 36. Thus, this action causes the angle β defined by the pivots 36, 40, 50 to increase. The rotation of the upper swing link 38 drives the forward end of the backpost 48 upward. As a consequence, the backpost 48 rotates counterclockwise about the pivot 58 relative to the sequencer plate 54 (and to the seat adapter 66 and the seat frame 62). Rotation of the backpost 48 and, in turn, the backrest 15, continues until (a) the pin 48b on the backpost 48 reaches the upper end of the slot 54a in the sequencer plate 54 and (b) the pin 52b of the coupling link 52 reaches the rear end of the slot 32a of the mounting bracket 32. In this position, the backrest 15 has reclined relative to the seat 13 at a greater angle than in the upright and TV positions.

The rotation of the lower swing link 34 and of the coupling link 52 also causes the seat 15 to rise relative to the base unit 11.

As can also be seen in FIG. 4, the locking mechanism 130 continues to prevent the chair 10 from gliding as the chair 10 moves to the fully reclined position. It can also be seen that the seat 13 rises during movement from the TV position to the fully reclined position.

The chair 10 can be returned to its TV position from the fully reclined position by the occupant operating the actuator in the reverse direction. Because of the over-center condition of the pivots 82, 84, 88, the ottomans 17a, 17b remain extended, such that retraction of the rod 118 draws the backrest 15 to its upright position. Once the motion of the backrest 15 is complete, the over-center condition of the pivots 82, 84 and 88 can be overcome, and the ottomans 17a, 17b are able to retract to the upright position.

Those skilled in this art will recognize that other variations of the chair 10 are contemplated in connection with the present invention. For example, the power unit 112 may be reversed, such that the motor 114 is at the rear end of the power unit 112 and the rod 118 is at the front end. In such a case, the rod 118 would extend forwardly relative to the base unit 11 as the chair 10 moved from the upright position to the TV position, and the motor 114 would move rearwardly relative to the base unit 11 when the chair moved from the TV position to the fully reclined position. In either instance, the front end of the power unit 112 moves relative to the base unit 11 during the movement from the upright to the TV position,

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and the rear end of the power unit moves relative to the base unit 11 during the movement from the TV to the fully reclined position.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as recited in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

That which is claimed is:

1. A seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the actuating unit is attached to an actuating mechanism, the actuating mechanism comprising: a cross-member extending transversely across the seating unit;

first and second projections fixed to the cross-member;

wherein the first projection is pivotally attached to the actuating unit; and

wherein the second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism.

2. The seating unit defined in claim 1, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

3. The seating unit defined in claim 1, further comprising a locking mechanism that allows the seating unit to reciprocate while in the upright position but prevents reciprocating of the seating unit while in the TV and fully reclined positions.

4. The seating unit defined in claim 1, wherein the seat rises in moving from the TV position to the fully reclined position.

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5. The seating unit defined in claim 1, wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the reciprocating mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

6. The seating unit defined in claim 1, wherein the reciprocating mechanism is a gliding mechanism.

7. The seating unit defined in claim 1, wherein the power actuating unit includes opposed first and second ends, and wherein the first end of the power actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the power actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

8. The seating unit defined in claim 7, wherein the upper swing link is pivotally attached with the backrest.

9. The seating unit defined in claim 7, wherein the power actuating unit includes a motor and an extendable member at the first and second ends.

10. The seating unit defined in claim 9, wherein the lower swing link is angled, and wherein a segment of the lower swing link attached to the upper swing link is generally parallel to an axis defined by the extendable member of the actuating unit when the seating unit is in the upright and TV positions.

11. The seating unit defined in claim 10, wherein a first pivot defined by the second projection and the lower swing link and a second pivot defined by the lower swing link and the upper swing link are below the axis defined by the extendable member of the actuating unit when the seating unit is in the upright and TV positions.

12. The seating unit defined in claim 11, wherein the lower swing link is attached to the mounting bracket at a third pivot, and wherein in the upright and TV positions, the first, second and third pivots define a first angle, and in the fully reclined position, the first, second and third pivots define a second angle that is larger than the first angle.

13. A reciprocating and reclining seating unit, comprising: a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a power linear actuating unit attached to the reclining mechanism, the actuating unit including a motor and an extendable rod and configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

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wherein the actuating unit is attached to an actuating mechanism, the actuating mechanism comprising:
 a cross-member extending transversely across the seating unit;
 first and second projections fixed to the cross-member;
 wherein the first projection is pivotally attached to the actuating unit; and
 wherein the second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism; and
 wherein the actuating unit includes opposed first and second ends, and wherein the first end of the actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the

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second end of the actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

14. The seating unit defined in claim 13, wherein a first pivot defined by the second projection and the lower swing link and a second pivot defined by the lower swing link and the upper swing link are below an axis defined by the extendable rod of the actuating unit when the seating unit is in the upright and TV positions.

15. The seating unit defined in claim 14, wherein in the upright and TV positions, the lower swing link is attached to the mounting bracket at a third pivot, and wherein in the upright and TV positions, the first, second and third pivots define a first angle, and in the fully reclined position, the first, second and third pivots define a second angle that is larger than the first angle.

* * * * *

PUBLIC VERSION

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 8,016,348 B2
APPLICATION NO. : 12/508901
DATED : September 13, 2011
INVENTOR(S) : Hoffinan et al.

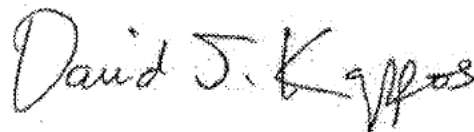
Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims:

Column 10, Claim 13, Line 60: Please correct "ottoman are is"
to read -- ottoman is --

Signed and Sealed this
Thirteenth Day of December, 2011

A handwritten signature in black ink, reading "David J. Kappos". The signature is written in a cursive, flowing style with a large initial "D".

David J. Kappos
Director of the United States Patent and Trademark Office



US007997644B2

(12) **United States Patent**
Hoffman et al.

(10) **Patent No.:** US 7,997,644 B2
(45) **Date of Patent:** Aug. 16, 2011

(54) **GLIDING-RECLINING SEATING UNIT WITH POWER ACTUATOR**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 275 days.

Primary Examiner — Milton Nelson, Jr.

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(51) Int. Cl.

A47C 1/031 (2006.01)

A47C 1/00 (2006.01)

(52) U.S. Cl. 297/85 M; 297/344.21

(58) Field of Classification Search 297/85 M, 297/68, 342, 330, 344.21, 344.24, 344.26

See application file for complete search history.

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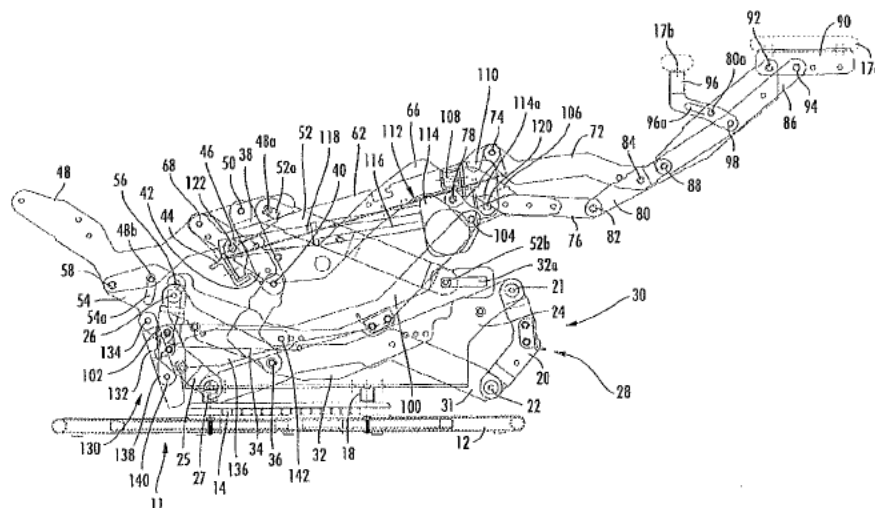
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ABSTRACT

A gliding and reclining seating unit includes: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

23 Claims, 5 Drawing Sheets



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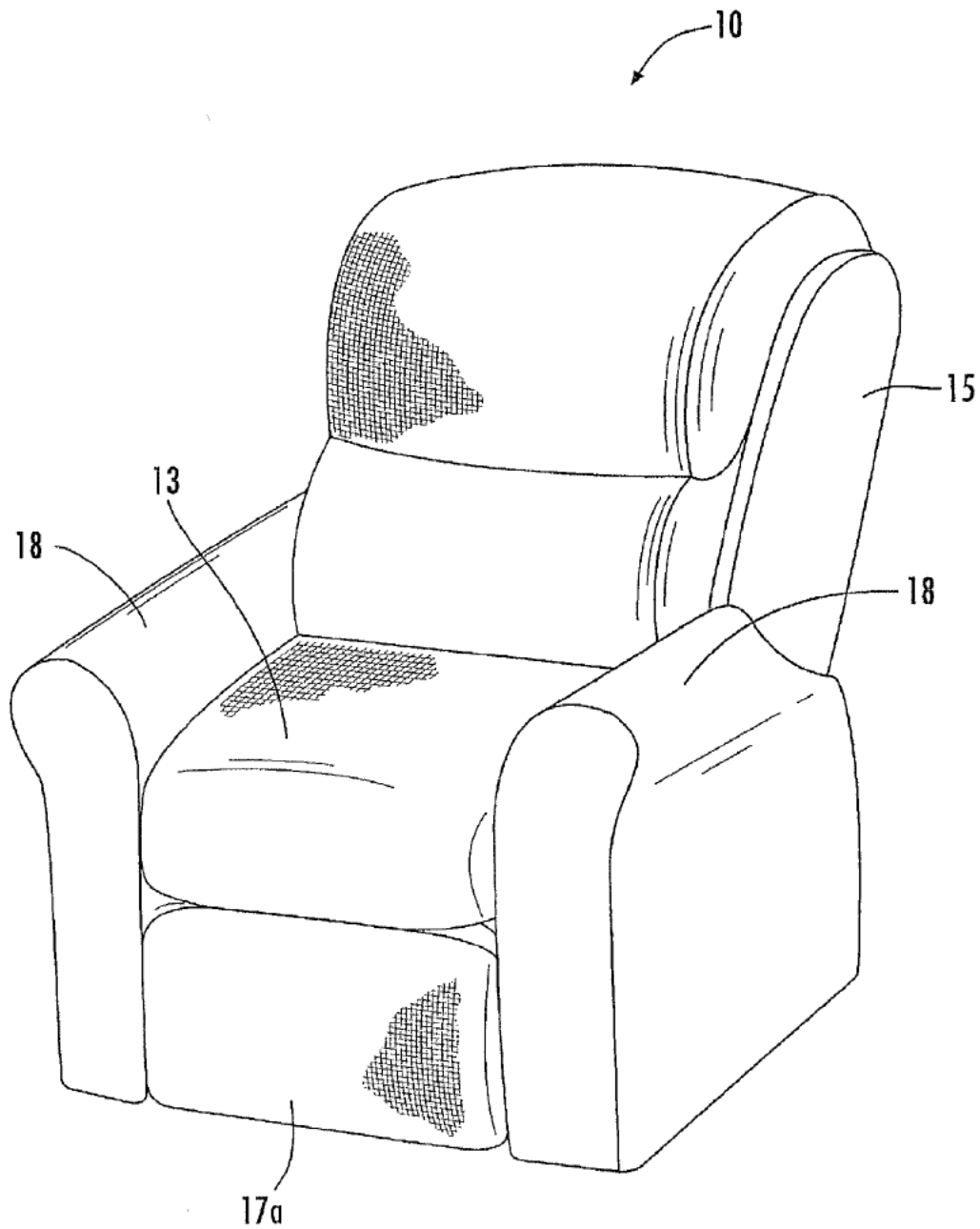


FIG. 1

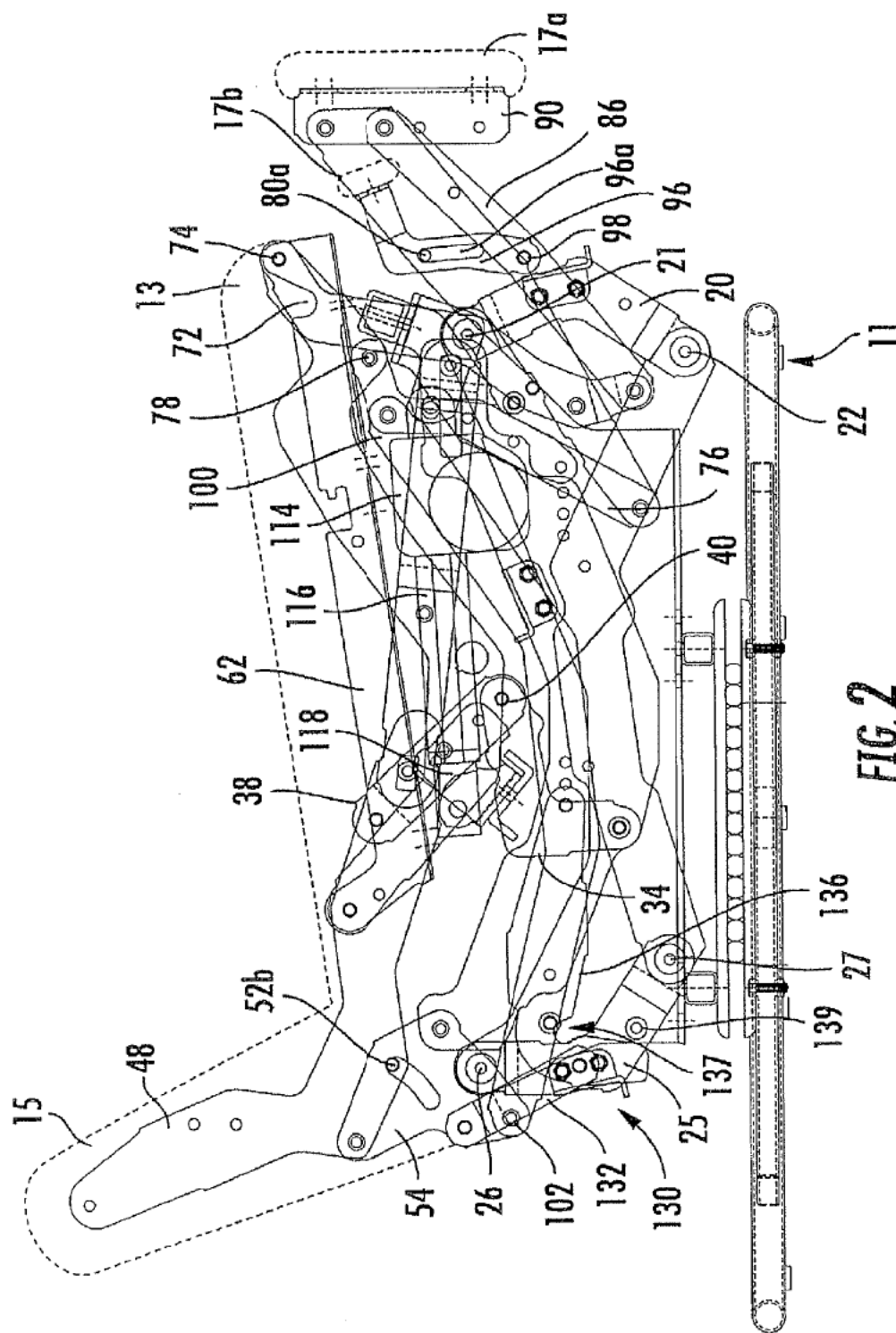


FIG. 2

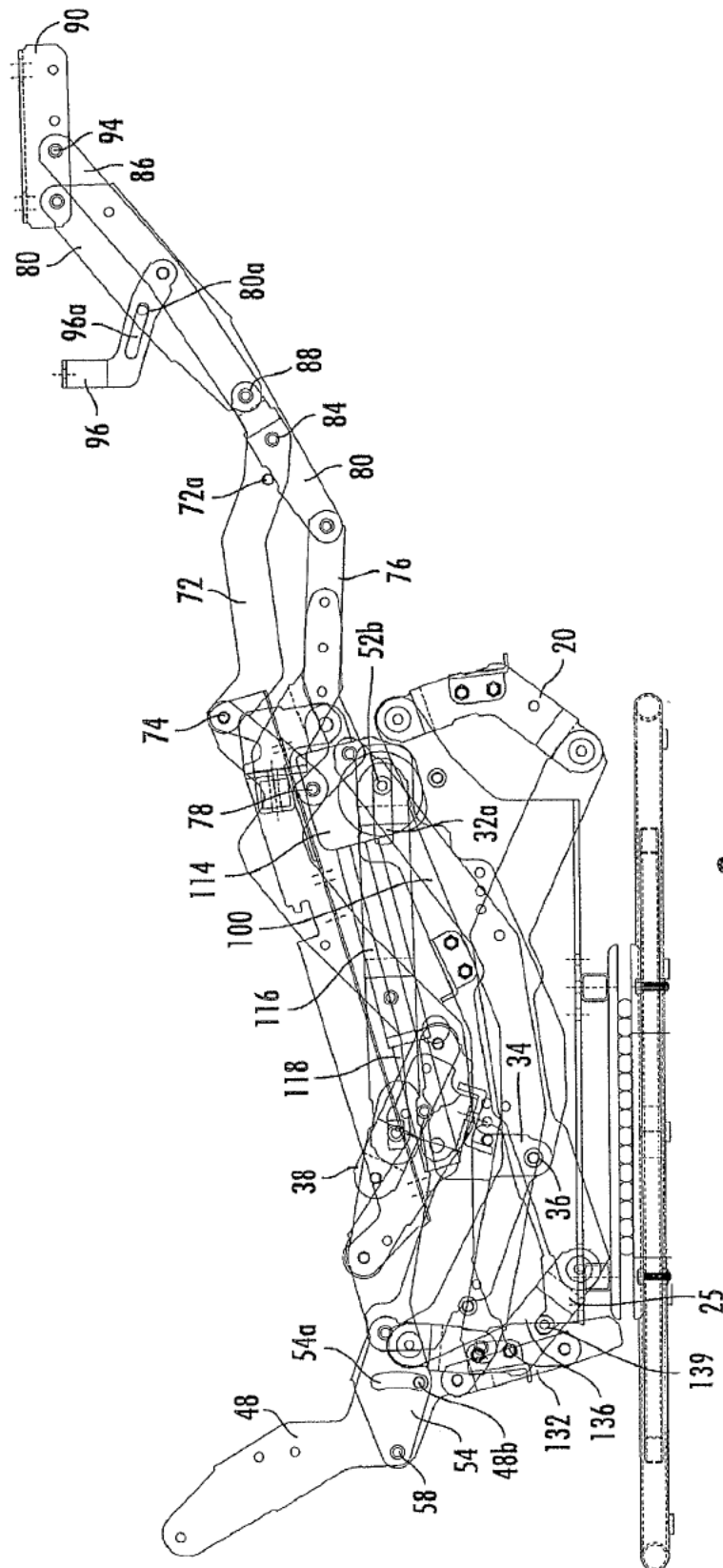
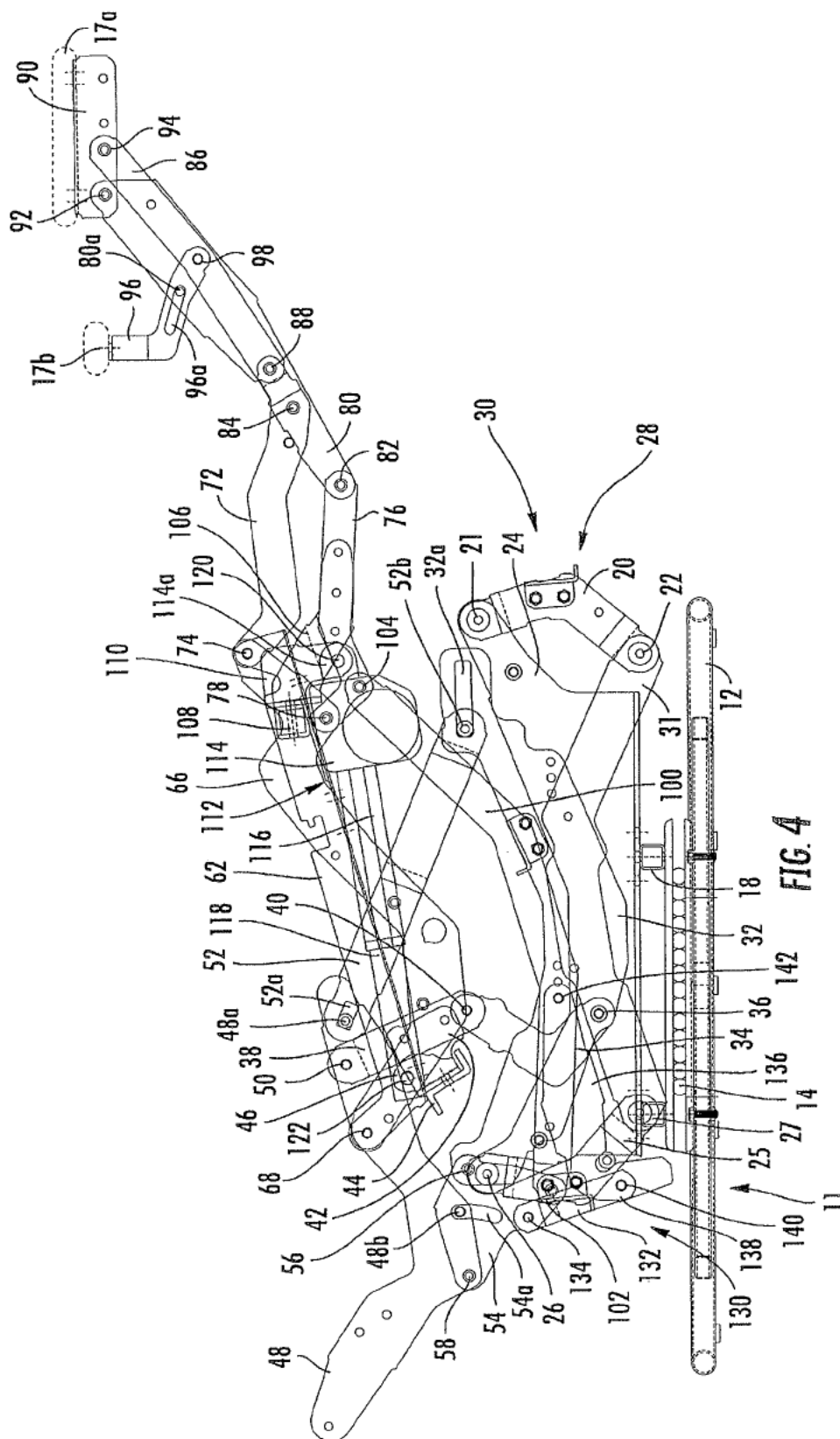
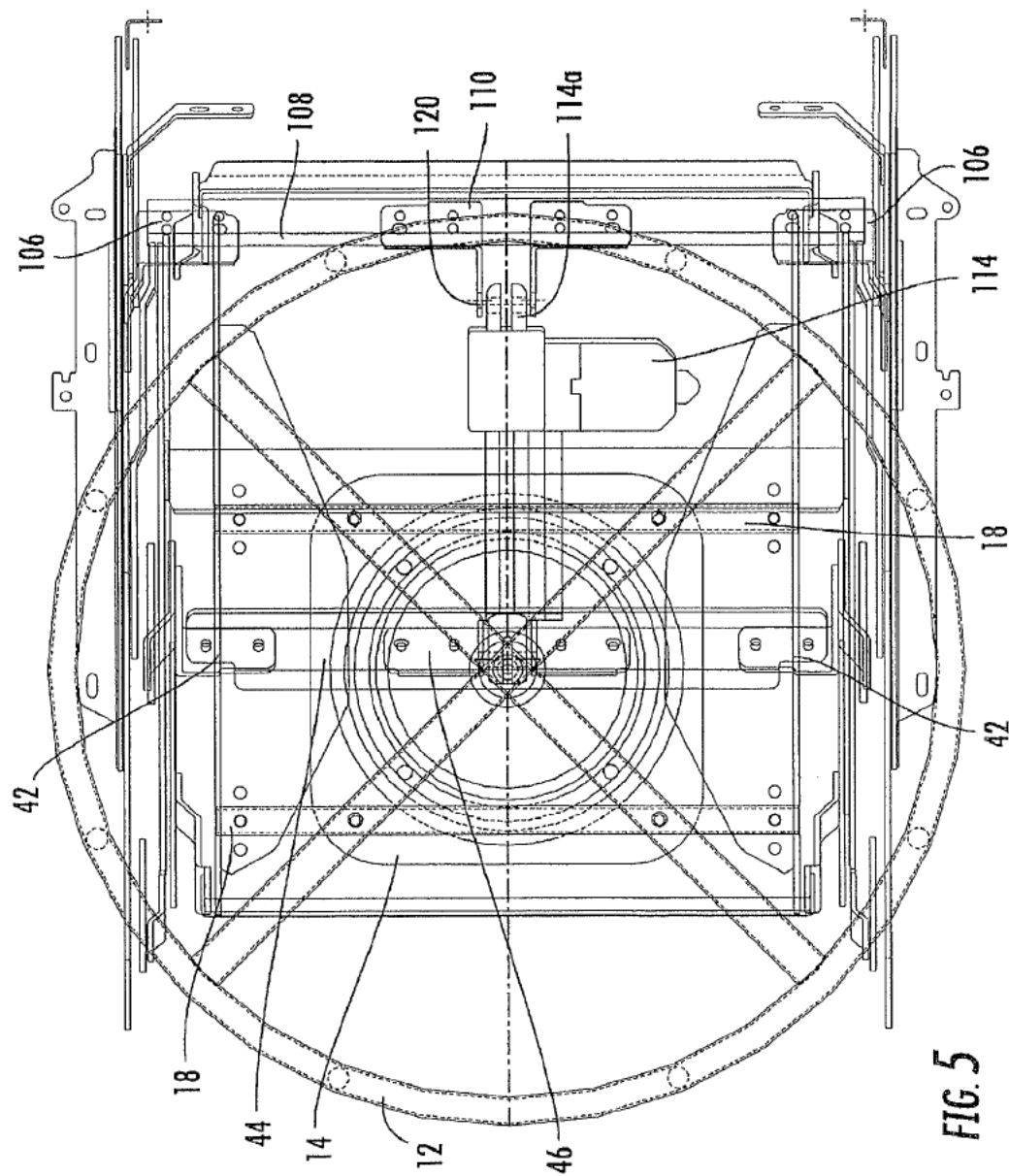


FIG. 3





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**GLIDING-RECLINING SEATING UNIT WITH
POWER ACTUATOR****FIELD OF THE INVENTION**

This invention relates generally to seating units, and relates more particularly to reclining seating units with rocking capability.

BACKGROUND OF THE INVENTION

Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

In recent years, furniture designers have looked for alternatives to rocking chairs that can provide a similarly relaxing repetitive motion. One alternative has been the gliding chair, or "glider", which includes structure that enables the seat portion of the chair to "glide" forwardly and rearwardly relative to its base to mimic generally the rocking motion of a rocking chair. Often the gliding structure comprises a set of swing links (usually two at the front of the chair, and two at the rear) that are pivotally attached at their upper ends to the base and extend downwardly therefrom to attach to a structure, such as a mounting bracket, that is attached to the seat. In this configuration, the seat is suspended from the base and is free to swing forwardly and rearwardly in a double pendulum-type motion in response to a forwardly or rearwardly-directed force applied by a seated occupant. The gliding path of the chair is controlled by the configuration and mounting of the swing links. These chairs can be constructed to resemble traditional rocking chairs and thus are quite popular.

Reclining capability has been combined with gliding capability in a single unit to provide a chair that both reclines and glides. This chair includes a reclining mechanism that enables it to move between upright and one or more reclined positions, and further includes the aforementioned swing links attached between the base and the seat, armrests, or mechanism itself to enable the chair to glide. Examples of such chairs are illustrated and described in U.S. Pat. Nos. 4,536,029 and 4,544,201, both to Rogers, Jr., the disclosures of which are hereby incorporated herein by reference in their entireties.

Although they are already popular seating units, it may be desirable to provide additional functionality to glider-recliners.

SUMMARY OF THE INVENTION

As a first aspect, embodiments of the present invention are directed to a gliding and reclining seating unit. The glider-recliner comprises: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the

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base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a second aspect, embodiments of the present invention are directed to a gliding and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a third aspect, embodiments of the present invention are directed to a gliding and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a pantographic linkage on which the ottoman is mounted. In the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position. The reclining mechanism also includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed rela-

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tive to the rocker cams and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

As a fourth aspect, embodiments of the present invention are directed to a gliding and reclining seating unit comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a backpost fixed relative to the backrest and a sequencer plate pivotally interconnected with the backrest and with the remainder of the reclining mechanism, and wherein the sequencer plate rotates in a first rotative direction relative to the backrest when the seating unit moves from the upright position to the TV position, and wherein the backrest rotates relative to the sequencer plate in the first rotative direction when the seating unit moves from the TV position to the fully reclined position.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a perspective view of a glider-recliner chair according to embodiments of the present invention.

FIG. 2 is a side section view of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 3 is a side section view of the chair of FIG. 1 with the chair in its TV position.

FIG. 4 is a side section view of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 5 is a top cutaway view of the frame of the chair of FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

The present invention now is described more fully herein after with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

Like numbers refer to like elements throughout. In the figures, the thickness of certain lines, layers, components, elements or features may be exaggerated for clarity. Broken lines illustrate optional features or operations unless specified otherwise.

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The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as "between X and Y" and "between about X and Y" should be interpreted to include X and Y. As used herein, phrases such as "between about X and Y" mean "between about X and about Y." As used herein, phrases such as "from about X to Y" mean "from about X to about Y."

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well-known functions or constructions may not be described in detail for brevity and/or clarity.

It will be understood that when an element is referred to as being "on", "attached" to, "connected" to, "coupled" with, "contacting", etc., another element, it can be directly on, attached to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, "directly on", "directly attached" to, "directly connected" to, "directly coupled" with or "directly contacting" another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed "adjacent" another feature may have portions that overlap or underlie the adjacent feature.

This invention is directed to seating units that have a stationary base, a seat portion, and a backrest. As used herein, the terms "forward", "forwardly", and "front" and derivatives thereof refer to the direction defined by a vector extending from the backrest toward the seat parallel to the underlying surface. Conversely, the terms "rearward", "rearwardly", and derivatives thereof refer to the direction directly opposite the forward direction; the rearward direction is defined by a vector that extends from the seat toward the backrest parallel to the underlying surface. The terms "lateral," "laterally", and derivatives thereof refer to the direction parallel with the floor, perpendicular to the forward and rearward directions, and extending away from a plane bisecting the seating units between their armrests. The terms "medial," "inward," "inboard," and derivatives thereof refer to the direction that is the converse of the lateral direction, i.e., the direction parallel with the floor, perpendicular to the forward direction, and extending from the periphery of the seating units toward the aforementioned bisecting plane.

The seating units illustrated and described herein comprise a plurality of pivotally interconnected links. Those skilled in this art will appreciate that the pivots between links can take a variety of configurations, such as pivot pins, rivets, bolt and nut combinations, and the like, any of which would be suitable for use with the present invention. Also, the shapes of the

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links may vary as desired, as may the locations of certain of the pivots. Moreover, in some instances combinations of pivot points may be replaced by equivalent structures, such as "slider-crank" configurations, like those described in B. Paul, *Kinematics and Dynamics of Planar Machinery* 4-21 (1979).

Referring now to the figures, a swiveling glider-recliner chair, designated broadly at 10, is illustrated in FIGS. 1-5. The chair 10 includes a base unit 11, a seat 13 that is generally horizontally disposed above the base unit 11, a backrest 15 that is generally vertically and disposed substantially above a rear portion of the seat 13, and two ottomans 17a, 17b, which, in the upright position of FIG. 1, are generally vertically disposed below a front portion of the seat 13. Arms 18 are positioned on either side of the seat 13 and move in concert with the seat 13.

A pair of mirror image reclining mechanisms 30 (only one of which is shown herein in FIGS. 2-4) are attached to the backrest 15, the seat 13, and the ottomans 17a, 17b. The reclining mechanisms 30, which comprise a plurality of interconnected links, move the chair 10 between (a) an upright position (FIGS. 1 and 2), in which the seat 13 is generally horizontally disposed, the backrest 15 is generally vertically disposed, and the ottomans 17a, 17b are generally vertically disposed and positioned below the seat 13, (b) an intermediate TV position (FIG. 3), in which the ottomans 17a, 17b are generally horizontally disposed in front of the seat 13 and the backrest 15 and the seat 13 substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position (FIG. 4), in which the angle between the backrest 15 and the seat 13 increases. These components are described in greater detail below.

The base unit 11 includes a circular lower base 12. A swivel unit 14 is mounted onto the top of the lower base 12. Cross-members 18 are mounted atop the swivel unit 14. Those skilled in this art will recognize that the base unit 11 may take other forms that provide mounting locations for the remaining components of the chair 10; for example, the swivel unit 14 may be omitted.

Glide foundation plates 24 are mounted to the top surfaces of the cross-members 18. A front glide link 20 is attached at a pivot 21 to the front end of the glide foundation plate 24, and a rear glide link 25 is attached at a pivot 26 to the rear end of the glide foundation plate 24. The front and rear glide links 20, 25 are suspended from the glide foundation plate 24 and together form a gliding mechanism 28 that provides a gliding motion to the chair 10 when it is in its upright position (FIG. 2). Those skilled in this art will appreciate that the gliding mechanism may take other forms; it may include glide links of different shapes, or it may include a "track"-based gliding mechanism.

Because the reclining mechanisms 30 are mirror images of each other, only one reclining mechanism will be described in detail herein, with the understanding that the discussion is equally applicable to the mirror image mechanism 30. Also, for clarity the mechanism will be described first with reference to FIG. 4 (with the mechanism 30 in its reclined position), then will be described in its upright and TV positions.

Turning now to FIG. 4, the reclining mechanism 30 includes a generally arcuate foundation link 31, which is attached at either end to the front and rear glide links 20, 25 at pivots 22, 27, respectively. A generally crescent-shaped mounting bracket 32 is fixed to the foundation link 31. A slot 32a is present at the front end of the mounting bracket 32. An L-shaped lower swing link 34 is attached to a lower region of the mounting bracket 32 at a pivot 36 and extends upwardly and rearwardly, then upwardly and forwardly therefrom. A

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straight upper swing link 38 is attached at its lower end to the upper end of the lower swing link 34 at a pivot 40 and extends upwardly therefrom.

A backpost 48 is fixed to the backrest 15 and extends generally forwardly from the lower front end of the backrest 15. The upper swing link 38 is attached to a forward portion of the backpost 48 at a pivot 50. A straight coupling link 52 having a slot 52a is attached to the backpost 48 via a pin 48a located forwardly of the pivot 50. The coupling link 52 extends forwardly and downwardly from the pin 48a to attach to the front end of the mounting bracket 32 via a pin 52b inserted into a slot 32a in the mounting bracket 32. A trap-ezoidal sequencer plate 54 is attached to the backpost 48 at a pivot 58, and also interacts with the backpost 48 via a slot 54a that receives a pin 48b. The sequencer plate 54 is also attached to the rear end of the mounting bracket 32 at a pivot 56.

Still referring to FIG. 4, a seat frame 62, to which the arms 18 are mounted, underlies the seat 13. A seat adapter 66 is fixed to the seat frame 62. The seat adapter 66 is attached to the backpost 48 at a pivot 68 that is located just rearward of the pivot 50. An upper ottoman swing link 72 is attached to a front region of the seat adapter 66 at a pivot 74 and extends downwardly and forwardly therefrom. A tripartite lower ottoman swing link 76 is attached to a pivot 78 that is located rearwardly and downwardly from the pivot 74; the lower ottoman swing link 76 extends generally forwardly from the pivot 78. An upper ottoman extension link 80 is attached to the forward end of the lower ottoman swing link 76 at a pivot 82 and extends forwardly and upwardly therefrom. Also the upper ottoman extension link 80 is attached to the upper ottoman swing link at a pivot 84. A lower ottoman extension link 86 is attached to the forward end of the upper ottoman swing link at a pivot 88 that is positioned above and forward of the pivot 84 and extends upwardly and forwardly therefrom generally parallel with the upper ottoman extension link. An outer ottoman bracket 90 is generally horizontally disposed and is attached to the upper and lower ottoman extension links 80, 86 at pivots 92, 94 respectively. The ottoman 17a is mounted on the outer ottoman bracket 90.

An L-shaped inner ottoman bracket 96 is attached at its lower, forward end to the lower ottoman extension link 86 at a pivot 98. At its opposite end, the inner ottoman bracket 96 supports the ottoman 17b. The inner ottoman bracket 96 also includes a slot 96a that receives a pin 80a located on the upper ottoman extension link 80.

A connecting link 100 is attached to the sequencer plate 54 at a pivot 102. The connecting link 100 extends forwardly and slightly upwardly from the pivot 102 to a pivot 104 with the lower ottoman swing link 76.

A locking mechanism 130 is attached to the reclining mechanism 30 to prevent gliding of the chair 10 when it is in the TV or fully reclined positions. The locking mechanism 130 includes a drive link 132 that is pivotally interconnected at one end to the sequencer plate 54 at a pivot 134. The drive link 132 is a straight link that slopes downwardly and slightly forwardly from the pivot 134. The opposite end of the drive link 132 is pivotally interconnected with a downwardly-extending projection 138 of a hook-shaped locking link 136 at a pivot 140. The locking link 136 is attached to the mounting bracket 32 at a pivot 142.

The chair 10 includes a power unit 112 that drives the chair 10 between its upright, TV and fully reclined positions. The power unit 112 includes a motor unit 114, to which is attached a sleeve 116. The sleeve extends rearwardly from the motor unit 114 and receives a retractable rod 118. The motor unit

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114 is electrically connected with an actuator (not shown), such as a toggle switch or the like, that energizes the motor unit 114 upon actuation.

The motor unit 114 is attached to a mounting bracket 110 at a pivot 120 via a mounting tab 114a (see FIG. 5). The mounting bracket 110 is then attached to a cross-member 108 that extends between the reclining mechanisms 30. A bracket 106 is fixed to each of the lower ottoman swing links 76 and to the cross-member 108.

The rod 118 of the power unit 112 is attached to a bracket 46 (FIG. 5) at a pivot 122. The bracket 46 is then attached to a cross-member 44 that extends between the reclining mechanisms 30; the cross-member 44 is fixed to a bracket 42 that is in turn fixed to the upper swing link 38 of each reclining mechanism 30.

As can be seen in FIG. 2, in the upright position, the rod 118 of the power unit 112 is retracted into the sleeve 116. As a result, a pantographic linkage formed by the upper and lower ottoman swing links 72, 76 and the upper and lower ottoman extension links 80, 86 are folded under the seat frame 62, which positions the ottoman 17b underneath a forward portion of the seat frame 62 and the ottoman 17a just forward of and below the seat frame 62 in a vertical orientation. Also, the lower and upper swing links 34, 38 are oriented such that the upper segment of the lower swing link 34 is generally horizontal, which positions the backpost 48 and, in turn, the backrest 15, in a generally upright position.

Additionally, the sequencer plate 54 is oriented such that the pivot 102 is located downwardly and rearwardly, such that the locking link 136 of the locking mechanism 130 is raised, with a pocket 137 facing downwardly and positioned above a post 139 on the rear glide link 25. In this position, the chair 10 is free to glide as the front and rear glide links 20, 25 are free to swing on the pivots 21, 26, thereby enabling the reclining mechanisms 30, the seat 13, the backrest 15 and the ottomans 17a, 17b to move fore-and-aft relative to the base unit 11. During the gliding motion, the majority of the locking mechanism 130 is stationary relative to the reclining mechanisms 30. Because the locking link 136 does not move relative to the reclining mechanism 30 as the chair 10 glides, it does not interact with the post 139.

To move the chair 10 from the upright position to the TV position of FIG. 3, an occupant of the chair 10 actuates the actuator, which causes the rod 118 to begin to extend from the sleeve 116. Extension of the rod 118 forces the upper swing link 38 to rotate slightly counterclockwise (from the vantage point of FIGS. 2 and 3) about the pivot 40. This movement draws the backpost 48 and rear end of the seat frame 62 downwardly, which causes the seat 13 to increase in pitch angle relative to the underlying surface. However, the weight of the seated occupant prevents further rotation of the upper swing link 38 at this point. Because there is little to no fore-and-aft movement of the rear end of the rod 118, the motor unit 114 moves forwardly relative to the base unit 11, thereby driving the lower ottoman swing link 76 counterclockwise about the pivot 78. Rotation of the lower ottoman swing link 76 forces the upper ottoman extension link 80 forward, which in turn draws the upper ottoman swing link 72 counterclockwise around the pivot 74. Also, the lower ottoman extension link 86 moves forwardly more than the upper ottoman extension link 80, such that the outer ottoman bracket 90 and the inner ottoman bracket 96 rotate counterclockwise (rotation of the inner ottoman bracket 96 causes the pin 80a to move in the slot 96a toward the pivot 98). The rotation of the outer and inner ottoman brackets 90, 96 induces the ottomans 17a, 17b to rotate from a vertical orientation to a horizontal orientation. Motion of the ottomans 17a, 17b ceases when a pin 72a on the

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upper ottoman swing link 72 strikes the upper edge of the upper ottoman extension link 80.

In addition, the rotation of the lower ottoman swing link 76 draws the connecting link 100 forward. Movement of the connecting link 100 rotates the sequencer plate 54 counterclockwise about the pivot 58 with the backpost 48. The drive link 132 of the locking mechanism 130 is driven down and rotates the locking link 136 counterclockwise. The movement of the locking link 136 ceases when the pocket 137 of the locking link 136 descends sufficiently to receive the post 139 on the rear glide link 25. In this position, relative movement between the front and rear glide links 20, 25 and the reclining mechanisms is restrained, thereby preventing the reclining mechanisms 30, and in turn the chair 10, from gliding relative to the base unit 11.

To move the chair 10 from the TV position of FIG. 3 to the fully reclined position of FIG. 4, the occupant can continue to operate the actuator, which extends the rod 118 farther out of the sleeve 116. When the ottomans 17a, 17b are fully extended, the pivots 84, 88 and 94 are in a near "over-center" condition that helps to lock the ottomans 17a, 17b in an extended position. Thus, as the rod 118 continues to extend from the sleeve 116 when the chair is in the TV position, the motor unit 114 cannot move forward relative to the base unit 11 any farther, so the rear end of the rod 118 begins to move rearwardly relative to the base unit 11, resulting in clockwise rotation of the upper swing link 38 about the pivot 40. This rotation both drives the lower swing link 34 counterclockwise about the pivot 36 and drives the forward end of the backpost 48 upward. As a consequence, the backpost 48 rotates counterclockwise about the pivot 58 relative to the sequencer plate 54 (and to the seat adapter 66 and the seat frame 62). Rotation of the backpost 48 and, in turn, the backrest 15, continues until (a) the pin 48b on the backpost 48 reaches the upper end of the slot 54a in the sequencer plate 54 and the pin 52b of the coupling link 52 reaches the rear end of the slot 32a of the mounting bracket 32. In this position, the backrest 15 has reclined relative to the seat 13 at a greater angle than in the upright and TV positions.

As can also be seen in FIG. 4, the locking mechanism 130 continues to prevent the chair 10 from gliding as the chair 10 moves to the fully reclined position. It can also be seen that the seat 13 rises during movement from the TV position to the fully reclined position.

The chair 10 can be returned to its TV position from the fully reclined position by the occupant operating the actuator in the reverse direction. Because of the over-center condition of the pivots 82, 84, 88, the ottomans 17a, 17b remain extended, such that retraction of the rod 118 draws the backrest 15 to its upright position. Once the motion of the backrest 15 is complete, the over-center condition of the pivots 84, 88 and 94 can be overcome, and the ottomans 17a, 17b are able to retract to the upright position.

Those skilled in this art will recognize that other variations of the chair 10 are contemplated in connection with the present invention. For example, the power unit 112 may be reversed, such that the motor 114 is at the rear end of the power unit 112 and the rod 118 is at the front end. In such a case, the rod 118 would extend forwardly relative to the base unit 11 as the chair 10 moved from the upright position to the TV position, and the motor 114 would move rearwardly relative to the base unit 11 when the chair moved from the TV position to the fully reclined position. In either instance, the front end of the power unit 112 moves relative to the base unit 11 during the movement from the upright to the TV position,

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and the rear end of the power unit moves relative to the base unit 11 during the movement from the TV to the fully reclined position.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as recited in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

That which is claimed is:

1. A gliding and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

2. The seating unit defined in claim 1, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

3. The seating unit defined in claim 1, further comprising a locking mechanism that allows the seating unit to glide while in the upright position but prevents gliding of the seating unit while in the TV and fully reclined positions.

4. The seating unit defined in claim 1, wherein the power actuating unit includes opposed first and second ends, and wherein the first end of the power actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the power actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

5. The seating unit defined in claim 4, wherein the reclining mechanism includes a lower swing link pivotally attached with the base unit and an upper swing link pivotally attached with the lower swing link and with the backrest, and wherein the second end of the power actuating unit is attached with the upper swing link.

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6. The seating unit defined in claim 4, wherein the power actuating unit includes a motor and an extendable member at the first and second ends.

7. The seating unit defined in claim 1, wherein the seat rises in moving from the TV position to the fully reclined position.

8. The seating unit defined in claim 1, wherein the base unit further comprises a swivel unit.

9. The seating unit defined in claim 1, wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the gliding mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

10. A gliding and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a linear actuator attached to the reclining mechanism, the linear actuator configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

11. The seating unit defined in claim 10, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

12. The seating unit defined in claim 10, further comprising a locking mechanism that allows the seating unit to glide on the gliding mechanism while in the upright position but prevents gliding of the seating unit while in the TV and fully reclined positions.

13. The seating unit defined in claim 10, further comprising a swivel unit attached to the base unit.

14. The seating unit defined in claim 10, wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the gliding mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat pivotally attached to the backpost.

15. The seating unit defined in claim 10, wherein the seat rises in moving from the TV position to the fully reclined position.

16. A gliding and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

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an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a linear power actuating unit attached to the reclining mechanism, the actuating unit configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the linear actuator includes opposed first and second ends, and wherein the first end of the linear actuator moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the linear actuator moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

17. The seating unit defined in claim 16, wherein the reclining mechanism includes a lower swing link pivotally attached with the base unit and an upper swing link pivotally attached with the lower swing link and with the backrest, and wherein the second end of the linear actuator is attached with the upper swing link.

18. A gliding and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a linear actuator attached to the reclining mechanism, the linear actuator configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between

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links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position; and wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the gliding mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

19. The seating unit defined in claim 18, wherein the linear actuator includes opposed first and second ends, and wherein the first end of the linear actuator moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the linear actuator moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

20. The seating unit defined in claim 19, wherein the reclining mechanism includes a lower swing link pivotally attached with the base unit and an upper swing link pivotally attached with the lower swing link and with the backrest, and wherein the second end of the linear actuator is attached with the upper swing link.

21. A gliding and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a gliding mechanism attached to the base unit and the reclining mechanism, the gliding mechanism being configured to enable the seat, backrest and reclining mechanism to glide relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the reclining mechanism includes a backpost fixed relative to the backrest and a sequencer plate pivotally interconnected with the backrest and with others of the plurality of pivotally interconnected links of the reclining mechanism, and wherein the sequencer plate rotates in a first rotative direction relative to the backrest when the seating unit moves from the upright position to the TV position, and wherein the backrest rotates relative to the sequencer plate in the first rotative direction when the seating unit moves from the TV position to the fully reclined position.

22. The seating unit defined in claim 21, wherein the backrest includes one of a pin and a slot, and the sequencer plate includes the other of a pin and a slot, the pin being received in the slot.

23. The seating unit defined in claim 22, wherein the backrest includes the pin, and the sequencer plate includes the slot.

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(54) **RECIPROCATING SEATING UNIT WITH
POWER ACTUATOR**

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(52) **U.S. Cl.** **297/85 M**

(58) **Field of Classification Search** **297/85 M,**
297/69, DIG. 7

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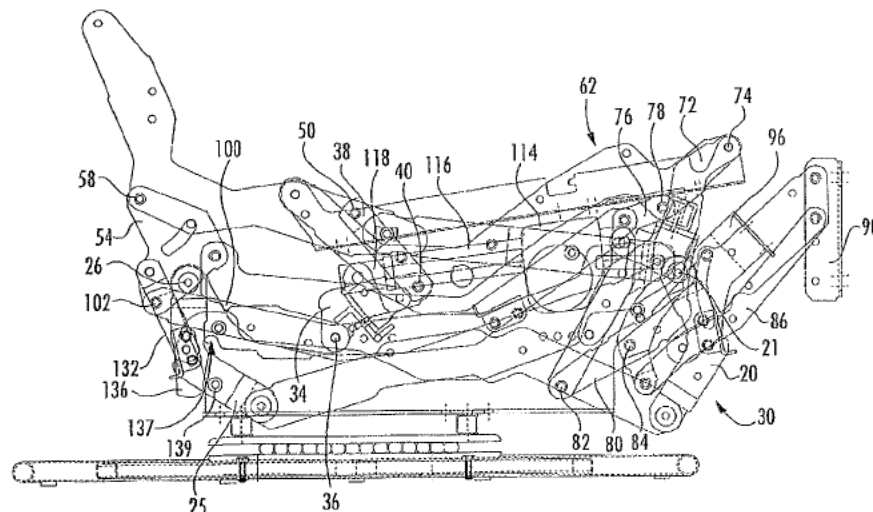
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(57) **ABSTRACT**

A reciprocating and reclining seating unit includes: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between an upright position, an intermediate TV position, and a fully reclined position.

14 Claims, 8 Drawing Sheets



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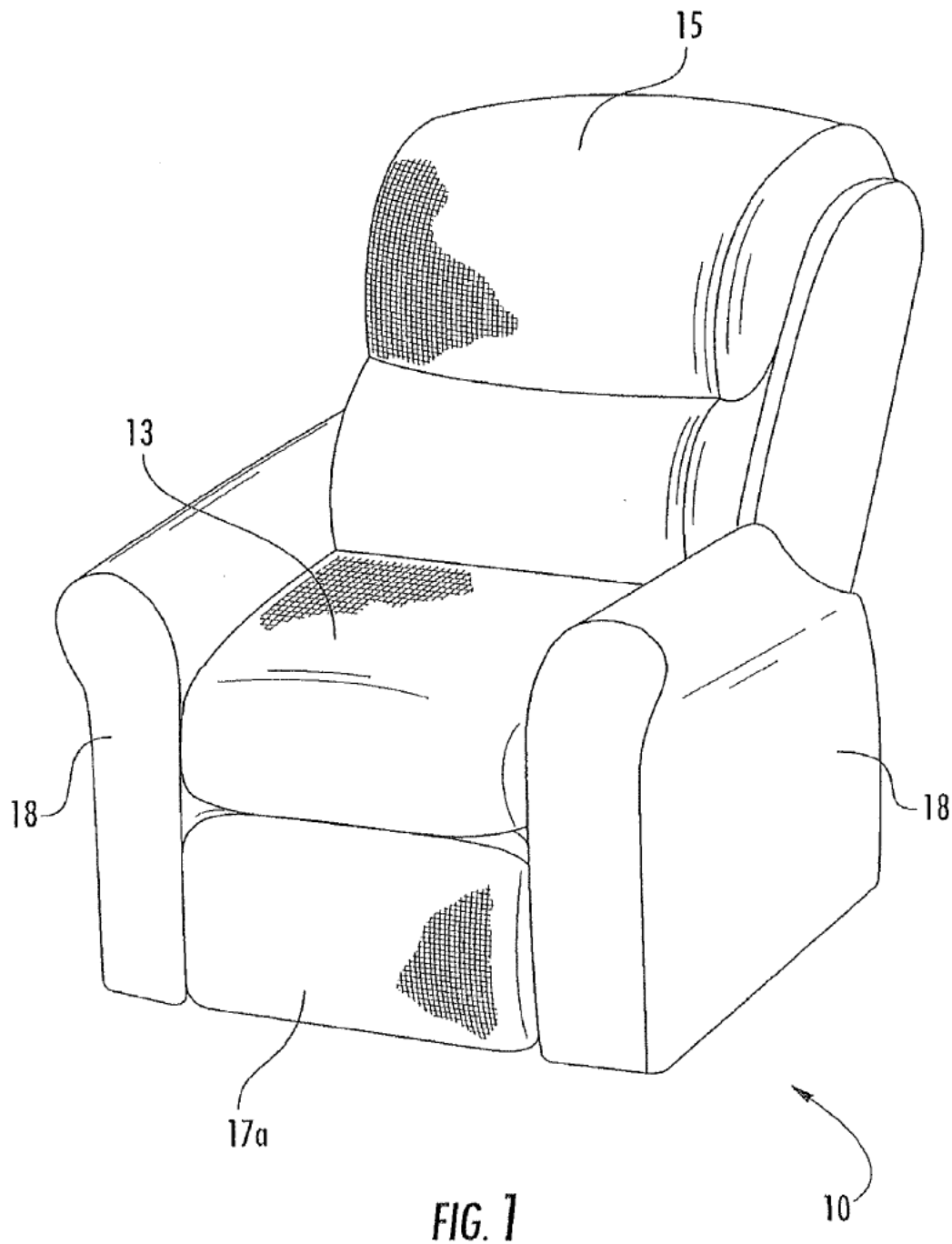
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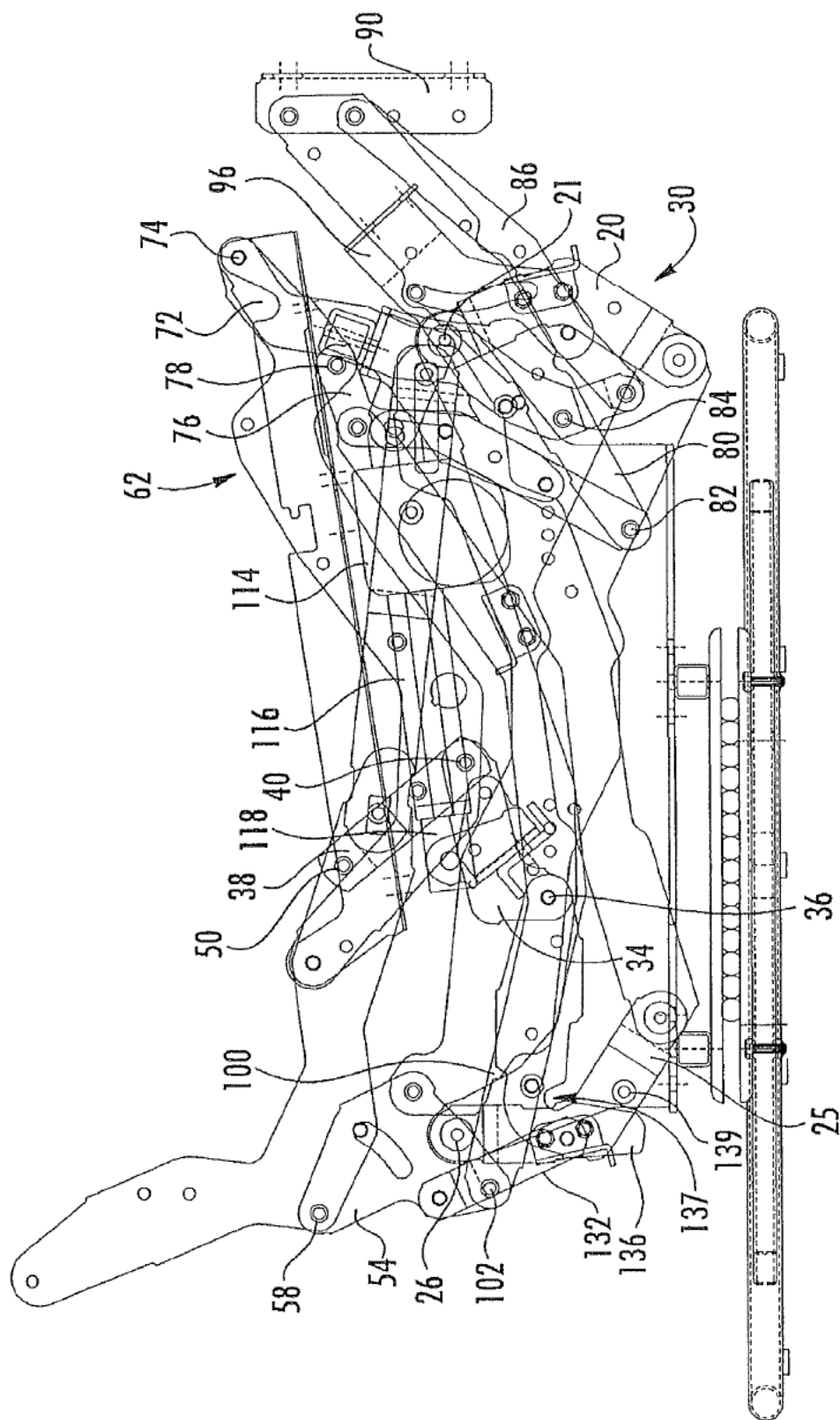
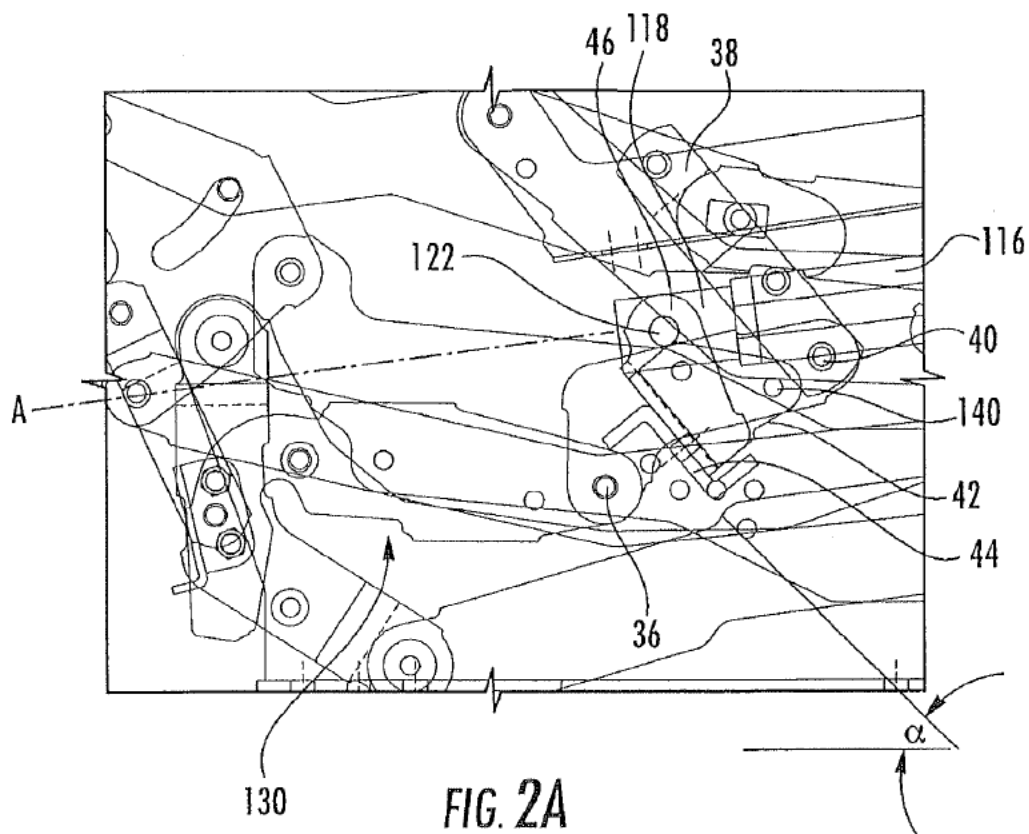
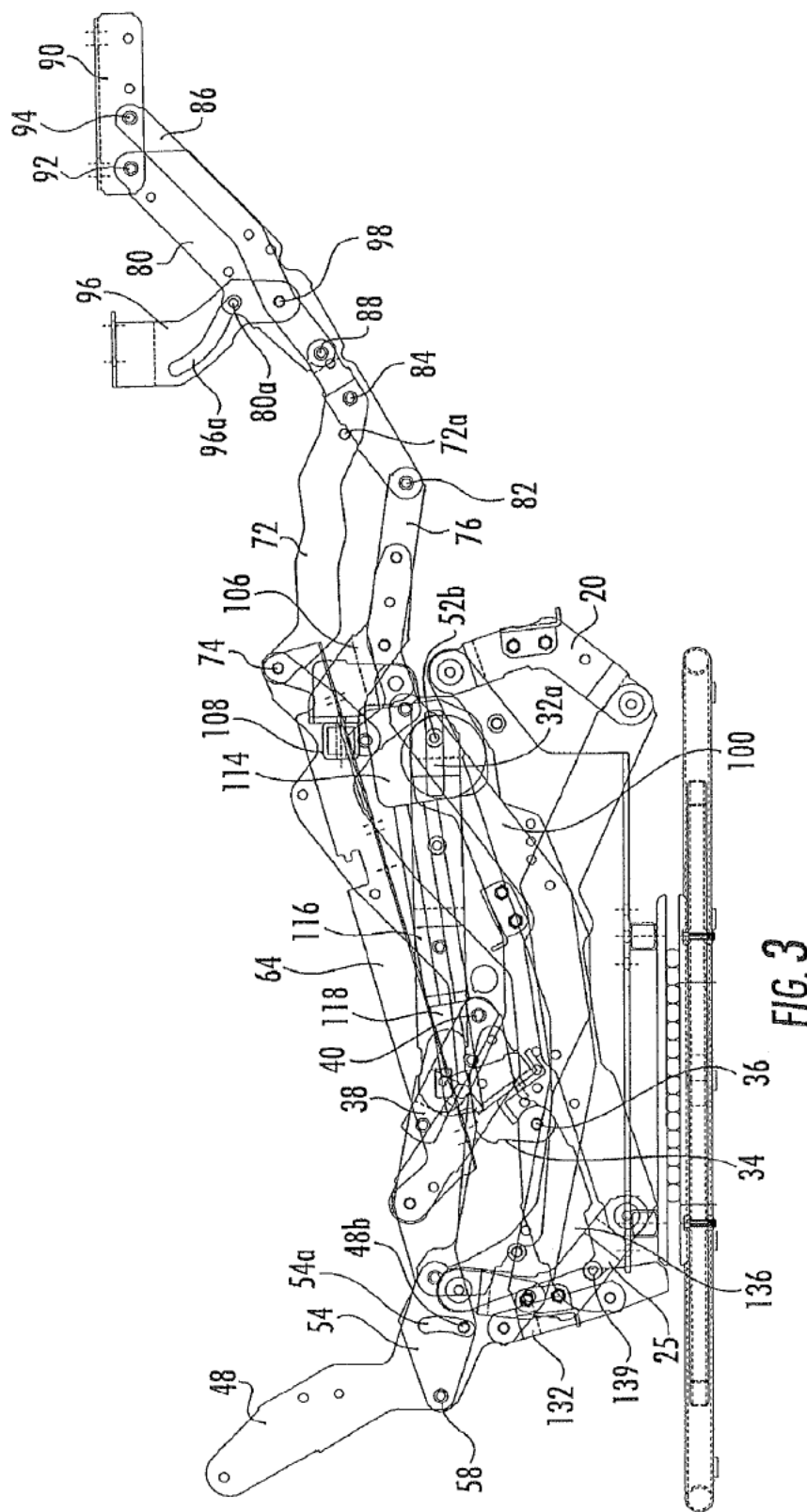
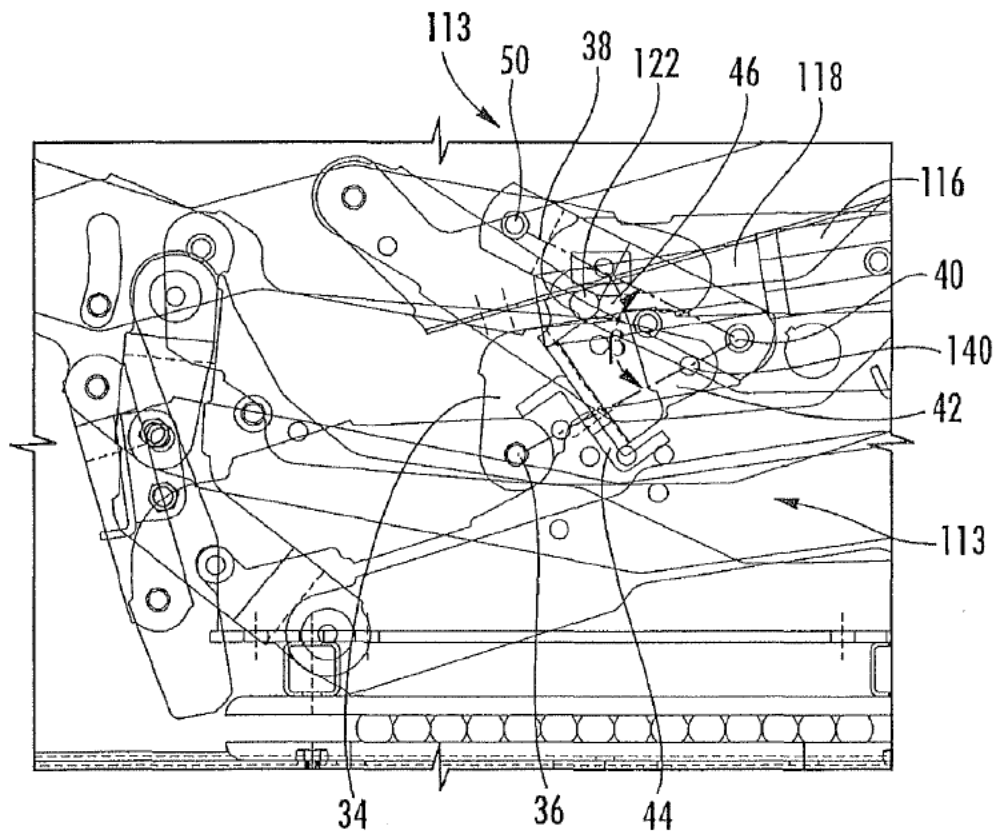
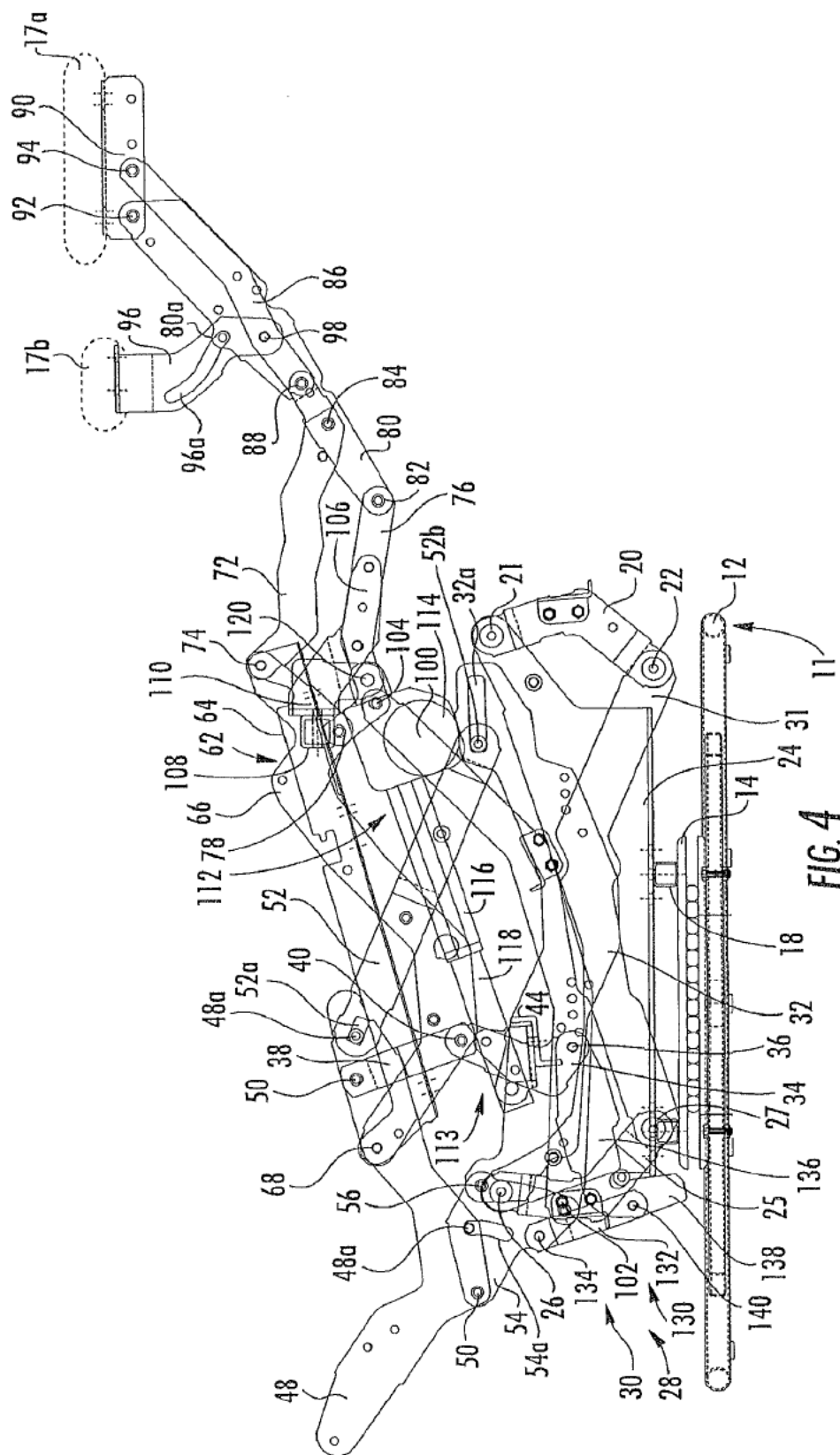


FIG. 2









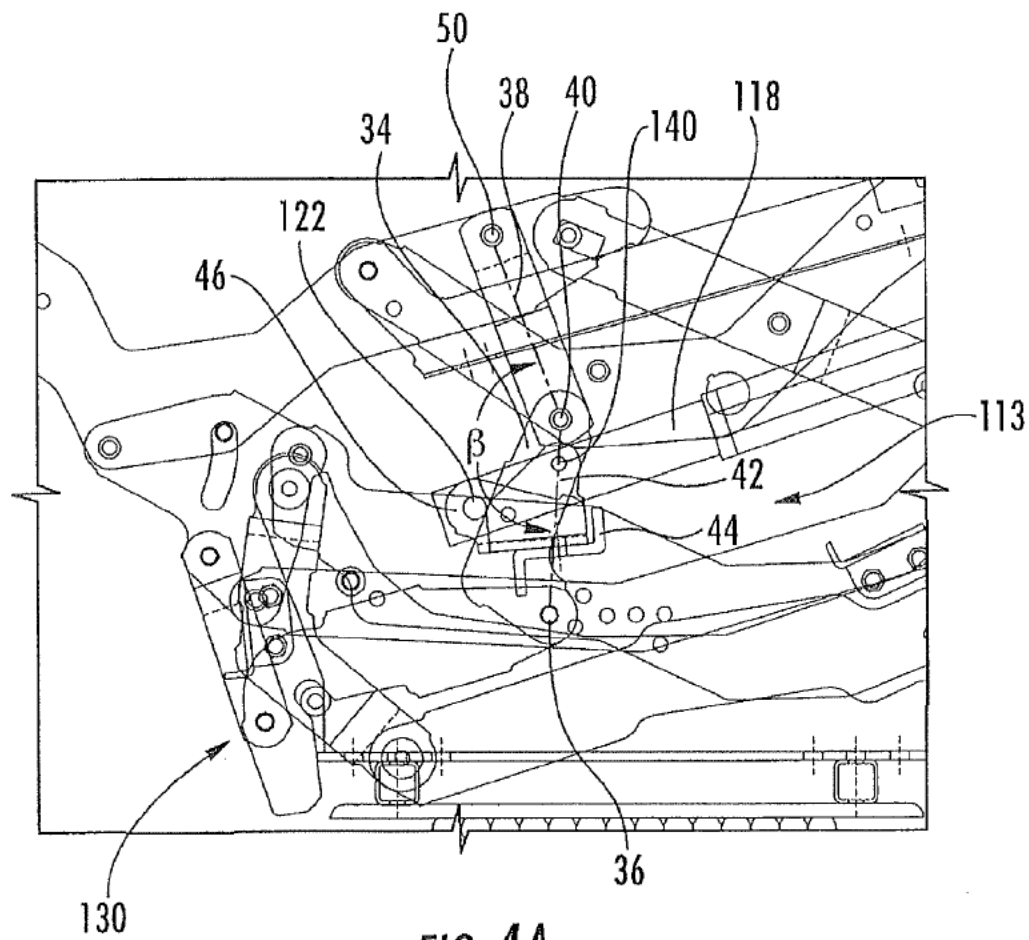


FIG. 4A

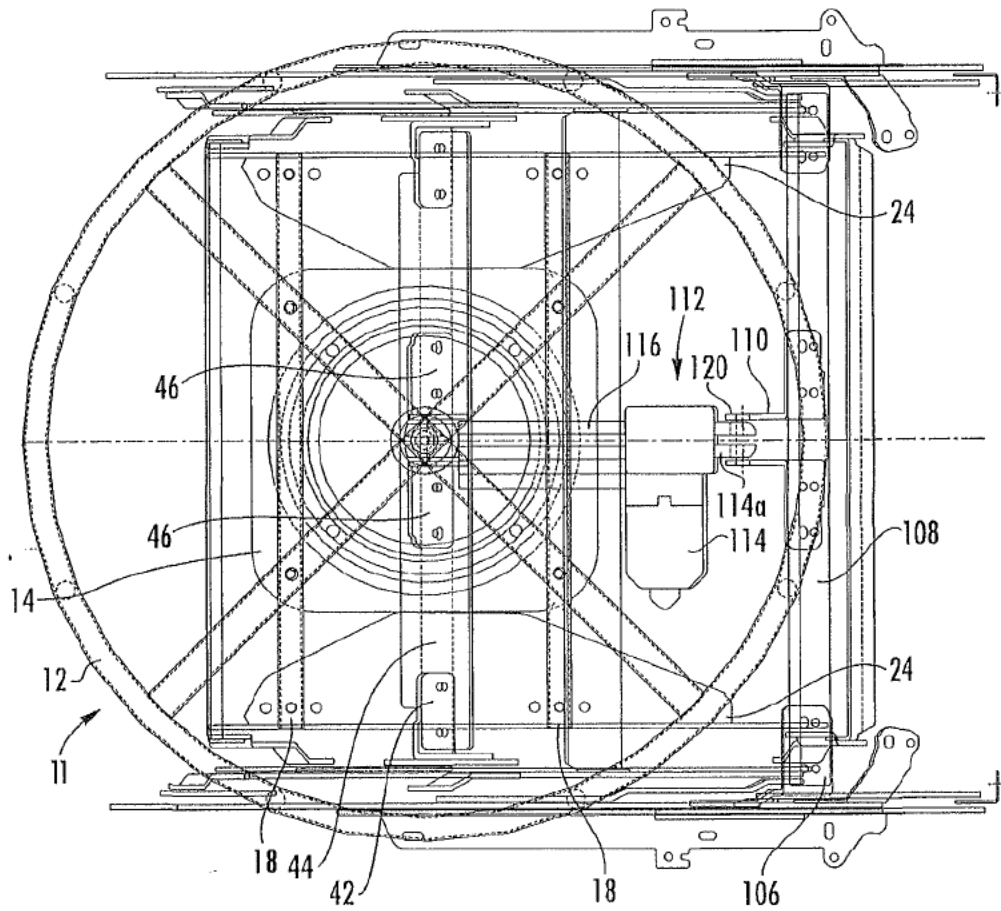


FIG. 5

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RECIPROCATING SEATING UNIT WITH POWER ACTUATOR

RELATED APPLICATION

This application is a continuation application of U.S. patent application Ser. No. 12/508,901, filed Jul. 24, 2009, now U.S. Pat. No. 8,016,348, which is a continuation-in-part of U.S. patent application Ser. No. 12/276,567, filed Nov. 24, 2008 (now U.S. Pat. No. 7,997,644, issued Aug. 16, 2011, the disclosure of each of which is hereby incorporated herein in its entirety.

FIELD OF THE INVENTION

This invention relates generally to seating units, and relates more particularly to reclining seating units with rocking capability.

BACKGROUND OF THE INVENTION

Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

In recent years, furniture designers have looked for alternatives to rocking chairs that can provide a similarly relaxing repetitive motion. One alternative has been the gliding chair, or "glider", which includes structure that enables the seat portion of the chair to "glide" forwardly and rearwardly relative to its base to mimic generally the rocking motion of a rocking chair. Often the gliding structure comprises a set of swing links (usually two at the front of the chair, and two at the rear) that are pivotally attached at their upper ends to the base and extend downwardly therefrom to attach to a structure, such as a mounting bracket, that is attached to the seat. In this configuration, the seat is suspended from the base and is free to swing forwardly and rearwardly in a double pendulum-type motion in response to a forwardly or rearwardly-directed force applied by a seated occupant. The gliding path of the chair is controlled by the configuration and mounting of the swing links. These chairs can be constructed to resemble traditional rocking chairs and thus are quite popular.

Reclining capability has been combined with gliding capability in a single unit to provide a chair that both reclines and glides. This chair includes a reclining mechanism that enables it to move between upright and one or more reclined positions, and further includes the aforementioned swing links attached between the base and the seat, armrests, or mechanism itself to enable the chair to glide. Examples of such chairs are illustrated and described in U.S. Pat. Nos. 4,536,029 and 4,544,201, both to Rogers, Jr., the disclosures of which are hereby incorporated herein by reference in their entireties.

Although they are already popular seating units, it may be desirable to provide additional functionality to glider-recliners.

SUMMARY OF THE INVENTION

As a first aspect, embodiments of the present invention are directed to a reciprocating and reclining seating unit. The

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seating unit comprises: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The actuating unit is attached to an actuating mechanism, the actuating mechanism comprising a cross-member extending transversely across the seating unit and first and second projections fixed to the cross-member. The first projection is pivotally attached to the actuating unit. The second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism.

As a second aspect, embodiments of the present invention are directed to a reciprocating and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and a power linear actuating unit attached to the reclining mechanism, the actuating unit including a motor and an extendable rod and configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The actuating unit is attached to an actuating mechanism, the actuating mechanism comprising: a cross-member extending transversely across the seating unit; and first and second projections fixed to the cross-member. The first projection is pivotally attached to the actuating unit. The second projection is pivotally attached to a lower swing link of the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism. The actuating unit includes

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opposed first and second ends, wherein the first end of the actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a perspective view of a glider-recliner chair according to embodiments of the present invention.

FIG. 2 is a side section view of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 2A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 3 is a side section view of the chair of FIG. 1 with the chair in its TV position.

FIG. 3A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its TV position.

FIG. 4 is a side section view of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 4A is an enlarged side section view of the actuating mechanism of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 5 is a top cutaway view of the frame of the chair of FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

The present invention now is described more fully herein after with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

Like numbers refer to like elements throughout. In the figures, the thickness of certain lines, layers, components, elements or features may be exaggerated for clarity. Broken lines illustrate optional features or operations unless specified otherwise.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as "between X and Y" and "between about X and Y" should be interpreted to include X and Y. As used herein, phrases such as "between about X and about Y" mean "between about X and about Y." As used herein, phrases such as "from about X to Y" mean "from about X to about Y."

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood

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that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well-known functions or constructions may not be described in detail for brevity and/or clarity.

It will be understood that when an element is referred to as being "on", "attached" to, "connected" to, "coupled" with, "contacting", etc., another element, it can be directly on, attached to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, "directly on", "directly attached" to, "directly connected" to, "directly coupled" with or "directly contacting" another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed "adjacent" another feature may have portions that overlap or underlie the adjacent feature.

This invention is directed to seating units that have a stationary base, a seat portion, and a backrest. As used herein, the terms "forward", "forwardly", and "front" and derivatives thereof refer to the direction defined by a vector extending from the backrest toward the seat parallel to the underlying surface. Conversely, the terms "rearward", "rearwardly", and derivatives thereof refer to the direction directly opposite the forward direction; the rearward direction is defined by a vector that extends from the seat toward the backrest parallel to the underlying surface. The terms "lateral," "laterally", and derivatives thereof refer to the direction parallel with the floor, perpendicular to the forward and rearward directions, and extending away from a plane bisecting the seating units between their armrests. The terms "medial," "inward," "inboard," and derivatives thereof refer to the direction that is the converse of the lateral direction, i.e., the direction parallel with the floor, perpendicular to the forward direction, and extending from the periphery of the seating units toward the aforementioned bisecting plane.

The seating units illustrated and described herein comprise a plurality of pivotally interconnected links. Those skilled in this art will appreciate that the pivots between links can take a variety of configurations, such as pivot pins, rivets, bolt and nut combinations, and the like, any of which would be suitable for use with the present invention. Also, the shapes of the links may vary as desired, as may the locations of certain of the pivots. Moreover, in some instances combinations of pivot points may be replaced by equivalent structures, such as "slider-crank" configurations, like those described in B. Paul, *Kinematics and Dynamics of Planar Machinery* 4-21 (1979). In other embodiments, another mechanism that provides reciprocating motion along a longitudinal path, such as a rocking unit like that discussed in co-assigned and co-pending U.S. patent application Ser. No. 12/276,559, the disclosure of which is hereby incorporated herein in its entirety, may also be employed.

Referring now to the figures, a swiveling glider-recliner chair, designated broadly at 10, is illustrated in FIGS. 1-5. The chair 10 includes a base unit 11, a seat 13 that is generally horizontally disposed above the base unit 11, a backrest 15 that is generally vertically and disposed substantially above a rear portion of the seat 13, and two ottomans 17a, 17b, which, in the upright position of FIG. 1, are generally vertically disposed below a front portion of the seat 13. Arms 18 are positioned on either side of the seat 13 and move in concert with the seat 13.

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A pair of mirror image reclining mechanisms 30 (only one of which is shown herein in FIGS. 2-4) are attached to the backrest 15, the seat 13, and the ottomans 17a, 17b. The reclining mechanisms 30, which comprise a plurality of interconnected links, move the chair 10 between (a) an upright position (FIGS. 1 and 2), in which the seat 13 is generally horizontally disposed, the backrest 15 is generally vertically disposed, and the ottomans 17a, 17b are generally vertically disposed and positioned below the seat 13, (b) an intermediate TV position (FIG. 3), in which the ottomans 17a, 17b are generally horizontally disposed in front of the seat 13 and the backrest 15 and the seat 13 substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position (FIG. 4), in which the angle between the backrest 15 and the seat 13 increases. These components are described in greater detail below.

The base unit 11 includes a circular lower base 12. A swivel unit 14 is mounted onto the top of the lower base 12. Cross-members 18 are mounted atop the swivel unit 14. Those skilled in this art will recognize that the base unit 11 may take other forms that provide mounting locations for the remaining components of the chair 10; for example, the swivel unit 14 may be omitted.

Glide foundation plates 24 are mounted to the top surfaces of the cross-members 18. A front glide link 20 is attached at a pivot 21 to the front end of the glide foundation plate 24, and a rear glide link 25 is attached at a pivot 26 to the rear end of the glide foundation plate 24. The front and rear glide links 20, 25 are suspended from the glide foundation plate 24 and together form a gliding mechanism 28 that provides a gliding motion to the chair 10 when it is in its upright position (FIG. 2). Those skilled in this art will appreciate that the gliding mechanism may take other forms; it may include glide links of different shapes, or it may include a "track"—based gliding mechanism.

Because the reclining mechanisms 30 are mirror images of each other, only one reclining mechanism will be described in detail herein, with the understanding that the discussion is equally applicable to the mirror image mechanism 30. Also, for clarity the mechanism will be described first with reference to FIG. 4 (with the mechanism 30 in its reclined position), then will be described in its upright and TV positions.

Turning now to FIG. 4, the reclining mechanism 30 includes a generally arcuate foundation link 31, which is attached at either end to the front and rear glide links 20, 25 at pivots 22, 27, respectively. A generally crescent-shaped mounting bracket 32 is fixed to the foundation link 31. A slot 32a is present at the front end of the mounting bracket 32. An L-shaped lower swing link 34 is attached to a lower region of the mounting bracket 32 at a pivot 36 and extends upwardly and rearwardly, then upwardly and forwardly therefrom. A straight upper swing link 38 is attached at its lower end to the upper end of the lower swing link 34 at a pivot 40 and extends upwardly therefrom.

A backpost 48 is fixed to the backrest 15 and extends generally forwardly from the lower front end of the backrest 15. The upper swing link 38 is attached to a forward portion of the backpost 48 at a pivot 50. A straight coupling link 52 having a slot 52a is attached to the backpost 48 via a pin 48a located forwardly of the pivot 50. The coupling link 52 extends forwardly and downwardly from the pin 48a to attach to the front end of the mounting bracket 32 via a pin 52b inserted into a slot 32a in the mounting bracket 32. A trapezoidal sequencer plate 54 is attached to the backpost 48 at a pivot 58, and also interacts with the backpost 48 via a slot 54a that receives a pin 48b. The sequencer plate 54 is also attached to the rear end of the mounting bracket 32 at a pivot 56.

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Still referring to FIG. 4, a seat frame 62, to which the arms 18 are mounted, underlies the seat 13. A seat adapter 66 is fixed to the seat frame 62. The seat adapter 66 is attached to the backpost 48 at a pivot 68 that is located just rearward of the pivot 50. An upper ottoman swing link 72 is attached to a front region of the seat adapter 66 at a pivot 74 and extends downwardly and forwardly therefrom. A tripartite lower ottoman swing link 76 is attached to a pivot 78 that is located rearwardly and downwardly from the pivot 74; the lower ottoman swing link 76 extends generally forwardly from the pivot 78. An upper ottoman extension link 80 is attached to the forward end of the lower ottoman swing link 76 at a pivot 82 and extends forwardly and upwardly therefrom. Also the upper ottoman extension link 80 is attached to the upper ottoman swing link at a pivot 84. A lower ottoman extension link 86 is attached to the forward end of the upper ottoman swing link at a pivot 88 that is positioned above and forward of the pivot 84 and extends upwardly and forwardly therefrom generally parallel with the upper ottoman extension link. An outer ottoman bracket 90 is generally horizontally disposed and is attached to the upper and lower ottoman extension links 80, 86 at pivots 92, 94 respectively. The ottoman 17a is mounted on the outer ottoman bracket 90.

A bi-angled inner ottoman bracket 96 is attached at its lower, forward end to the lower ottoman extension link 86 at a pivot 98. At its opposite end, the inner ottoman bracket 96 supports the ottoman 17b. The inner ottoman bracket 96 also includes a slot 96a that receives a pin 80a located on the upper ottoman extension link 80.

A connecting link 100 is attached to the sequencer plate 54 at a pivot 102. The connecting link 100 extends forwardly and slightly upwardly from the pivot 102 to a pivot 104 with the lower ottoman swing link 76.

A locking mechanism 130 is attached to the reclining mechanism 30 to prevent gliding of the chair 10 when it is in the TV or fully reclined positions. The locking mechanism 130 includes a drive link 132 that is pivotally interconnected at one end to the sequencer plate 54 at a pivot 134. The drive link 132 is a straight link that slopes downwardly and slightly forwardly from the pivot 134. The opposite end of the drive link 132 is pivotally interconnected with a downwardly-extending projection 138 of a hook-shaped locking link 136 at a pivot 140. The locking link 136 is attached to the mounting bracket 32 at the pivot 36.

Referring to FIG. 4A, The chair 10 includes a power unit 112 that drives the chair 10 between its upright, TV and fully reclined positions. The power unit 112 includes a motor unit 114, to which is attached a sleeve 116. The sleeve extends rearwardly from the motor unit 114 and receives a retractable rod 118. The motor unit 114 is electrically connected with an actuator (not shown), such as a toggle switch or the like, that energizes the motor unit 114 upon actuation.

The motor unit 114 is attached to an actuating mechanism 113 that includes a mounting bracket 110 at a pivot 120 via a mounting tab 114a (see FIG. 5). The mounting bracket 110 is then attached to a cross-member 108 that extends between the reclining mechanisms 30. A bracket 106 is fixed to each of the lower ottoman swing links 76 and to the cross-member 108.

The rod 118 of the power unit 112 is attached to a projecting bracket 46 of the actuating unit 113 (FIGS. 4, 4A and 5) at a pivot 122. The bracket 46 is then attached to a Z-shaped cross-member 44 that extends between the reclining mechanisms 30. The cross-member 44 is fixed to a second projecting bracket 42 that is pivotally attached to the lower swing link 34 of each reclining mechanism 30 at a pivot 140.

As can be seen in FIG. 2A, in the upright position, the rod 118 of the power unit 112 is retracted into the sleeve 116. In

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this position, the "internal leg" of the Z-shaped cross-member 44 is disposed at an angle α of approximately 45 degrees to horizontal, such that both of the pivots 40, 140 are positioned below the axis A defined by the rod 118 of the power unit 112. The upper segment of the lower swing link 34 is generally parallel with the axis A.

Because the rod 118 is in its retracted position, a pantographic linkage formed by the upper and lower ottoman swing links 72, 76 and the upper and lower ottoman extension links 80, 86 is folded under the seat frame 62, which positions the ottoman 17b underneath a forward portion of the seat frame 62 and the ottoman 17a just forward of and below the seat frame 62 in a vertical orientation. Also, the lower and upper swing links 34, 38 are oriented such that the upper segment of the lower swing link 34 is generally horizontal, which positions the backpost 48 and, in turn, the backrest 15, in a generally upright position.

Additionally, the sequencer plate 54 is oriented such that the pivot 102 is located downwardly and rearwardly, such that the locking link 136 of the locking mechanism 130 is raised, with a pocket 137 formed at the vertex of the locking link 136 facing downwardly and positioned above a post 139 on the rear glide link 25. In this position, the front and rear glide links 20, 25 are free to swing on the pivots 21, 26, thereby enabling the reclining mechanisms 30, the seat 13, the backrest 15 and the ottomans 17a, 17b to move fore-and-aft relative to the base unit 11. During the gliding motion, the majority of the locking mechanism 130 is stationary relative to the reclining mechanisms 30. Because the locking link 136 does not move relative to the reclining mechanism 30 as the chair 10 glides, it does not interact with the post 139 to impede the gliding motion.

To move the chair 10 from the upright position to the TV position of FIG. 3, an occupant of the chair 10 actuates the actuator, which causes the rod 118 to begin to extend from the sleeve 116. Extension of the rod 118 attempts to force the lower swing link 34 to rotate slightly counterclockwise (from the vantage point of FIGS. 2 and 3) about the pivot 36. However, the weight of the seated occupant and the geometry of the lower swing link 34, the upper swing link 38, and the actuation mechanism 113 prevent any substantial rotation of the lower swing link 34 at this point. Because there is little to no fore-and-aft movement of the rear end of the rod 118, the motor unit 114 moves forwardly relative to the base unit 11, thereby driving the lower ottoman swing link 76 counterclockwise about the pivot 78. Rotation of the lower ottoman swing link 76 forces the upper ottoman extension link 80 forward, which in turn draws the upper ottoman swing link 72 counterclockwise around the pivot 74. Also, the lower ottoman extension link 86 moves forwardly more than the upper ottoman extension link 80, such that the outer ottoman bracket 90 and the inner ottoman bracket 96 rotate counterclockwise (rotation of the inner ottoman bracket 96 causes the pin 80a to move in the slot 96a toward the pivot 98). The rotation of the outer and inner ottoman brackets 90, 96 induces the ottomans 17a, 17b to rotate from a vertical orientation to a horizontal orientation. Motion of the ottomans 17a, 17b ceases when a pin 72a on the upper ottoman swing link 72 strikes the upper edge of the upper ottoman extension link 80.

In addition, the rotation of the lower ottoman swing link 76 draws the connecting link 100 forward. Movement of the connecting link 100 rotates the sequencer plate 54 counterclockwise about the pivot 58 with the backpost 48. The drive link 132 of the locking mechanism 130 is driven down and rotates the locking link 136 counterclockwise. The movement of the locking link 136 ceases when the pocket 137 of the

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locking link 136 descends sufficiently to receive the post 139 on the rear glide link 25. In this position, relative movement between the front and rear glide links 20, 25 and the reclining mechanisms is restrained, thereby preventing the reclining mechanisms 30, and in turn the chair 10, from gliding relative to the base unit 11.

Rotation of the sequencer plate 54 and movement of the connecting link 100 also lower the backpost 48 relative to the base unit 11, which forces the seat link 64 to increase in pitch angle relative to its underlying surface. The upper swing link 38 rotates counterclockwise about the pivot 48 as a result.

To move the chair 10 from the TV position of FIG. 3 to the fully reclined position of FIG. 4, the occupant can continue to operate the actuator, which extends the rod 118 farther out of the sleeve 116. When the ottomans 17a, 17b are fully extended, the pivots 82, 84 and 88 are in a near "over-center" condition that helps to lock the ottomans 17a, 17b in an extended position. Thus, as the rod 118 continues to extend from the sleeve 116 when the chair is in the TV position, the motor unit 114 cannot move forward relative to the base unit 11 any farther, so the rear end of the rod 118 overcomes the resistance provided by the occupant's weight and the geometry of the actuating mechanism 113 and the lower and upper swing links 34, 38 and begins to move rearwardly relative to the base unit 11, resulting in counterclockwise rotation of the lower swing link 34 about the pivot 36. This rotation drives the upper swing link 38 clockwise about the pivot 36. Thus, this action causes the angle β defined by the pivots 36, 40, 50 to increase. The rotation of the upper swing link 38 drives the forward end of the backpost 48 upward. As a consequence, the backpost 48 rotates counterclockwise about the pivot 58 relative to the sequencer plate 54 (and to the seat adapter 66 and the seat frame 62). Rotation of the backpost 48 and, in turn, the backrest 15, continues until (a) the pin 48b on the backpost 48 reaches the upper end of the slot 54a in the sequencer plate 54 and (b) the pin 52b of the coupling link 52 reaches the rear end of the slot 32a of the mounting bracket 32. In this position, the backrest 15 has reclined relative to the seat 13 at a greater angle than in the upright and TV positions. The rotation of the lower swing link 34 and of the coupling link 52 also causes the seat 15 to rise relative to the base unit 11.

As can also be seen in FIG. 4, the locking mechanism 130 continues to prevent the chair 10 from gliding as the chair 10 moves to the fully reclined position. It can also be seen that the seat 13 rises during movement from the TV position to the fully reclined position.

The chair 10 can be returned to its TV position from the fully reclined position by the occupant operating the actuator in the reverse direction. Because of the over-center condition of the pivots 82, 84, 88, the ottomans 17a, 17b remain extended, such that retraction of the rod 118 draws the backrest 15 to its upright position. Once the motion of the backrest 15 is complete, the over-center condition of the pivots 82, 84 and 88 can be overcome, and the ottomans 17a, 17b are able to retract to the upright position.

Those skilled in this art will recognize that other variations of the chair 10 are contemplated in connection with the present invention. For example, the power unit 112 may be reversed, such that the motor 114 is at the rear end of the power unit 112 and the rod 118 is at the front end. In such a case, the rod 118 would extend forwardly relative to the base unit 11 as the chair 10 moved from the upright position to the TV position, and the motor 114 would move rearwardly relative to the base unit 11 when the chair moved from the TV position to the fully reclined position. In either instance, the front end of the power unit 112 moves relative to the base unit

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11 during the movement from the upright to the TV position, and the rear end of the power unit moves relative to the base unit 11 during the movement from the TV to the fully reclined position.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as recited in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

That which is claimed is:

1. A seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base unit;

a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached to the base unit and the reclining mechanism, the reciprocating mechanism being configured to enable the seat, backrest and reclining mechanism to reciprocate relative to the base unit along a longitudinal path responsive to a longitudinally-directed force; and

a linear actuating unit attached to the reclining mechanism, the linear actuating unit configured to move the seating unit between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the linear actuating unit includes opposed first and second ends, and wherein the first end of the linear actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the linear actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

2. The seating unit defined in claim 1, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

3. The seating unit defined in claim 1, further comprising a locking mechanism that allows the seating unit to reciprocate while in the upright position but prevents reciprocating of the seating unit while in the TV and fully reclined positions.

4. The seating unit defined in claim 1, wherein the linear actuating unit is attached to an actuating mechanism, the actuating mechanism comprising a cross-member extending transversely across the seating unit, wherein the cross-member is pivotally attached to the actuating unit; and wherein the cross-member is pivotally attached to a lower swing link of

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the reclining mechanism, the lower swing link being pivotally attached to an upper swing link of the reclining mechanism and to a mounting bracket that is attached to the reciprocating mechanism.

5. The seating unit defined in claim 4, wherein the upper swing link is pivotally attached with the backrest.

6. The seating unit defined in claim 1, wherein the seat rises in moving from the TV position to the fully reclined position.

7. The seating unit defined in claim 1, wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the reciprocating mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

8. The seating unit defined in claim 7, wherein the lower swing link is angled, and wherein a segment of the lower swing link attached to the upper swing link is generally parallel to an axis defined by an extendable member of the linear actuating unit when the seating unit is in the upright and TV positions.

9. The seating unit defined in claim 8, wherein a first pivot defined by the cross-member and the lower swing link and a second pivot defined by the lower swing link and the upper swing link are below the axis defined by the extendable member of the actuating unit when the seating unit is in the upright and TV positions.

10. The seating unit defined in claim 9, wherein the lower swing link is attached to the mounting bracket at a third pivot, and wherein in the upright and TV positions, the first, second and third pivots define a first angle, and in the fully reclined position, the first, second and third pivots define a second angle that is larger than the first angle.

11. The seating unit defined in claim 1, wherein the reciprocating mechanism is a gliding mechanism.

12. A reciprocating seating unit, comprising:

a base;

a generally horizontally-disposed seat positioned above the base;

a generally upright backrest positioned above the base and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base, the reclining mechanism comprising a plurality of pivotally interconnected links;

a reciprocating mechanism attached with the base unit and the reclining mechanism, the reciprocating mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed reciprocating motion relative to the base unit; and

a linear actuator attached to the reclining mechanism, the linear actuator having first and second movable members, wherein the relative movement of the first and second movable members moves the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the reclining mechanism includes:

a backpost that is fixed relative to the backrest;

a mounting bracket that is attached to the second movable member of the linear actuator;

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a lower swing link that is attached to the mounting bracket;
and

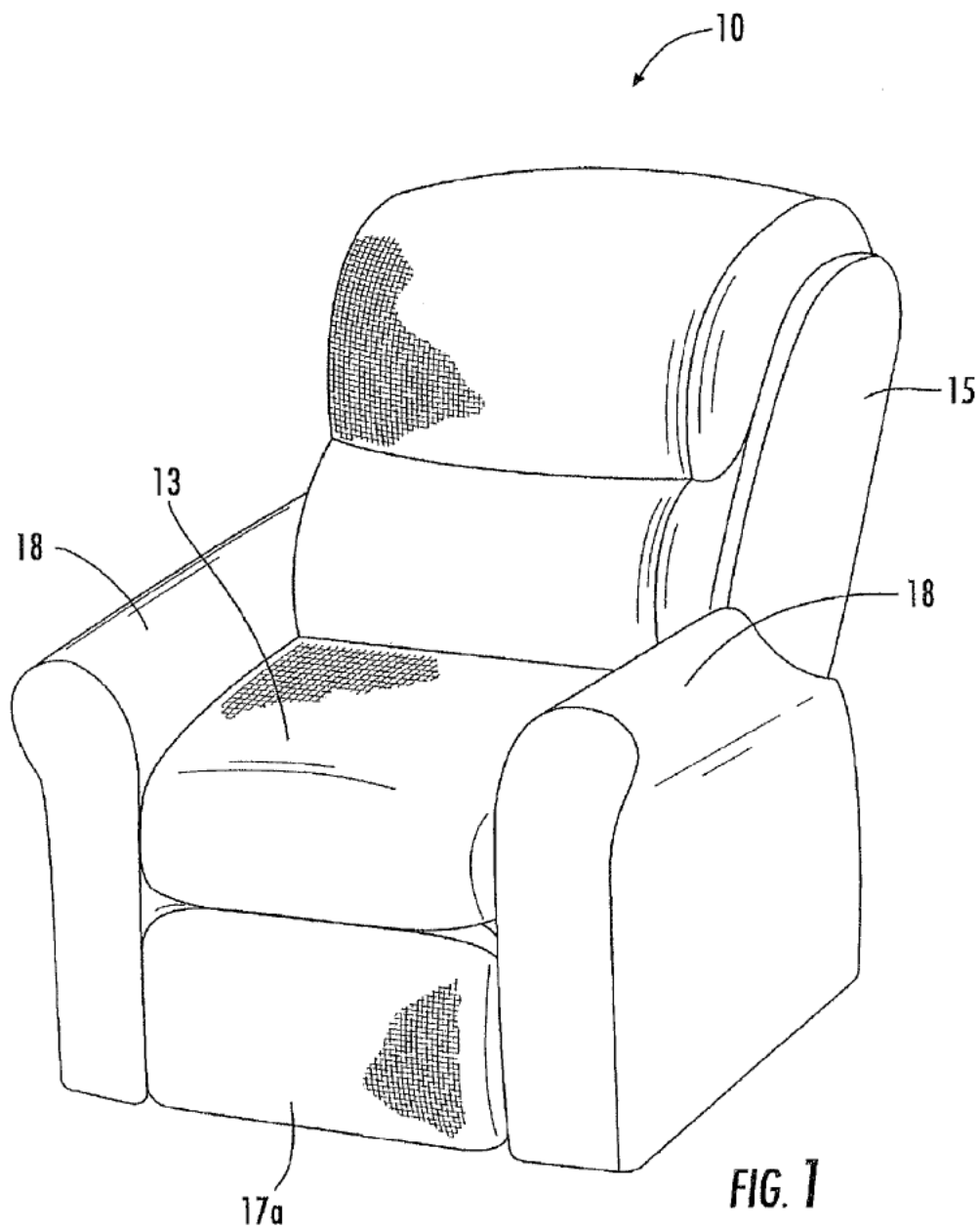
an upper swing link that is pivotally attached to the lower
swing link and to the backpost.

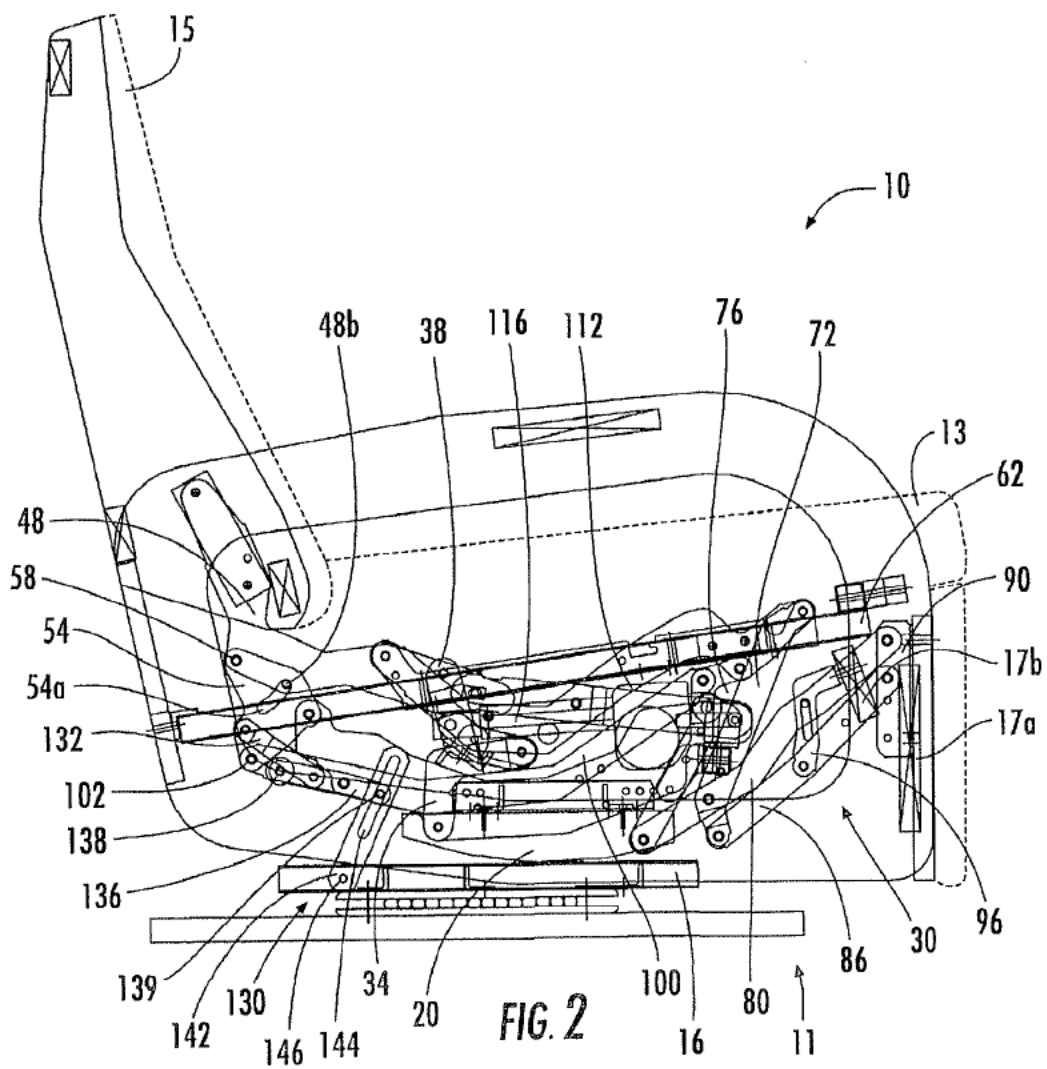
13. The seating unit defined in claim 12, further comprising
a seat adapter that is fixed relative to the seat and directly
pivotally attached to the backpost.

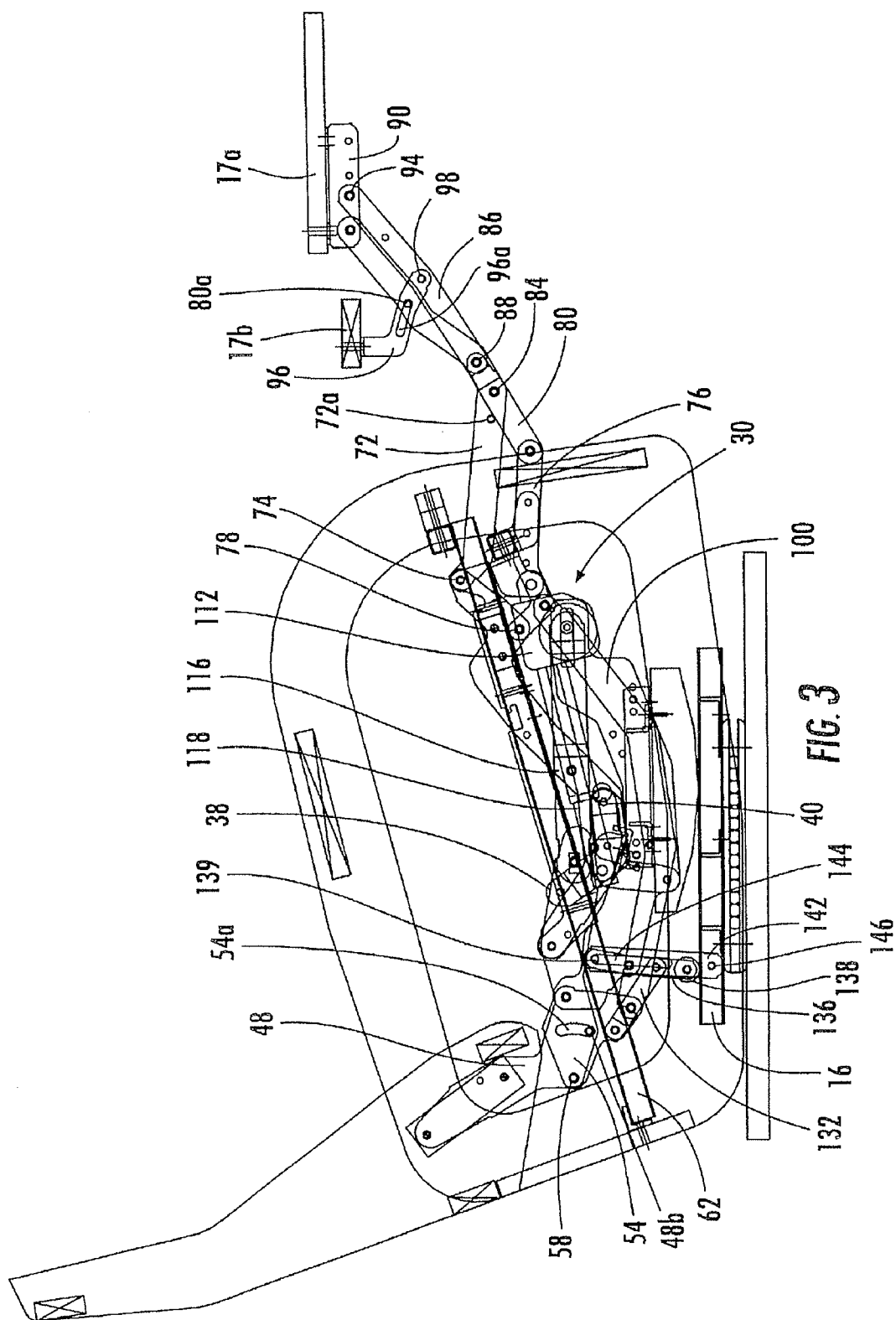
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14. The reciprocating seating unit defined in claim 13,
wherein the reclining mechanism includes a pantographic
linkage that extends the footrest in the TV and fully reclined
positions, and wherein the pantographic linkage comprises
upper and lower ottoman swing links that are directly pivot-
ally attached to the seat adapter.

* * * * *







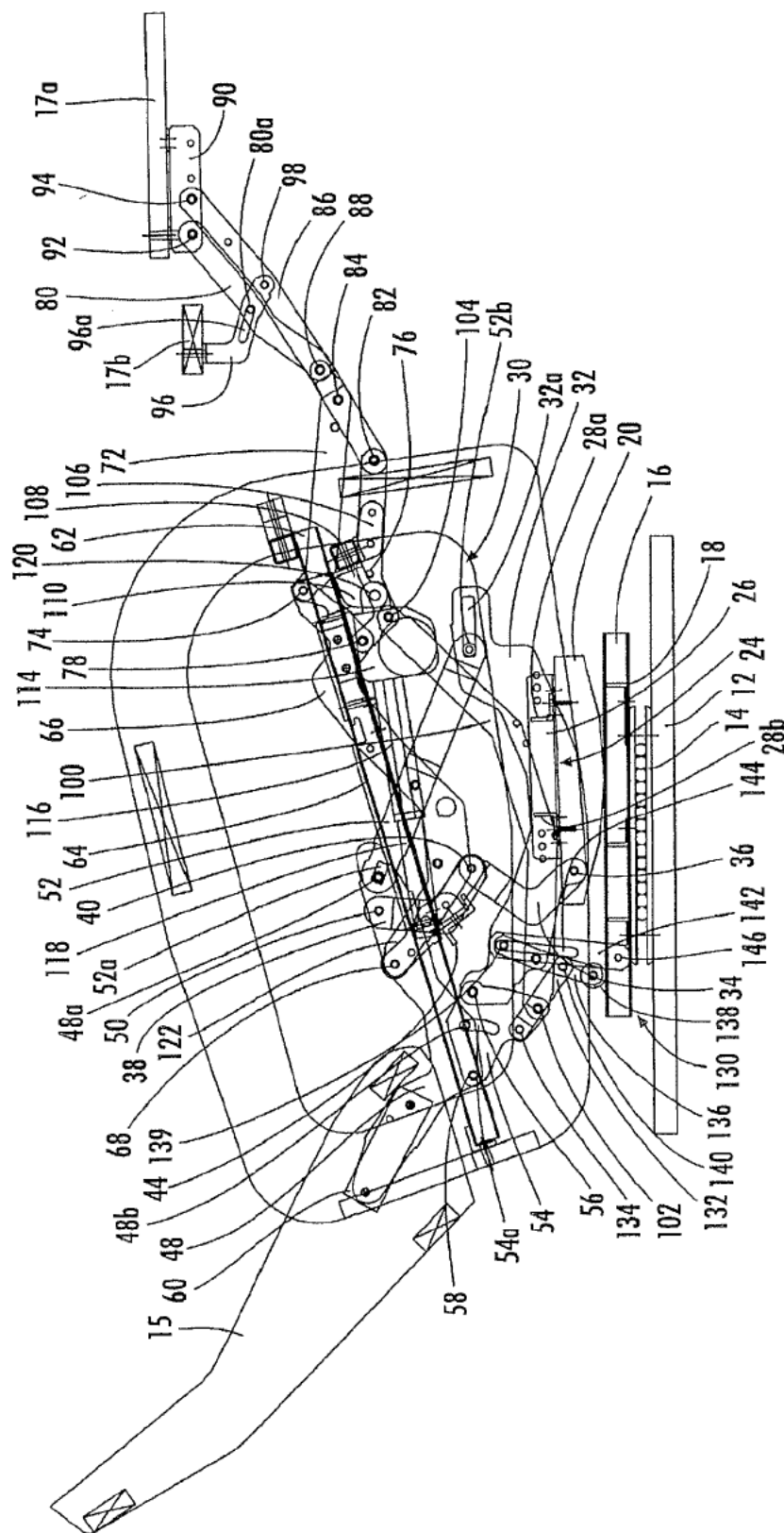


FIG. 4

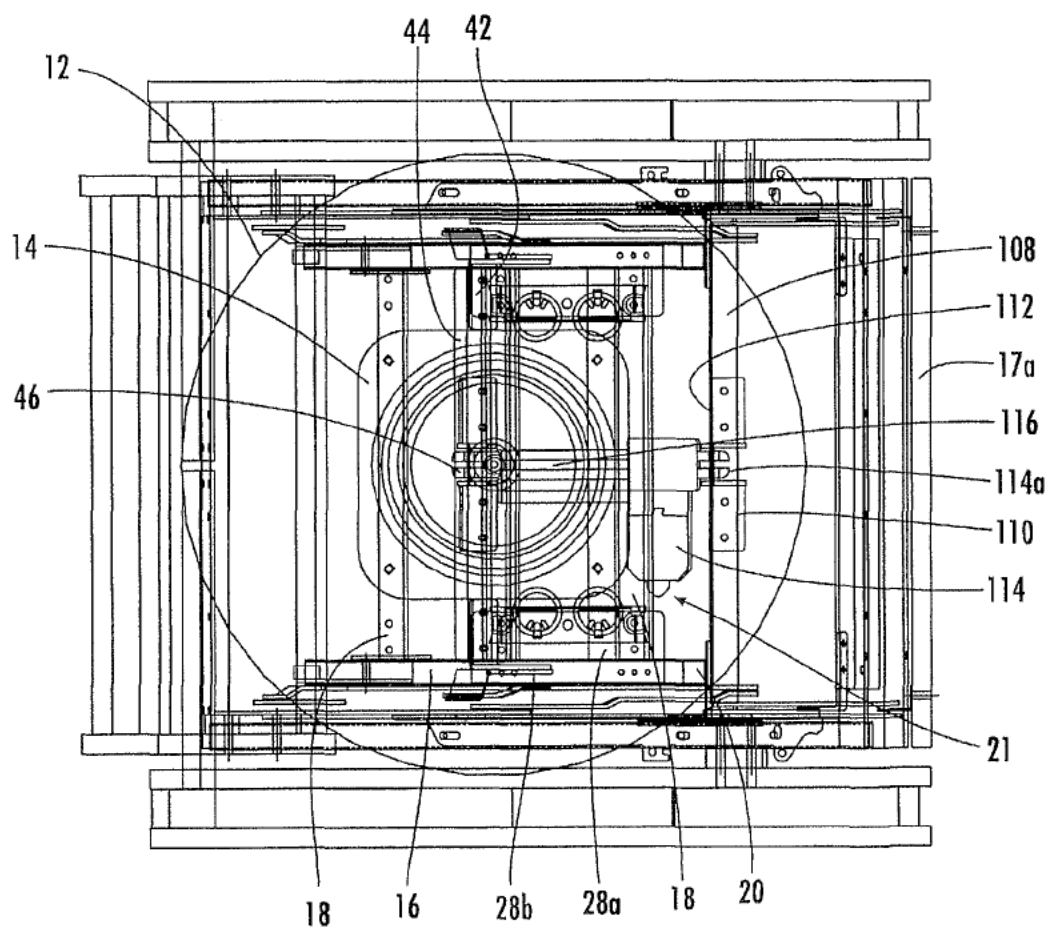


FIG. 5

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ROCKING-RECLINING SEATING UNIT WITH POWER ACTUATOR

FIELD OF THE INVENTION

This invention relates generally to seating units, and relates more particularly to reclining seating units with rocking capability.

BACKGROUND OF THE INVENTION

Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

One particularly popular reclining chair is the so-called "rocker-recliner," which can, when in the upright position, rock with a forward and rearward motion similar to that of a traditional rocking chair. A typical rocker recliner, one of which is illustrated in U.S. Pat. No. 4,519,647 to Rogers, includes an arcuate rocker cam that is attached with the lower portion of each mechanism, with the lower convex surface of the rocker cam contacting a level bearing surface of the base. Also, a spring assembly is mounted to the base of the chair and to each rocker cam. Each spring assembly includes two quite stiff, vertically-oriented helical springs attached to mounting brackets that are in turn fixed to the base and to the rocker cam. When the chair is in its upright position and is unoccupied, the seat, backrest and reclining mechanisms reside above the base, the rocker springs are deflected only along their longitudinal axes, and the rocker cams rest on a level portion of the base. When an occupant sits on the chair and applies a forwardly- or rearwardly-directed force to the seat or backrest, the seat and backrest move relative to the base. The path of movement is defined by the convex shape of the rocker cams as they rock on the level bearing surface of the base, with the result that the seat and backrest simulate the rocking motion of a rocking chair. During the rocking movement, the rocker springs deflect such that their top portions bend away from their longitudinal axes as the chair rocks forward and back. The deflection in the springs urges the springs (and, in turn, the seat and backrest) to return to their original positions as the chair returns to and through the upright position. In this manner, the chair is capable of providing a controlled rocking motion when in the upright position.

Although they are already popular seating units, it may be desirable to provide additional functionality to rocker-recliners.

SUMMARY OF THE INVENTION

As a first aspect, embodiments of the present invention are directed to a rocking and reclining seating unit. The rocker-recliner comprises: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining

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mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a second aspect, embodiments of the present invention are directed to a rocking and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a third aspect, embodiments of the present invention are directed to a rocking and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a pantographic linkage on which the ottoman is mounted. In the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position. The reclining mechanism also includes a backpost that is fixed relative to the backrest, a

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mounting bracket that is fixed relative to the rocker cams and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

As a fourth aspect, embodiments of the present invention are directed to a rocking and reclining seating unit comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a backpost fixed relative to the backrest and a sequencer plate pivotally interconnected with the backrest and with the remainder of the reclining mechanism, and wherein the sequencer plate rotates in a first rotative direction relative to the backrest when the seating unit moves from the upright position to the TV position, and wherein the backrest rotates relative to the sequencer plate in the first rotative direction when the seating unit moves from the TV position to the fully reclined position.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a perspective view of a rocker-recliner chair according to embodiments of the present invention.

FIG. 2 is a side section view of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 3 is a side section view of the chair of FIG. 1 with the chair in its TV position.

FIG. 4 is a side section view of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 5 is a top cutaway view of the frame of the chair of FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

The present invention now is described more fully herein-after with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

Like numbers refer to like elements throughout. In the figures, the thickness of certain lines, layers, components, elements or features may be exaggerated for clarity. Broken lines illustrate optional features or operations unless specified otherwise.

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The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as "between X and Y" and "between about X and Y" should be interpreted to include X and Y. As used herein, phrases such as "between about X and Y" mean "between about X and about Y." As used herein, phrases such as "from about X to Y" mean "from about X to about Y."

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well-known functions or constructions may not be described in detail for brevity and/or clarity.

It will be understood that when an element is referred to as being "on", "attached" to, "connected" to, "coupled" with, "contacting", etc., another element, it can be directly on, attached to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, "directly on", "directly attached" to, "directly connected" to, "directly coupled" with or "directly contacting" another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed "adjacent" another feature may have portions that overlap or underlie the adjacent feature.

This invention is directed to seating units that have a stationary base, a seat portion, and a backrest. As used herein, the terms "forward", "forwardly", and "front" and derivatives thereof refer to the direction defined by a vector extending from the backrest toward the seat parallel to the underlying surface. Conversely, the terms "rearward", "rearwardly", and derivatives thereof refer to the direction directly opposite the forward direction; the rearward direction is defined by a vector that extends from the seat toward the backrest parallel to the underlying surface. The terms "lateral," "laterally", and derivatives thereof refer to the direction parallel with the floor, perpendicular to the forward and rearward directions, and extending away from a plane bisecting the seating units between their armrests. The terms "medial," "inward," "inboard," and derivatives thereof refer to the direction that is the converse of the lateral direction, i.e., the direction parallel with the floor, perpendicular to the forward direction, and extending from the periphery of the seating units toward the aforementioned bisecting plane.

The seating units illustrated and described herein comprise a plurality of pivotally interconnected links. Those skilled in this art will appreciate that the pivots between links can take a variety of configurations, such as pivot pins, rivets, bolt and nut combinations, and the like, any of which would be suitable for use with the present invention. Also, the shapes of the

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links may vary as desired, as may the locations of certain of the pivots. Moreover, in some instances combinations of pivot points may be replaced by equivalent structures, such as "slider-crank" configurations, like those described in B. Paul, *Kinematics and Dynamics of Planar Machinery* 4-21 (1979).

Referring now to the figures, a swiveling rocker-recliner chair, designated broadly at 10, is illustrated in FIGS. 1-5. The chair 10 includes a base unit 11, a seat 13 that is generally horizontally disposed above the base unit 11, a backrest 15 that is generally vertically and disposed substantially above a rear portion of the seat 13, and two ottomans 17a, 17b, which, in the upright position of FIG. 1, are generally vertically disposed below a front portion of the seat 13. Arms 18 are positioned on either side of the seat 13 and move in concert with the seat 13.

A pair of mirror image reclining mechanisms 30 (only one of which is shown herein in FIGS. 2-4) are attached to the backrest 15, the seat 13, and the ottomans 17a, 17b. The reclining mechanisms 30, which comprise a plurality of interconnected links, move the chair 10 between (a) an upright position (FIGS. 1 and 2), in which the seat 13 is generally horizontally disposed, the backrest 15 is generally vertically disposed, and the ottomans 17a, 17b are generally vertically disposed and positioned below the seat 13, (b) an intermediate TV position (FIG. 3), in which the ottomans 17a, 17b are generally horizontally disposed in front of the seat 13 and the backrest 15 and the seat 13 substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position (FIG. 4), in which the angle between the backrest 15 and the seat 13 increases. These components are described in greater detail below.

The base unit 11 includes a circular lower base 12. A swivel unit 14 is mounted onto the top of the lower base 12. Rocker rails 16 are mounted atop the swivel unit 14 and are reinforced by cross-members 18. Rocker cams 20 rest on the upper bearing surfaces of the legs of the rails 16. A recliner foundation 24 is mounted to the top surfaces of the rocker cams 20; the foundation 24 includes longitudinal brackets 26 and cross-members 28a, 28b that span the longitudinal brackets 26. The arcuate lower surfaces of the cams 20 are configured for rolling contact with the bearing surfaces and enable the chair 10 to have a fore-to-aft rocking motion. This motion is controlled by rocker spring assemblies 22 that are attached to the cross-members 18 and to the cross-members 28a, 28b. The rocker cams 20, the rocker spring assemblies 22, and the components to which they are attached form a rocker mechanism 21. It is to be understood other rocker mechanisms may be employed; for example, the rocker spring assemblies 22 may also be attached directly to the reclining mechanisms 30 in other embodiments, such as in the manner illustrated in U.S. Pat. No. 5,876,094 to Hoffman, and the reclining mechanism can be mounted directly onto the rocker cams 20 as illustrated in the U.S. Pat. No. 6,000,754 to Lawson.

Because the reclining mechanisms 30 are mirror images of each other, only one reclining mechanism will be described in detail herein, with the understanding that the discussion is equally applicable to the mirror image mechanism 30. Also, for clarity the mechanism will be described first with reference to FIG. 4 (with the mechanism 30 in its reclined position), then will be described in its upright and TV positions.

Turning now to FIG. 4, the reclining mechanism 30 includes a generally crescent-shaped mounting bracket 32 that is fixed to the longitudinal rail 26. A slot 32a is present at the front end of the mounting bracket 30. An L-shaped lower swing link 34 is attached to a lower region of the mounting bracket 32 at a pivot 36 and extends upwardly and rearwardly, then upwardly and forwardly therefrom. An L-shaped upper

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swing link 38 is attached at its lower end to the upper end of the lower swing link 34 at a pivot 40 and extends upwardly and rearwardly, then upwardly therefrom.

A backpost 48 is fixed to the backrest 15 via a spacer 60 and extends generally forwardly from the lower front end of the backrest 15. The upper swing link 38 is attached to a forward portion of the backpost 48 at a pivot 50. A straight coupling link 52 having a slot 52a is attached to the backpost 48 via a pin 48a located forwardly of the pivot 50. The coupling link 52 extends forwardly and downwardly from the pin 48a to attach to the front end of the mounting bracket 32 via a pin 52b inserted into a slot 32a in the mounting bracket 32. A trapezoidal sequencer plate 54 is attached to the backpost 48 at a pivot 58, and also interacts with the backpost 48 via a slot 54a that receives a pin 48b. The sequencer plate 54 is also attached to the rear end of the mounting bracket 32 at a pivot 56.

Still referring to FIG. 4, a seat frame 62, to which the arms 18 are mounted, underlies the seat 13. A seat bracket 64 is fixed to the seat frame 62. In turn, a seat adapter 66 is fixed to the seat bracket 64. The seat adapter 66 is attached to the backpost 48 at a pivot 68 that is located just rearward of the pivot 50. An upper ottoman swing link 72 is attached to a front region of the seat adapter 66 at a pivot 74 and extends downwardly and forwardly therefrom. A tripartite lower ottoman swing link 76 is attached to a pivot 78 that is located rearwardly and downwardly from the pivot 74; the lower ottoman swing link 76 extends generally forwardly from the pivot 78. An upper ottoman extension link 80 is attached to the forward end of the lower ottoman swing link 76 at a pivot 82 and extends forwardly and upwardly therefrom. Also the upper ottoman extension link 80 is attached to the upper ottoman swing link at a pivot 84. A lower ottoman extension link 86 is attached to the forward end of the upper ottoman swing link at a pivot 88 that is positioned above and forward of the pivot 84 and extends upwardly and forwardly therefrom generally parallel with the upper ottoman extension link. An outer ottoman bracket 90 is generally horizontally disposed and is attached to the upper and lower ottoman extension links 80, 86 at pivots 92, 94 respectively. The ottoman 17a is mounted on the outer ottoman bracket 90.

An L-shaped inner ottoman bracket 96 is attached at its lower, forward end to the lower ottoman extension link 86 at a pivot 98. At its opposite end, the inner ottoman bracket 96 supports the ottoman 17b. The inner ottoman bracket 96 also includes a slot 96a that receives a pin 80a located on the upper ottoman extension link 80.

A connecting link 100 is attached to the sequencer plate 54 at a pivot 102. The connecting link 100 extends forwardly and slightly upwardly from the pivot 102 to a pivot 104 with the lower ottoman swing link 76.

A locking mechanism 130 is attached to the reclining mechanism 30 to prevent rocking of the chair 10 when it is in the TV or fully reclined positions. The locking mechanism 130 includes a drive link 132 that is pivotally interconnected at one end to the sequencer plate 54 at a pivot 134. The drive link 132 is a straight link that slopes downwardly and forwardly from the pivot 134. The opposite end of the drive link 132 is pivotally interconnected with the intermediate portion of a straight bracing link 136 at a pivot 140. The bracing link 136 includes a wheel 138 or other engagement structure at its rearward (or engagement) end, and further includes a pin 139 at its forward end. A straight control link 142 is pivotally interconnected with the longitudinal member 26 of the base unit 11 at a pivot 146; the control link 142 extends upwardly and slightly forwardly therefrom. The control link 142 includes in its upper half a slot 144 within which the pin 139 of the bracing link 136 resides.

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The chair 10 includes a power unit 112 that drives the chair 10 between its upright, TV and fully reclined positions. The power unit 112 includes a motor unit 114, to which is attached a sleeve 116. The sleeve extends rearwardly from the motor unit 114 and receives a retractable rod 118. The motor unit 114 is electrically connected with an actuator (not shown), such as a toggle switch or the like, that energizes the motor unit 114 upon actuation.

The motor unit 114 is attached to a mounting bracket 110 at a pivot 120 via a mounting tab 114a (see FIG. 5). The mounting bracket 110 is then attached to a cross-member 108 that extends between the reclining mechanisms 30. A bracket 106 is fixed to each of the lower ottoman swing links 76 and to the cross-member 108.

The rod 118 of the power unit 112 is attached to a bracket 46 (FIG. 5) at a pivot 122. The bracket 46 is then attached to a cross-member 44 that extends between the reclining mechanisms 30; the cross-member 44 is fixed to a bracket 42 that is in turn fixed to the upper swing link 38 of each reclining mechanism 30.

As can be seen in FIG. 2, in the upright position, the rod 118 of the power unit 112 is retracted into the sleeve 116. As a result, a pantographic linkage formed by the upper and lower ottoman swing links 72, 76 and the upper and lower ottoman extension links 80, 86 are folded under the seat frame 62, which positions the ottoman 17b underneath a forward portion of the seat frame 62 and the ottoman 17a just forward of and below the seat frame 62 in a vertical orientation. Also, the lower and upper swing links 34, 38 are oriented such that the segments thereof adjacent the pivot 36 partially overlap, which positions the backpost 48 and, in turn, the backrest 15, in a generally upright position.

Additionally, the sequencer plate 54 is oriented such that the pivot 102 is located downwardly and rearwardly, such that the bracing link 136 of the locking mechanism 130 is raised, with the wheel 138 facing rearwardly. In this position, the chair 10 is free to rock as the rocker cams 20 engage in rolling motion relative to the rocker rails 16. During the rocking motion, the majority of the locking mechanism 130 is stationary relative to the reclining mechanisms 30. Because the bracing link 136 does not move relative to the reclining mechanism 30 as the chair 10 rocks, similarly the pin 139 that resides in the slot 144 in the control link 142 does not move relative to the reclining mechanism 30 as the chair 10 rocks. However, the presence of the slot 144 permits the pin 139 to slide and/or reciprocate therein when the chair 10 is rocking. In the illustrated embodiment, the control link 142 moves very little during the rocking motion; however, in other embodiments, the control link 142 may pivot about the pivot 146 during rocking.

To move the chair 10 from the upright position to the TV position of FIG. 3, an occupant of the chair 10 actuates the actuator, which causes the rod 118 to begin to extend from the sleeve 116. Extension of the rod 118 forces the upper swing link 38 to rotate slightly counterclockwise (from the vantage point of FIGS. 2 and 3) about the pivot 40. This movement draws the backpost 48 and rear end of the seat frame 62 downwardly, which causes the seat 13 to increase in pitch angle relative to the underlying surface. However, the weight of the seated occupant prevents further rotation of the upper swing link 38 at this point. Because there is little to no fore-and-aft movement of the rear end of the rod 118, the motor unit 114 moves forwardly relative to the base unit 11, thereby driving the lower ottoman swing link 76 counterclockwise about the pivot 78. Rotation of the lower ottoman swing link 76 forces the upper ottoman extension link 80 forward, which in turn draws the upper ottoman swing link 72 counterclock-

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wise around the pivot 74. Also, the lower ottoman extension link 86 moves forwardly more than the upper ottoman extension link 80, such that the outer ottoman bracket 90 and the inner ottoman bracket 96 rotate counterclockwise (rotation of the inner ottoman bracket 96 causes the pin 80a to move in the slot 96a toward the pivot 98. The rotation of the outer and inner ottoman brackets 90, 96 induces the ottomans 17a, 17b to rotate from a vertical orientation to a horizontal orientation. Motion of the ottomans 17a, 17b ceases when a pin 72a on the upper ottoman swing link 72 strikes the upper edge of the lower ottoman extension link 80.

In addition, the rotation of the lower ottoman swing link 76 draws the connecting link 100 forward. Movement of the connecting link 100 rotates the sequencer plate 54 counterclockwise about the pivot 58 with the backpost 48. The drive link 132 of the locking mechanism is driven forward and rotates the bracing link 136. The forward motion of the drive link 132 causes the bracing link 136 to rotate such that its forward end rises and moves rearwardly and such that its rearward end, on which the wheel 138 is mounted, descends and moves forwardly (this rotation is counterclockwise from the vantage point of FIGS. 2 and 3). Also, the elevation of the forward end of the bracing link 136 causes the pin 139 to rise to the top of the slot 144 of the control link 142. The control link 142 also rotates slightly about the pivot 146 to take a more vertical disposition. The movement of these links ceases when the wheel 138 engages a portion of the upper bearing surface of the rocker rail 16 rearward of the portion of the bearing surface beneath the rocker cams 20 (which substantially coincides with the pin 48b reaching the lower end of the slot 54a). In this position, the linkage between the bracing link 136, the drive link 132 and the sequencer plate 54 restrains the mounting bracket 32, thereby preventing the reclining mechanisms 30, and in turn the chair 10, from rocking relative to the base unit 11.

To move the chair 10 from the TV position of FIG. 3 to the fully reclined position of FIG. 4, the occupant can continue to operate the actuator, which extends the rod 118 farther out of the sleeve 116. When the ottomans 17a, 17b are fully extended, the pivots 84, 88 and 94 are in a near "over-center" condition that helps to lock the ottomans 17a, 17b in an extended position. Thus, as the rod 118 continues to extend from the sleeve 116 when the chair is in the TV position, the motor unit 114 cannot move forward relative to the base unit 11 any farther, so the rear end of the rod 118 begins to move rearwardly relative to the base unit 11, resulting in clockwise rotation of the upper swing link 38 about the pivot 40. This rotation both drives the lower swing link 34 counterclockwise about the pivot 36 and drives the forward end of the backpost 48 upward. As a consequence, the backpost 48 rotates counterclockwise about the pivot 58 relative to the sequencer plate 54 (and to the seat adapter 66, the seat frame 62 and the seat bracket 64). Rotation of the backpost 48 and, in turn, the backrest 15, continues until (a) the pin 48b on the backpost 48 reaches the upper end of the slot 54a in the sequencer plate 54 and the pin 52b of the coupling link 52 reaches the rear end of the slot 32a of the mounting bracket 32. In this position, the backrest 15 has reclined relative to the seat 13 at a greater angle than in the upright and TV positions.

It can also be seen in FIG. 4, the locking mechanism 130 continues to prevent the chair 10 from rocking as the chair 10 moves to the fully reclined position. It can also be seen that the seat 13 rises during movement from the TV position to the fully reclined position.

The chair 10 can be returned to its TV position from the fully reclined position by the occupant operating the actuator in the reverse direction. Because of the over-center condition

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of the pivots 82, 84, 88, the ottomans 17a, 17b remain extended, such that retraction of the rod 118 draws the backrest 15 to its upright position. Once the motion of the backrest 15 is complete, the over-center condition of the pivots 84, 88 and 94 can be overcome, and the ottomans 17a, 17b are able to retract to the upright position.

Those skilled in this art will recognize that other variations of the chair 10 are contemplated in connection with the present invention. For example, the power unit 112 may be reversed, such that the motor 114 is at the rear end of the power unit 112 and the rod 118 is at the front end. In such a case, the rod 118 would extend forwardly relative to the base unit 11 as the chair 10 moved from the upright position to the TV position, and the motor 114 would move rearwardly relative to the base unit 11 when the chair moved from the TV position to the fully reclined position. In either instance, the front end of the power unit 112 moves relative to the base unit 11 during the movement from the upright to the TV position, and the rear end of the power unit moves relative to the base unit 11 during the movement from the TV to the fully reclined position.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as recited in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

That which is claimed is:

1. A rocking and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base;

a generally upright backrest positioned above the base and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and

a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the reclining mechanism includes a mounting bracket fixed relative to the rocker mechanism, a lower swing link pivotally attached with the mounting bracket, and an upper swing link pivotally attached with the lower swing link and with the backrest, and wherein a rear end of the power actuating unit is attached with the upper swing link.

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2. The seating unit defined in claim 1, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

3. The seating unit defined in claim 1, further comprising a locking mechanism that allows the seating unit to rock while in the upright position but prevents rocking of the seating unit while in the TV and fully reclined positions.

4. The seating unit defined in claim 1, wherein the power actuating unit includes opposed first and second ends, the rear end being the second end, and wherein the first end of the power actuating unit moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the power actuating unit moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

5. The seating unit defined in claim 4, wherein the power actuating unit includes a motor and an extendable member at the first and second ends.

6. The seating unit defined in claim 1, wherein the seat rises in moving from the TV position to the fully reclined position.

7. The seating unit defined in claim 1, further comprising a swivel unit attached to the base.

8. The seating unit defined in claim 1, wherein the reclining mechanism includes a backpost fixed relative to the backrest and pivotally interconnected to the upper swing link and a sequencer plate pivotally interconnected with the backrest and with the remainder of the reclining mechanism, and wherein the sequencer plate rotates in a first rotative direction relative to the backrest when the seating unit moves from the upright position to the TV position, and wherein the backrest rotates relative to the sequencer plate in the first rotative direction when the seating unit moves from the TV position to the fully reclined position.

9. The seating unit defined in claim 8, wherein the backrest includes one of a pin and a slot, and the sequencer plate includes the other of a pin and a slot, the pin being received in the slot.

10. The seating unit defined in claim 9, wherein the backrest includes the pin, and the sequencer plate includes the slot.

11. A rocking and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base;

a generally upright backrest positioned above the base and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and

a linear actuator attached to the reclining mechanism, the linear actuator configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they

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have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the linear actuator includes opposed first and second ends, and wherein the first end of the linear actuator moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the linear actuator moves rearwardly when the seating unit moves from the TV position to the fully reclined position.

12. The seating unit defined in claim 11, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

13. The seating unit defined in claim 11, further comprising a locking mechanism that allows the seating unit to rock while in the upright position but prevents rocking of the seating unit while in the TV and fully reclined positions.

14. The seating unit defined in claim 11, wherein the seat rises in moving from the TV position to the fully reclined position.

15. The seating unit defined in claim 11, further comprising a swivel unit attached to the base.

16. A rocking and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base;

a generally upright backrest positioned above the base and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and

a power actuating unit attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

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wherein the reclining mechanism includes a mounting bracket fixed relative to the rocker mechanism, a lower swing link pivotally attached with the mounting bracket, and an upper swing link pivotally attached with the lower swing link and with the backrest, and wherein a rear end of the power actuating unit is attached with the upper swing link; and

wherein the rocker mechanism includes a rocker cam that contacts the first bearing surface, and wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, the mounting bracket, which is pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat pivotally attached to the backpost.

17. A rocking and reclining seating unit, comprising:

a base unit with a first bearing surface;

a generally horizontally-disposed seat positioned above the base;

a generally upright backrest positioned above the base and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links;

a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and

a linear actuator attached to the reclining mechanism, the actuating unit configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the linear actuator includes opposed first and second ends, and wherein the first end of linear actuator moves forwardly as the seating unit moves from the upright position to the TV position, and wherein the second end of the linear actuator moves rearwardly when the seating unit moves from the TV position to the fully reclined position;

wherein the reclining mechanism includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the rocker mechanism and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat pivotally attached to the backpost.

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(12) **United States Patent**
Hoffman et al.

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(45) **Date of Patent:** *Jun. 11, 2013

(54) **ROCKING-RECLINING SEATING UNIT
WITH POWER ACTUATOR**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 42 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **13/363,641**

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(65) **Prior Publication Data**

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Nov. 24, 2008, now Pat. No. 8,113,574.

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A47C 1/35 (2006.01)

A47C 1/38 (2006.01)

(52) **U.S. Cl.**
USPC 297/85 M; 297/362.11; 297/85 R;
297/362.14; 297/DIG. 7

(58) **Field of Classification Search**
USPC 297/85 R, 85 M, 362.11, 362.13,
297/362.14, DIG. 7

See application file for complete search history.

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Primary Examiner — David Dunn

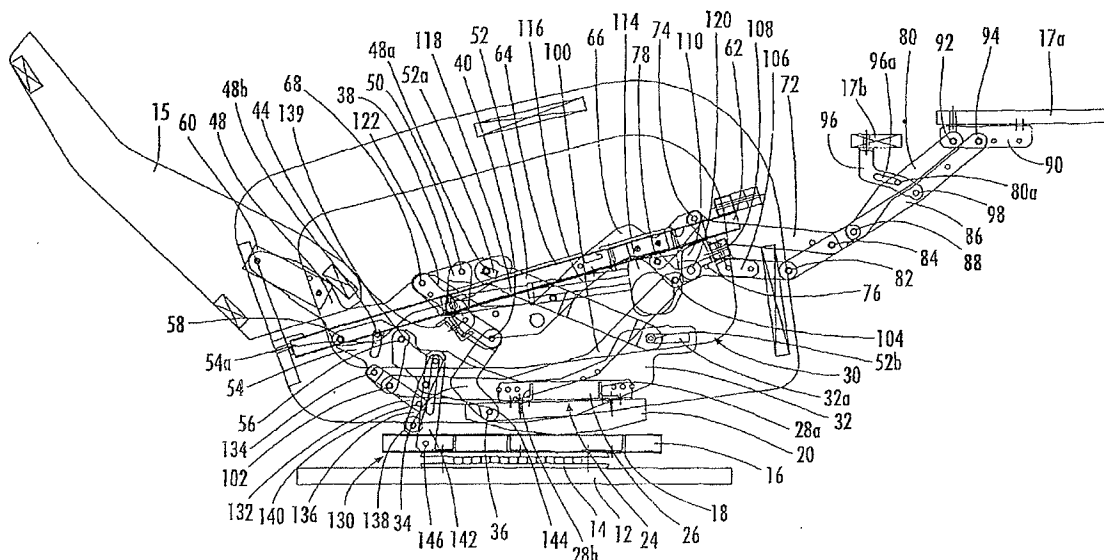
Assistant Examiner — Tania Abraham

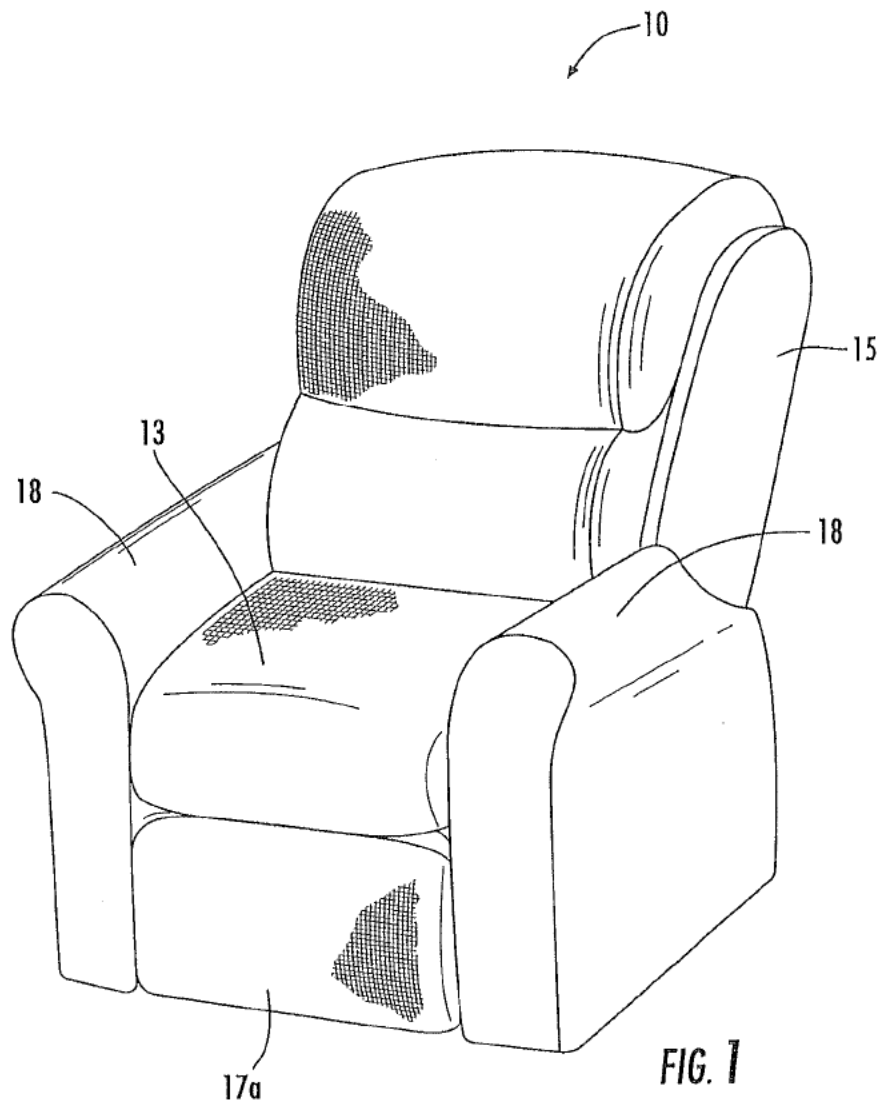
(74) *Attorney, Agent, or Firm* — Myers Bigel Sibley &
Sajovec, P.A.

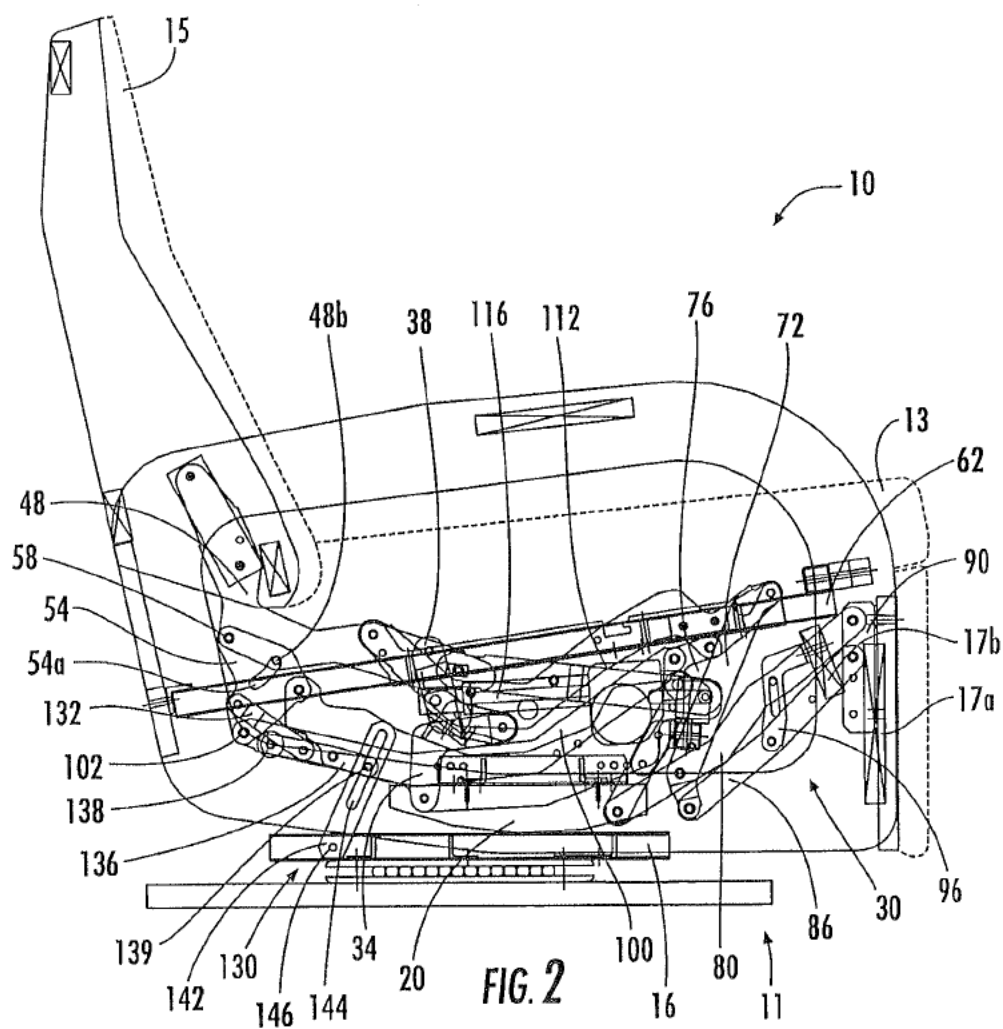
(57) **ABSTRACT**

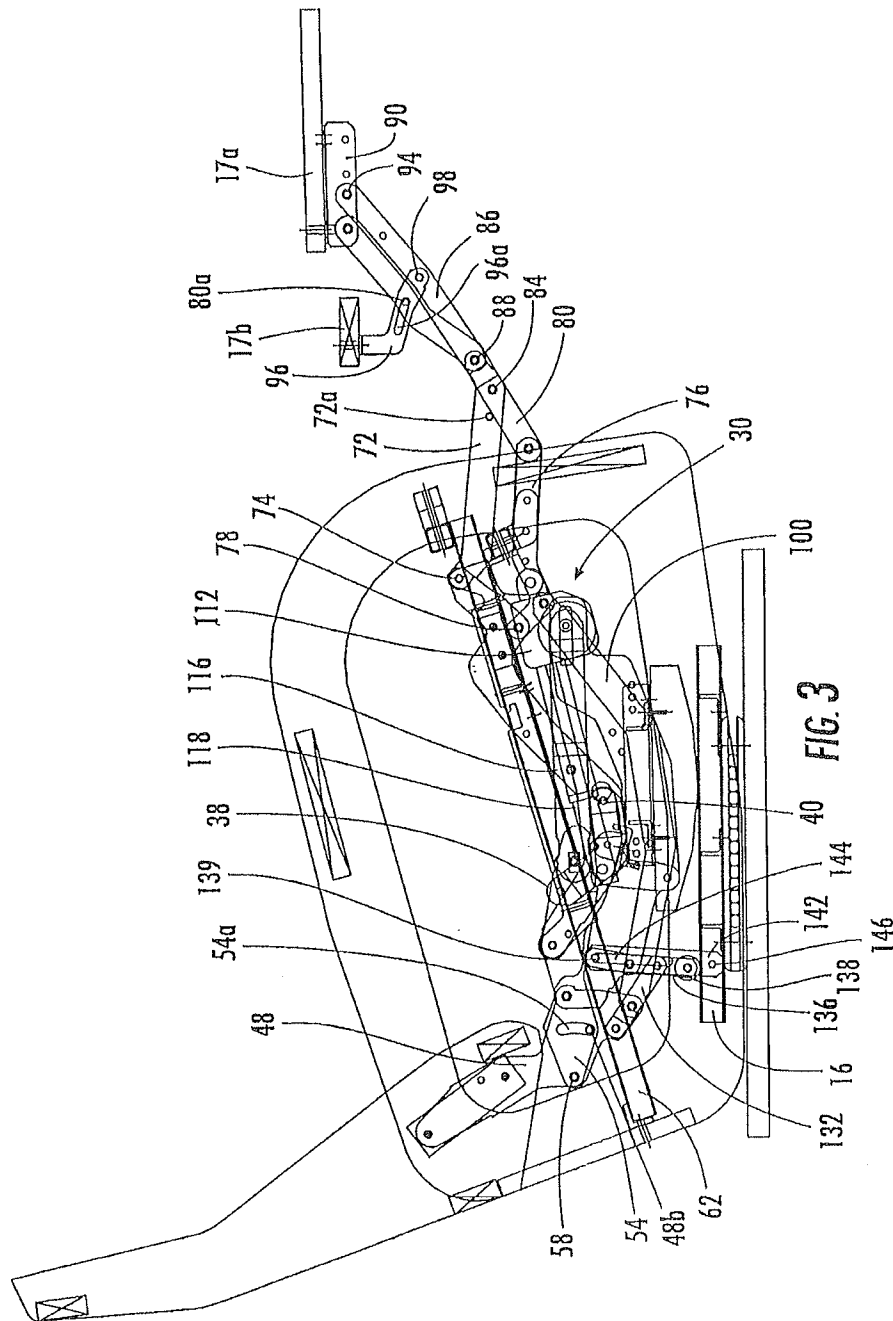
A rocking and reclining seating unit includes: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between an upright position, an intermediate TV position, and a fully reclined position.

5 Claims, 5 Drawing Sheets









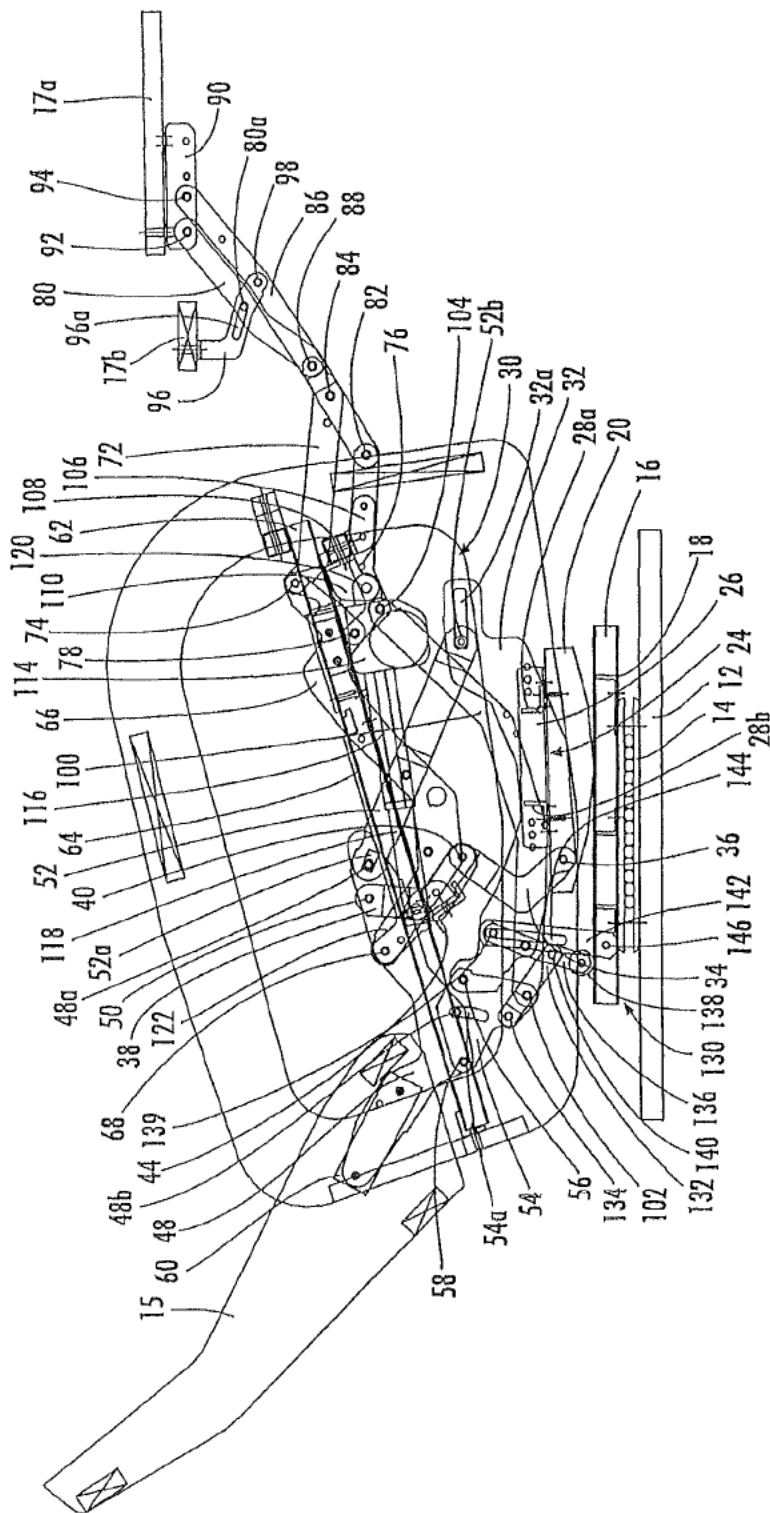


FIG. 4

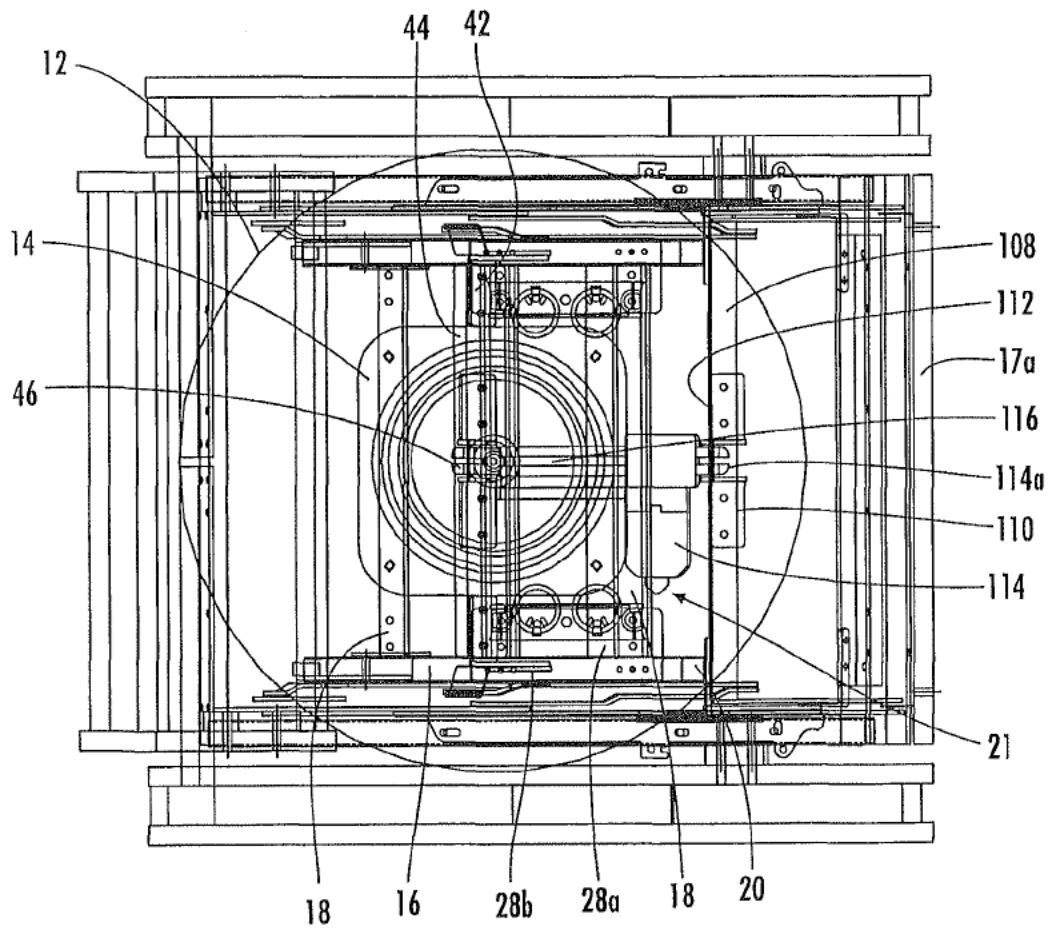


FIG. 5

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**ROCKING-RECLINING SEATING UNIT
WITH POWER ACTUATOR****RELATED APPLICATIONS**

This application is a continuation of prior U.S. patent application Ser. No. 12/276,559, filed Nov. 24, 2008 now U.S. Pat. No. 8,113,574 and entitled ROCKING-RECLINING SEATING UNIT WITH POWER ACTUATOR, the disclosure of which is hereby incorporated herein in its entirety.

FIELD OF THE INVENTION

This invention relates generally to seating units, and relates more particularly to reclining seating units with rocking capability.

BACKGROUND OF THE INVENTION

Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

One particularly popular reclining chair is the so-called "rocker-recliner," which can, when in the upright position, rock with a forward and rearward motion similar to that of a traditional rocking chair. A typical rocker recliner, one of which is illustrated in U.S. Pat. No. 4,519,647 to Rogers, includes an arcuate rocker cam that is attached with the lower portion of each mechanism, with the lower convex surface of the rocker cam contacting a level bearing surface of the base. Also, a spring assembly is mounted to the base of the chair and to each rocker cam. Each spring assembly includes two quite stiff, vertically-oriented helical springs attached to mounting brackets that are in turn fixed to the base and to the rocker cam. When the chair is in its upright position and is unoccupied, the seat, backrest and reclining mechanisms reside above the base, the rocker springs are deflected only along their longitudinal axes, and the rocker cams rest on a level portion of the base. When an occupant sits on the chair and applies a forwardly- or rearwardly-directed force to the seat or backrest, the seat and backrest move relative to the base. The path of movement is defined by the convex shape of the rocker cams as they rock on the level bearing surface of the base, with the result that the seat and backrest simulate the rocking motion of a rocking chair. During the rocking movement, the rocker springs deflect such that their top portions bend away from their longitudinal axes as the chair rocks forward and back. The deflection in the springs urges the springs (and, in turn, the seat and backrest) to return to their original positions as the chair returns to and through the upright position. In this manner, the chair is capable of providing a controlled rocking motion when in the upright position.

Although they are already popular seating units, it may be desirable to provide additional functionality to rocker-recliners.

SUMMARY OF THE INVENTION

As a first aspect, embodiments of the present invention are directed to a rocking and reclining seating unit. The rocker-

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recliner comprises: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a second aspect, embodiments of the present invention are directed to a rocking and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases.

As a third aspect, embodiments of the present invention are directed to a rocking and reclining seating unit, comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a linear actuator attached to the reclining mechanism. The linear actuator is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relation-

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ship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a pantographic linkage on which the ottoman is mounted. In the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position. The reclining mechanism also includes a backpost that is fixed relative to the backrest, a mounting bracket that is fixed relative to the rocker cams and pivotally attached to the backpost, and a seat adapter that is fixed relative to the seat and pivotally attached to the backpost.

As a fourth aspect, embodiments of the present invention are directed to a rocking and reclining seating unit comprising: a base unit with a first bearing surface; a generally horizontally-disposed seat positioned above the base; a generally upright backrest positioned above the base and substantially rearward of the seat; an extendable ottoman; a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a rocker mechanism attached with the base unit and the reclining mechanism, the rocker mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed rocking motion relative to the base unit; and a power actuating unit attached to the reclining mechanism. The actuating unit is configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman are generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases. The reclining mechanism includes a backpost fixed relative to the backrest and a sequencer plate pivotally interconnected with the backrest and with the remainder of the reclining mechanism, and wherein the sequencer plate rotates in a first rotative direction relative to the backrest when the seating unit moves from the upright position to the TV position, and wherein the backrest rotates relative to the sequencer plate in the first rotative direction when the seating unit moves from the TV position to the fully reclined position.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a perspective view of a rocker-recliner chair according to embodiments of the present invention.

FIG. 2 is a side section view of the chair of FIG. 1 with the chair in its fully upright position.

FIG. 3 is a side section view of the chair of FIG. 1 with the chair in its TV position.

FIG. 4 is a side section view of the chair of FIG. 1 with the chair in its fully reclined position.

FIG. 5 is a top cutaway view of the frame of the chair of FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

The present invention now is described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure

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will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

Like numbers refer to like elements throughout. In the figures, the thickness of certain lines, layers, components, elements or features may be exaggerated for clarity. Broken lines illustrate optional features or operations unless specified otherwise.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as "between X and Y" and "between about X and Y" should be interpreted to include X and Y. As used herein, phrases such as "between about X and Y" mean "between about X and about Y." As used herein, phrases such as "from about X to Y" mean "from about X to about Y."

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well-known functions or constructions may not be described in detail for brevity and/or clarity.

It will be understood that when an element is referred to as being "on", "attached" to, "connected" to, "coupled" with, "contacting", etc., another element, it can be directly on, attached to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, "directly on", "directly attached" to, "directly connected" to, "directly coupled" with or "directly contacting" another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed "adjacent" another feature may have portions that overlap or underlie the adjacent feature.

This invention is directed to seating units that have a stationary base, a seat portion, and a backrest. As used herein, the terms "forward", "forwardly", and "front" and derivatives thereof refer to the direction defined by a vector extending from the backrest toward the seat parallel to the underlying surface. Conversely, the terms "rearward", "rearwardly", and derivatives thereof refer to the direction directly opposite the forward direction; the rearward direction is defined by a vector that extends from the seat toward the backrest parallel to the underlying surface. The terms "lateral," "laterally", and derivatives thereof refer to the direction parallel with the floor, perpendicular to the forward and rearward directions, and extending away from a plane bisecting the seating units between their armrests. The terms "medial," "inward," "inboard," and derivatives thereof refer to the direction that is the converse of the lateral direction, i.e., the direction parallel with the floor, perpendicular to the forward direction, and

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extending from the periphery of the seating units toward the aforementioned bisecting plane.

The seating units illustrated and described herein comprise a plurality of pivotally interconnected links. Those skilled in this art will appreciate that the pivots between links can take a variety of configurations, such as pivot pins, rivets, bolt and nut combinations, and the like, any of which would be suitable for use with the present invention. Also, the shapes of the links may vary as desired, as may the locations of certain of the pivots. Moreover, in some instances combinations of pivot points may be replaced by equivalent structures, such as "slider-crank" configurations, like those described in B. Paul, *Kinematics and Dynamics of Planar Machinery* 4-21 (1979).

Referring now to the figures, a swiveling rocker-recliner chair, designated broadly at 10, is illustrated in FIGS. 1-5. The chair 10 includes a base unit 11, a seat 13 that is generally horizontally disposed above the base unit 11, a backrest 15 that is generally vertically and disposed substantially above a rear portion of the seat 13, and two ottomans 17a, 17b, which, in the upright position of FIG. 1, are generally vertically disposed below a front portion of the seat 13. Arms 18 are positioned on either side of the seat 13 and move in concert with the seat 13.

A pair of mirror image reclining mechanisms 30 (only one of which is shown herein in FIGS. 2-4) are attached to the backrest 15, the seat 13, and the ottomans 17a, 17b. The reclining mechanisms 30, which comprise a plurality of interconnected links, move the chair 10 between (a) an upright position (FIGS. 1 and 2), in which the seat 13 is generally horizontally disposed, the backrest 15 is generally vertically disposed, and the ottomans 17a, 17b are generally vertically disposed and positioned below the seat 13, (b) an intermediate TV position (FIG. 3), in which the ottomans 17a, 17b are generally horizontally disposed in front of the seat 13 and the backrest 15 and the seat 13 substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position (FIG. 4), in which the angle between the backrest 15 and the seat 13 increases. These components are described in greater detail below.

The base unit 11 includes a circular lower base 12. A swivel unit 14 is mounted onto the top of the lower base 12. Rocker rails 16 are mounted atop the swivel unit 14 and are reinforced by cross-members 18. Rocker cams 20 rest on the upper bearing surfaces of the legs of the rails 16. A recliner foundation 24 is mounted to the top surfaces of the rocker cams 20; the foundation 24 includes longitudinal brackets 26 and cross-members 28a, 28b that span the longitudinal brackets 26. The arcuate lower surfaces of the cams 20 are configured for rolling contact with the bearing surfaces and enable the chair 10 to have a fore-to-aft rocking motion. This motion is controlled by rocker spring assemblies 22 that are attached to the cross-members 18 and to the cross-members 28a, 28b. The rocker cams 20, the rocker spring assemblies 22, and the components to which they are attached form a rocker mechanism 21. It is to be understood other rocker mechanisms may be employed; for example, the rocker spring assemblies 22 may also be attached directly to the reclining mechanisms 30 in other embodiments, such as in the manner illustrated in U.S. Pat. No. 5,876,094 to Hoffman, and the reclining mechanism can be mounted directly onto the rocker cams 20 as illustrated in the U.S. Pat. No. 6,000,754 to Lawson.

Because the reclining mechanisms 30 are mirror images of each other, only one reclining mechanism will be described in detail herein, with the understanding that the discussion is equally applicable to the mirror image mechanism 30. Also, for clarity the mechanism will be described first with refer-

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ence to FIG. 4 (with the mechanism 30 in its reclined position), then will be described in its upright and TV positions.

Turning now to FIG. 4, the reclining mechanism 30 includes a generally crescent-shaped mounting bracket 32 that is fixed to the longitudinal rail 26. A slot 32a is present at the front end of the mounting bracket 30. An L-shaped lower swing link 34 is attached to a lower region of the mounting bracket 32 at a pivot 36 and extends upwardly and rearwardly, then upwardly and forwardly therefrom. An L-shaped upper swing link 38 is attached at its lower end to the upper end of the lower swing link 34 at a pivot 40 and extends upwardly and rearwardly, then upwardly therefrom.

A backpost 48 is fixed to the backrest 15 via a spacer 60 and extends generally forwardly from the lower front end of the backrest 15. The upper swing link 38 is attached to a forward portion of the backpost 48 at a pivot 50. A straight coupling link 52 having a slot 52a is attached to the backpost 48 via a pin 48a located forwardly of the pivot 50. The coupling link 52 extends forwardly and downwardly from the pin 48a to attach to the front end of the mounting bracket 32 via a pin 52b inserted into a slot 32a in the mounting bracket 32. A trapezoidal sequencer plate 54 is attached to the backpost 48 at a pivot 58, and also interacts with the backpost 48 via a slot 54a that receives a pin 48b. The sequencer plate 54 is also attached to the rear end of the mounting bracket 32 at a pivot 56.

Still referring to FIG. 4, a seat frame 62, to which the arms 18 are mounted, underlies the seat 13. A seat bracket 64 is fixed to the seat frame 62. In turn, a seat adapter 66 is fixed to the seat bracket 64. The seat adapter 66 is attached to the backpost 48 at a pivot 68 that is located just rearward of the pivot 50. An upper ottoman swing link 72 is attached to a front region of the seat adapter 66 at a pivot 74 and extends downwardly and forwardly therefrom. A tripartite lower ottoman swing link 76 is attached to a pivot 78 that is located rearwardly and downwardly from the pivot 74; the lower ottoman swing link 76 extends generally forwardly from the pivot 78. An upper ottoman extension link 80 is attached to the forward end of the lower ottoman swing link 76 at a pivot 82 and extends forwardly and upwardly therefrom. Also the upper ottoman extension link 80 is attached to the upper ottoman swing link at a pivot 84. A lower ottoman extension link 86 is attached to the forward end of the upper ottoman swing link at a pivot 88 that is positioned above and forward of the pivot 84 and extends upwardly and forwardly therefrom generally parallel with the upper ottoman extension link. An outer ottoman bracket 90 is generally horizontally disposed and is attached to the upper and lower ottoman extension links 80, 86 at pivots 92, 94 respectively. The ottoman 17a is mounted on the outer ottoman bracket 90.

An L-shaped inner ottoman bracket 96 is attached at its lower, forward end to the lower ottoman extension link 86 at a pivot 98. At its opposite end, the inner ottoman bracket 96 supports the ottoman 17b. The inner ottoman bracket 96 also includes a slot 96a that receives a pin 80a located on the upper ottoman extension link 80.

A connecting link 100 is attached to the sequencer plate 54 at a pivot 102. The connecting link 100 extends forwardly and slightly upwardly from the pivot 102 to a pivot 104 with the lower ottoman swing link 76.

A locking mechanism 130 is attached to the reclining mechanism 30 to prevent rocking of the chair 10 when it is in the TV or fully reclined positions. The locking mechanism 130 includes a drive link 132 that is pivotally interconnected at one end to the sequencer plate 54 at a pivot 134. The drive link 132 is a straight link that slopes downwardly and forwardly from the pivot 134. The opposite end of the drive link 132 is pivotally interconnected with the intermediate portion

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of a straight bracing link 136 at a pivot 140. The bracing link 136 includes a wheel 138 or other engagement structure at its rearward (or engagement) end, and further includes a pin 139 at its forward end. A straight control link 142 is pivotally interconnected with the longitudinal member 26 of the base unit 11 at a pivot 146; the control link 142 extends upwardly and slightly forwardly therefrom. The control link 142 includes in its upper half a slot 144 within which the pin 139 of the bracing link 136 resides.

The chair 10 includes a power unit 112 that drives the chair 10 between its upright, TV and fully reclined positions. The power unit 112 includes a motor unit 114, to which is attached a sleeve 116. The sleeve extends rearwardly from the motor unit 114 and receives a retractable rod 118. The motor unit 114 is electrically connected with an actuator (not shown), such as a toggle switch or the like, that energizes the motor unit 114 upon actuation.

The motor unit 114 is attached to a mounting bracket 110 at a pivot 120 via a mounting tab 114a (see FIG. 5). The mounting bracket 110 is then attached to a cross-member 108 that extends between the reclining mechanisms 30. A bracket 106 is fixed to each of the lower ottoman swing links 76 and to the cross-member 108.

The rod 118 of the power unit 112 is attached to a bracket 46 (FIG. 5) at a pivot 122. The bracket 46 is then attached to a cross-member 44 that extends between the reclining mechanisms 30; the cross-member 44 is fixed to a bracket 42 that is in turn fixed to the upper swing link 38 of each reclining mechanism 30.

As can be seen in FIG. 2, in the upright position, the rod 118 of the power unit 112 is retracted into the sleeve 116. As a result, a pantographic linkage formed by the upper and lower ottoman swing links 72, 76 and the upper and lower ottoman extension links 80, 86 are folded under the seat frame 62, which positions the ottoman 17b underneath a forward portion of the seat frame 62 and the ottoman 17a just forward of and below the seat frame 62 in a vertical orientation. Also, the lower and upper swing links 34, 38 are oriented such that the segments thereof adjacent the pivot 36 partially overlap, which positions the backpost 48 and, in turn, the backrest 15, in a generally upright position.

Additionally, the sequencer plate 54 is oriented such that the pivot 102 is located downwardly and rearwardly, such that the bracing link 136 of the locking mechanism 130 is raised, with the wheel 138 facing rearwardly. In this position, the chair 10 is free to rock as the rocker cams 20 engage in rolling motion relative to the rocker rails 16. During the rocking motion, the majority of the locking mechanism 130 is stationary relative to the reclining mechanisms 30. Because the bracing link 136 does not move relative to the reclining mechanism 30 as the chair 10 rocks, similarly the pin 139 that resides in the slot 144 in the control link 142 does not move relative to the reclining mechanism 30 as the chair 10 rocks. However, the presence of the slot 144 permits the pin 139 to slide and/or reciprocate therein when the chair 10 is rocking. In the illustrated embodiment, the control link 142 moves very little during the rocking motion; however, in other embodiments, the control link 142 may pivot about the pivot 146 during rocking.

To move the chair 10 from the upright position to the TV position of FIG. 3, an occupant of the chair 10 actuates the actuator, which causes the rod 118 to begin to extend from the sleeve 116. Extension of the rod 118 forces the upper swing link 38 to rotate slightly counterclockwise (from the vantage point of FIGS. 2 and 3) about the pivot 40. This movement draws the backpost 48 and rear end of the seat frame 62 downwardly, which causes the seat 13 to increase in pitch

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angle relative to the underlying surface. However, the weight of the seated occupant prevents further rotation of the upper swing link 38 at this point. Because there is little to no fore-and-aft movement of the rear end of the rod 118, the motor unit 114 moves forwardly relative to the base unit 11, thereby driving the lower ottoman swing link 76 counterclockwise about the pivot 78. Rotation of the lower ottoman swing link 76 forces the upper ottoman extension link 80 forward, which in turn draws the upper ottoman swing link 72 counterclockwise around the pivot 74. Also, the lower ottoman extension link 86 moves forwardly more than the upper ottoman extension link 80, such that the outer ottoman bracket 90 and the inner ottoman bracket 96 rotate counterclockwise (rotation of the inner ottoman bracket 96 causes the pin 80a to move in the slot 96a toward the pivot 98. The rotation of the outer and inner ottoman brackets 90, 96 induces the ottomans 17a, 17b to rotate from a vertical orientation to a horizontal orientation. Motion of the ottomans 17a, 17b ceases when a pin 72a on the upper ottoman swing link 72 strikes the upper edge of the lower ottoman extension link 80.

In addition, the rotation of the lower ottoman swing link 76 draws the connecting link 100 forward. Movement of the connecting link 100 rotates the sequencer plate 54 counterclockwise about the pivot 58 with the backpost 48. The drive link 132 of the locking mechanism is driven forward and rotates the bracing link 136. The forward motion of the drive link 132 causes the bracing link 136 to rotate such that its forward end rises and moves rearwardly and such that its rearward end, on which the wheel 138 is mounted, descends and moves forwardly (this rotation is counterclockwise from the vantage point of FIGS. 2 and 3). Also, the elevation of the forward end of the bracing link 136 causes the pin 139 to rise to the top of the slot 144 of the control link 142. The control link 142 also rotates slightly about the pivot 146 to take a more vertical disposition. The movement of these links ceases when the wheel 138 engages a portion of the upper bearing surface of the rocker rail 16 rearward of the portion of the bearing surface beneath the rocker cams 20 (which substantially coincides with the pin 48b reaching the lower end of the slot 54a). In this position, the linkage between the bracing link 136, the drive link 132 and the sequencer plate 54 restrains the mounting bracket 32, thereby preventing the reclining mechanisms 30, and in turn the chair 10, from rocking relative to the base unit 11.

To move the chair 10 from the TV position of FIG. 3 to the fully reclined position of FIG. 4, the occupant can continue to operate the actuator, which extends the rod 118 farther out of the sleeve 116. When the ottomans 17a, 17b are fully extended, the pivots 84, 88 and 94 are in a near "over-center" condition that helps to lock the ottomans 17a, 17b in an extended position. Thus, as the rod 118 continues to extend from the sleeve 116 when the chair is in the TV position, the motor unit 114 cannot move forward relative to the base unit 11 any farther, so the rear end of the rod 118 begins to move rearwardly relative to the base unit 11, resulting in clockwise rotation of the upper swing link 38 about the pivot 40. This rotation both drives the lower swing link 34 counterclockwise about the pivot 36 and drives the forward end of the backpost 48 upward. As a consequence, the backpost 48 rotates counterclockwise about the pivot 58 relative to the sequencer plate 54 (and to the seat adapter 66, the seat frame 62 and the seat bracket 64). Rotation of the backpost 48 and, in turn, the backrest 15, continues until (a) the pin 48b on the backpost 48 reaches the upper end of the slot 54a in the sequencer plate 54 and the pin 52b of the coupling link 52 reaches the rear end of the slot 32a of the mounting bracket 32. In this position, the

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backrest 15 has reclined relative to the seat 13 at a greater angle than in the upright and TV positions.

It can also be seen in FIG. 4, the locking mechanism 130 continues to prevent the chair 10 from rocking as the chair 10 moves to the fully reclined position. It can also be seen that the seat 13 rises during movement from the TV position to the fully reclined position.

The chair 10 can be returned to its TV position from the fully reclined position by the occupant operating the actuator in the reverse direction. Because of the over-center condition of the pivots 82, 84, 88, the ottomans 17a, 17b remain extended, such that retraction of the rod 118 draws the backrest 15 to its upright position. Once the motion of the backrest 15 is complete, the over-center condition of the pivots 84, 88 and 94 can be overcome, and the ottomans 17a, 17b are able to retract to the upright position.

Those skilled in this art will recognize that other variations of the chair 10 are contemplated in connection with the present invention. For example, the power unit 112 may be reversed, such that the motor 114 is at the rear end of the power unit 112 and the rod 118 is at the front end. In such a case, the rod 118 would extend forwardly relative to the base unit 11 as the chair 10 moved from the upright position to the TV position, and the motor 114 would move rearwardly relative to the base unit 11 when the chair moved from the TV position to the fully reclined position. In either instance, the front end of the power unit 112 moves relative to the base unit 11 during the movement from the upright to the TV position, and the rear end of the power unit moves relative to the base unit 11 during the movement from the TV to the fully reclined position.

The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as recited in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

That which is claimed is:

1. A reciprocating seating unit, comprising:
a base unit;
a generally horizontally-disposed seat positioned above the base unit;

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a generally upright backrest positioned above the base unit and substantially rearward of the seat;

an extendable ottoman;

a reclining mechanism attached to the seat, the backrest, the ottoman and the base unit, the reclining mechanism comprising a plurality of pivotally interconnected links; a reciprocating mechanism attached with the base unit and the reclining mechanism, the reciprocating mechanism configured to enable the seat, backrest and reclining mechanism to experience a longitudinally-directed reciprocating motion relative to the base unit; and

a linear actuator attached to the reclining mechanism, the linear actuator configured to move the chair between (a) an upright position, in which the seat is generally horizontally disposed, the backrest is generally vertically disposed, and the ottoman is generally vertically disposed and positioned below the seat, (b) an intermediate TV position, in which the ottoman is generally horizontally disposed in front of the seat and the backrest and the seat substantially maintain the same relationship as they have in the upright position, and (c) a fully reclined position, in which the angle between the backrest and the seat increases;

wherein the linear actuator includes first and second members that move relative to each other, and wherein the first member moves forwardly relative to the base when the seating unit moves to from the upright to the TV position, and wherein the second member moves rearwardly relative to the base when the seating unit moves from the TV position to the fully reclined position.

2. The seating unit defined in claim 1, wherein the reclining mechanism includes a pantographic linkage on which the ottoman is mounted, and wherein in the TV and fully reclined positions, pivots between links of the pantographic linkage form a near over-center arrangement that locks the ottoman in position.

3. The seating unit defined in claim 2, further comprising a seat adapter fixed relative to the seat, and wherein the pantographic linkage is mounted to the seat adapter.

4. The seating unit defined in claim 1, further comprising a locking mechanism that allows the seating unit to rock while in the upright position but prevents rocking of the seating unit while in the TV and fully reclined positions.

5. The seating unit defined in claim 1, wherein the seat rises in moving from the TV position to the fully reclined position.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 8,459,733 B2
APPLICATION NO. : 13/363641
DATED : June 11, 2013
INVENTOR(S) : Hoffman et al.

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims:

Column 10, Claim 1, Line 28: Please correct "moves to from the upright"
to read -- moves from the upright --

Signed and Sealed this
Twentieth Day of May, 2014



Michelle K. Lee
Deputy Director of the United States Patent and Trademark Office

PUBLIC VERSION

IN THE UNITED STATES DISTRICT COURT
FOR THE MIDDLE DISTRICT OF NORTH CAROLINA

ULTRA-MEK, INC.,	)	
	)	
Plaintiff and	)	
Counter Defendant,	)	
	)	
v.	)	1:18-cv-281
	)	
UNITED FURNITURE INDUSTRIES,	)	
INC., OISEYS INTERNATIONAL,	)	
INC., MAN WAH HOLDINGS LTD.,	)	
JIANGSU YULONG SMART	)	
TECHNOLOGY CO., LTD.,	)	
REMACRO MACHINERY	)	
TECHNOLOGY CO., LTD.,	)	
TAIZHOU CHENGUANG VEHICLE CO.,	)	
LTD., and MAN WAH (USA), INC.,	)	
	)	
Defendants and	)	
Counter Claimants.	)	

CONSENT ORDER FOR PERMANENT INJUNCTION

This matter comes before the court on the Motion for Entry of Permanent Injunction on Consent, (Doc. 252), filed Plaintiff Ultra-Mek, Inc. ("Ultra-Mek"), and with the consent of Defendants United Furniture Industries, Inc., Oiseys International, Inc., Taizhou Chenguang Vehicle Co., Ltd., Man Wah (USA), Inc., Man Wah Holdings Ltd., Remacro Machinery

Technology Co., Ltd.¹, and Jiangsu Yulong Smart Technology Co., Ltd. (collectively "Defendants".) The parties have reached a confidential agreement in settlement of the dispute between them and have consented to the entry of this Permanent Injunction on Consent, based on the following stipulated findings of facts and conclusions of law. Upon agreement of the parties and for good cause shown,

IT IS HEREBY ORDERED that the parties' Motion for Entry of Permanent Injunction on Consent, (Doc. 252), is **GRANTED** and that:

FINDINGS OF FACT AND CONCLUSIONS OF LAW

1. Plaintiff Ultra-Mek, Inc. is a company organized under the laws of the State of North Carolina having its headquarters and principal place of business at 487 Bombay Road, Denton, North Carolina 27239.

2. Ultra-Mek is the owner of United States Patent No. 8,016,348 and United States Patent No. 8,297,693 (collectively "the Asserted Patents").

¹ Remacro Machinery Technology Co., Ltd. is now known as Remacro Technology Co., Ltd. and has also been formerly known or referred to as Remacro Machinery & Technology (Wujiang) Co., Ltd. and Remacro Machinery and Technology Co., Ltd. Any references herein to Remacro Machinery Technology Co., Ltd. or RMT refers to Remacro Technology Co., Ltd., its predecessors, successors, assigns, and other names, including Remacro Machinery & Technology (Wujiang) Co., Ltd. and Remacro Machinery and Technology Co., Ltd.

PUBLIC VERSION

3. United Furniture Industries, Inc. ("UFI") is a company organized under the laws of the State of Ohio with its principal place of business at 5380 Hwy 145 South, Tupelo, MS 38801. UFI is a furniture retailer that is engaged in the use, sales, offer for sale, and/or importation of various furniture items, including power-reclining rocking or reciprocating seating units.

4. Oiseys International, Inc. ("Oiseys") is a company organized under the laws of the State of North Carolina with its principal place of business at 3325 White Drive, Belden, MS 38826. Oiseys was at one time engaged in the use, sale, offer for sale, and/or importation of various furniture items, including mechanisms used in power-reclining rocking or reciprocating seating units.

5. Taizhou Chenguang Vehicle Co., Ltd. ("Taizhou") is a company organized under the laws of China with its principal place of business at No. 178, Longxiang Rd., Jiulong Industrial Park, Hailing District, Taizhou City, Jiangsu Province, China 225312. Taizhou was at one time engaged in the manufacture, use, sale, offer for sale, and/or importation of various furniture items, including mechanisms used in power-reclining rocking or reciprocating seating units.

6. Man Wah (USA), Inc. ("Man Wah USA") is a company organized under the laws of the State of Nevada with its principal place of business at 308 Lindsay Street, High Point, NC 27262. Man Wah (USA) is a furniture retailer that uses, sells, offers for sale, and imports various furniture items, including power-reclining rocking or reciprocating seating units.

7. Man Wah Holdings Ltd. ("Man Wah Holdings") is a company organized under the laws of Bermuda, with a principal place of business at 1st Floor, Wah Lai Industrial Center, 10-14 Kwei Tei Street, Fotan, New Territories, Hong Kong 999077. Man Wah Holdings Ltd. is the parent company of Man Wah USA and has an ownership interest in Man Wah USA.

8. Remacro Technology Co., Ltd. f/k/a Remacro Machinery Technology, Co., Ltd., and/or Remacro Machinery & Technology (Wujiang) Co., Ltd. ("RMT") is a company organized under the laws of China with a principal place of business at No.5555, Tongjin Road, Wujiang Economic Development Zone, Suzhou, Jiangsu, China 215200. Man Wah Holdings has an ownership interest in RMT.

9. Jiangsu Yulong Smart Technology Co., Ltd. ("Yulong") is a company organized under the laws of China with a principal place of business at Building 7, No.208, Yaojia Road, Jiulong

Town, Hailing District, Taizhou City, Jiangsu Province, China 225312. RMT has an ownership interest in Yulong. Yulong makes, uses, sells, offers for sale, and/or imports various furniture items, including mechanisms used in power-reclining rocking or reciprocating seating units.

10. On April 10, 2018, Ultra-Mek commenced the above-captioned action (the "Civil Action") by filing a complaint (the "Complaint") against UFI and Oiseys. An Amended Complaint that added the remaining defendants was filed on September 24, 2018. The Amended Complaint alleged claims for patent infringement of each of the Asserted Patents as to all Defendants (Counts I and II), breach of a settlement agreement by Man Wah USA, Man Wah Holdings, and RMT (Count III), and violation of a permanent injunction order and contempt as to all Defendants (Count IV). In particular, Ultra-Mek alleges in its infringement contentions and expert reports that the manufacture, sale, offer for sale, and/or importation of two power reclining mechanisms, referenced herein as the 4152 Mechanism and the 5151 Mechanism, and chairs containing those mechanisms, infringe its patents, breach the settlement agreement, and violate the consent injunction in Ultra-Mek, Inc. v. Man Wah (USA), Inc., Case No. 1:16-cv-00041 (M.D.N.C.). Defendants filed an Answer denying liability as to

each Count, as well as counterclaims for non-infringement and invalidity of each of the Asserted Patents.

11. The parties have agreed to resolve their dispute by entry of this Permanent Injunction on Consent, as well as additional terms subject to a confidential agreement. For purposes of this Permanent Injunction and the associated motion, the Defendants agree to the findings of fact and conclusions of law and remedies set forth in this Order. Ultra-Mek has been represented by Kilpatrick Townsend & Stockton LLP, and the Defendants have been represented by Mayer Brown LLP, Arch & Lake LLP, and Ellis & Winters LLP, for purposes of the confidential agreement, this Order, and associated motion.

12. Pursuant to 28 U.S.C. §§ 1331 and 1338(a), this Court has subject matter jurisdiction over Ultra-Mek's claims against the Defendants.

13. This Court has personal jurisdiction over each of the Defendants because each of the Defendants has a principal place of business in this District, regularly does business in this District, and/or has otherwise submitted itself to the jurisdiction of this Court for purposes of this injunction.

14. Venue is proper in this District pursuant to 28 U.S.C. §§ 1391 and 1400.

PERMANENT INJUNCTION AND ORDER

Based on the parties' stipulation and agreement hereto, it is hereby **ORDERED, ADJUDGED, and DECREED** as follows:

1. The Defendants and their agents, representatives, employees, successors and assigns, and all others in active concert or participation with them, are hereby **PERMANENTLY ENJOINED, and RESTRAINED** from: (i) making, using, selling, or offering for sale in the United States, (ii) importing into the United States, and (iii) actively inducing others in the United States or contributing to others in the United States to make, use, sell, or offer for sale in the United States or to import into the United States, (a) the 4152 Mechanism or the 5151 Mechanism, or any power reclining mechanisms not colorably different therefrom, or (b) power-seating units that incorporate the 4152 Mechanism or the 5151 Mechanism, or that incorporate power reclining mechanisms not colorably different therefrom.

2. This Permanent Injunction and Order shall not apply to a power-seating unit with a reclining mechanism, or to a reclining mechanism, that is authorized by Ultra-Mek, including without limitation, authorized through the sale by Ultra-Mek's licensee, Leggett & Platt, Incorporated, of a reclining mechanism to be included or incorporated into such power-seating units.

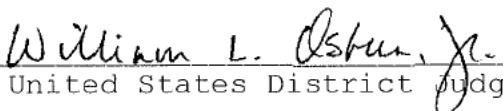
PUBLIC VERSION

3. The Defendants acknowledge that they have knowingly and voluntarily entered into this Permanent Injunction after reviewing the same with their counsel. The Defendants understand the undertakings, obligations, and terms of this Permanent Injunction.

4. This Court shall retain jurisdiction over this matter to enforce a violation of this injunction's terms.

5. The provisions of this injunction shall apply only so long as at least one claim of the Asserted Patents remains valid and enforceable. In addition, the provisions of this injunction shall no longer be in force as of the date both of the Asserted Patents have expired.

This the 17th day of November, 2022.


United States District Judge

PUBLIC VERSION

IN THE UNITED STATES DISTRICT COURT
FOR THE MIDDLE DISTRICT OF NORTH CAROLINA

ULTRA-MEK, INC.,	)	
	)	
Plaintiff and	)	
Counter Defendant,	)	
	)	
v.	)	1:18-cv-281
	)	
UNITED FURNITURE INDUSTRIES,	)	
INC., OISEYS INTERNATIONAL,	)	
INC., MAN WAH HOLDINGS LTD.,	)	
JIANGSU YULONG SMART	)	
TECHNOLOGY CO., LTD.,	)	
REMACRO MACHINERY	)	
TECHNOLOGY CO., LTD.,	)	
TAIZHOU CHENGUANG VEHICLE CO.,	)	
LTD., and MAN WAH (USA), INC.,	)	
	)	
Defendants and	)	
Counter Claimants.	)	

FINAL JUDGMENT

IT IS HEREBY ORDERED, ADJUDGED, AND DECREED that Defendants,¹ their agents, representatives, employees, successors and assigns, and all others in active concert or participation with them are **PERMANENTLY ENJOINED AND RESTRAINED** in the manner set forth in the Permanent Injunction and Order filed on the

¹ Remacro Machinery Technology Co., Ltd. is now known as Remacro Technology Co., Ltd. and has also been formerly known or referred to as Remacro Machinery & Technology (Wujiang) Co., Ltd. and Remacro Machinery and Technology Co., Ltd. Any references herein to Remacro Machinery Technology Co., Ltd. refers to Remacro Technology Co., Ltd., its predecessors, successors, assigns, and other names, including Remacro Machinery & Technology (Wujiang) Co., Ltd. and Remacro Machinery and Technology Co., Ltd.

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17th day of November, 2022. This court retains jurisdiction over this matter to enforce a violation of the injunction's terms.

Except as provided in the Permanent Injunction and Order filed on the 17th day of November, 2022 and this Judgment, any and all other claims and causes of action are dismissed as follows:

Ultra-Mek's claims against Defendants are hereby **DISMISSED WITH PREJUDICE**;

Defendants' counterclaims against Ultra-Mek relating to non-infringement are hereby **DISMISSED WITH PREJUDICE**; and

Defendants' counterclaims against Ultra-Mek relating to invalidity are hereby **DISMISSED WITHOUT PREJUDICE**.

Ultra-Mek shall not be responsible for Defendants' attorney's fees, expenses, and costs, nor shall Defendants be responsible for Ultra-Mek's attorney's fees, expenses, and costs.

The Clerk shall terminate all pending motions and close this case.

This the 13th day of February, 2023.


United States District Judge

IN THE UNITED STATES DISTRICT COURT
FOR THE MIDDLE DISTRICT OF NORTH CAROLINA

Ultra-Mek, Inc.,	)	
	)	
Plaintiff,	)	
	)	
v.	)	1:16CV41
	)	
Man Wah (USA), Inc.,	)	
	)	
Defendant.	)	
	)	

ENTRY OF PERMANENT INJUNCTION ON CONSENT

Plaintiff Ultra-Mek, Inc. (“Ultra-Mek”) and Defendant Man Wah (USA), Inc. (“Man Wah”), and its parent company Man Wah Holdings Ltd. and its affiliate Remacro Machinery & Technology, Co., Ltd., have reached a confidential agreement in settlement of the dispute between them and have consented to the entry of this Permanent Injunction on Consent, based on the following stipulated findings of facts and conclusions of law, which the Court hereby adopts for purposes of the entry of the Permanent Injunction.

FINDINGS OF FACT AND CONCLUSIONS OF LAW

1. Ultra-Mek is a company organized under the laws of the State of North Carolina having its headquarters and principal place of business at 487 Bombay Road, Denton, North Carolina 27239.
2. Ultra-Mek is the owner of United States Patent Number 7,997,644, United States Patent Number 8,016,348, and United States Patent Number 8,297,693 (collectively “the Asserted Patents”).

3. Man Wah is a company organized under the laws of the State of Nevada with its principal place of business at 688 North Main Street, High Point, NC 27260. Man Wah is a furniture retailer that uses, sells, offers for sale, and imports various furniture items, including power-reclining rocking, gliding, or reciprocating seating units.

4. Man Wah Holdings Ltd. is a Bermuda company, with a principal place of business at 1st Floor, Wah Lai Industrial Center, 10-14 Kwei Tei Street, Fotan, New Territories, Hong Kong. Man Wah Holdings Ltd. is the parent company of Man Wah and has an ownership interest in Man Wah.

5. Remacro Machinery & Technology, Co., Ltd., aka Remacro Machinery Technology (Wuijiang) Co., Ltd., (“RMT”), is a Peoples Republic of China company, with a principal place of business at No.5555, Tongjin Road, Wuijiang Economic Development Zone, Suzhou, Jiangsu, China, 215200. RMT is an affiliate company of Man Wah, and Man Wah Holdings Ltd. has an ownership interest in RMT. RMT makes, uses, sells, offers for sale, and/or imports reclining mechanisms, including reclining mechanisms that are incorporated into Man Wah’s products.

6. On January 15, 2016, Ultra-Mek commenced the above-captioned action (the “Civil Action”) by filing a complaint (the “Complaint”) against Man Wah. The Complaint alleged claims for patent infringement of each of the Asserted Patents.

7. The parties, Man Wah Holdings Ltd., and RMT have agreed to resolve their dispute by entry of this Permanent Injunction on Consent, as well as additional terms subject to a confidential agreement. For purposes of this Permanent Injunction and the associated motion, Man Wah Holdings Ltd. and RMT agree to the findings of fact and

conclusions of law and remedies set forth in this Order. Man Wah Holdings Ltd. and RMT have been represented by Nexsen Pruet, PLLC for purposes of the confidential agreement, this Order, and associated motion.

8. Pursuant to 28 U.S.C. §§ 1331 and 1338(a), this Court has subject matter jurisdiction over Ultra-Mek's claims against Man Wah.

9. This Court has personal jurisdiction over Man Wah because Man Wah's principal place of business is in this District and has otherwise submitted itself to the jurisdiction of this Court.

10. Venue is proper in this District pursuant to 28 U.S.C. §§ 1391 and 1400.

PERMANENT INJUNCTION AND ORDER

Based on the parties' stipulation and agreement hereto, it is hereby **ORDERED, ADJUDGED, and DECREED** as follows:

1. Man Wah and its members, agents, representatives, employees, successors and assigns, and all others in active concert or participation with them, including Man Wah Holdings Ltd. and RMT, are hereby **PERMANENTLY ENJOINED, and RESTRAINED** from: (i) making, using, selling, or offering for sale in the United States, (ii) importing into the United States, and (iii) actively inducing others in the United States or contributing to others in the United States (a) to make, use, sell, or offer for sale in the United States or (b) to import into the United States the following:

- a. Power seating units (chairs, recliners, and other seating units) designated, as of or prior to the date of this Order, with product numbers, references, or names K859, K859M, K860, K861, K862, K862M, K863 (35686), K863,

K863M, K865 (2529), K865, K877M K883M, K885, K885M, K886, K887, K887M, K895M, RUK887M, XW9788, XW9788 (3512/3515), UK861M, UK863, UK863M, UK865, UK865M, UK876M, UK877M, UK881, UK881M, UK882, UK882M, UK883, UK883M, UK885, UK885M, UK887, UK887M, UK888, UK888M or UK895M, and power-seating units that have each element and perform substantially the same function in substantially the same way to yield the substantially same result as such power seating units;

- b. Power-seating units (chairs, recliners, and other seating units) that incorporate reclining mechanisms designated, as of or prior to the date of this Order, with product numbers, references, or names RMT No. 661X or RMT No. 761X, and power-seating units that incorporate reclining mechanisms that have each element and perform substantially the same function in substantially the same way to yield the substantially same result as such reclining mechanisms; and
- c. Reclining mechanisms designated, as of or prior to the date of this Order, with product numbers, references, or names RMT No. 661X or RMT No. 761X, and reclining mechanisms that have each element and perform substantially the same function in substantially the same way to yield the substantially same result as such reclining mechanisms.

2. This Permanent Injunction and Order shall not apply to a power-seating unit with a reclining mechanism, or to a reclining mechanism, that is authorized by Ultra-

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Mek at Ultra-Mek's sole discretion, including without limitation, authorized through the sale by Ultra-Mek's licensee Leggett & Platt, Incorporated of a reclining mechanism to be included or incorporated into such power-seating units.

3. Ultra-Mek, Man Wah, Man Wah Holdings Ltd., and RMT acknowledge that they have knowingly and voluntarily entered into this Permanent Injunction after reviewing the same with their counsel. Ultra-Mek, Man Wah, Man Wah Holdings Ltd., and RMT understand the undertakings, obligations, and terms of this Permanent Injunction.

4. This Court shall retain jurisdiction over this matter to enforce a violation of this injunction's terms.

It is SO ORDERED, this the 17th day of March, 2017.

/s/ N. Carlton Tilley, Jr.
Senior United States District Judge